

5G RAN

High Speed Mobility Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Replaced the reserved-parameter bit that controls "service-based carrier selection enhancement in high-speed scenarios" with a formal-parameter option. For details, see 14.2 Uplink-Service-based Carrier Selection in High-Speed Scenarios .	Modified parameters: Added RSVDSWPARAM1_BIT11 of the reserved parameter NRCeIIrsvd.RsvdSwParam1 to the disuse list and replaced it with the newly added HIGH_SPD_SERV_CARR_SEL_ENH_SW option of the NRCeIIFeatureSwitch.MultiFreqAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added inter-TRP PDCCH interference avoidance for VoNR. For details, see 9.2 Inter-TRP PDCCH Interference Avoidance for VoNR .	Modified parameters: - Added the PDCCH_VO NR_INTF_AVOID_SW option to the NRDUCellPdcch.PdcchAlgoExtSwitch parameter. - Added the TOP_VO NR_U E_CCE_ALLOC_OPT_SW option to the NRDUCellPdcch.PdcchAlgoExtSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added "downlink priority enhancement for voice service UEs in a high-speed cell". For details, see 10.2 Downlink Priority Enhancement for Voice Service UEs in a High-Speed Cell .	Modified parameter: Added the VONR_DL_PRI_SCH_ENH_SW option to the NRDUCellDisch.VoiceDischSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Replaced the reserved-parameter bit that controls "uplink frequency offset range estimation optimization for high-speed cells" with a formal-parameter option. For details, see: • 8.1 Uplink Frequency Offset Range Estimation • 8.1.4 Operation and Maintenance	Modified parameters: Added RSVDSWPARAM1_BIT22 of the reserved parameter NRDUCellRsvedExt01.RsvedSwParam1 to the disuse list and replaced it with the newly added UL_FREQ_OFFSET_RANGE_EST_OPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added "SRS resource extension in a high-speed cell". For details, see 7.3 SRS Resource Extension in a High-Speed Cell .	Modified parameter: Added the SRS_RES_EXT_SW option to the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added inter-frequency directional handover for high-speed UEs. For details, see 13.2 Inter-Frequency Directional Handover for High-Speed UEs .	Modified parameter: Added the HS_INTER_FREQ_DIR_HO_SW option to the NRCellMobilityConfig.InterFreqHoExtSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between High-speed Railway Superior Experience and Uplink Boosting 2.0. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between High-speed Railway Superior Experience and L/NR Inter-band Deep Coordination. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between "interference coordination between public and dedicated networks" and L/NR Inter-band Deep Coordination. For details, see 12.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between High-speed Railway Superior Experience and blind SRS interference suppression enhancement. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Removed the mutually exclusive relationship between High-speed Railway Superior Experience and power saving BWP. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between High-speed Railway Superior Experience and RF module fast deep dormancy. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between High-speed Railway Superior Experience and "uplink 4-layer transmission for a single FWA user". For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added uplink-large-packet-based frequency priority. For details, see 14.2 Uplink-Service-based Carrier Selection in High-Speed Scenarios .	Modified parameter: Added the UL_LARGE_PKT_FREQ_PRI_SW option to the NRCellMobilityConfig.InterFreqHoExtSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added downlink-service-based carrier selection in high-speed scenarios. For details, see 14.3 Downlink-Service-based Carrier Selection in High-Speed Scenarios .	Modified parameter: Added the HIGH_SPD_DL_SERVICE_CARR_SELECTION option to the NRCellMobilityConfig . <i>InterFrequencyExtSwitch</i> parameter. Added the NRCellFrequencyRelation . <i>DLTrafficFrequencyPriority</i> parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added uplink channel densified equalization for high-speed UEs. For details, see 8.7 Uplink Channel Densified Equalization for High-Speed UEs .	Modified parameter: Added the HS_UL_LOW_COST_DENSE_EQUALIZATION option to the NRDUCellHighSpeedMobility . <i>HighSpeedAlgorithmOptionSwitch</i> parameter. Added parameters: • NRDUCellHighSpeedMobility. <i>ULRank1DenseEqualMcsOffset</i> • NRDUCellHighSpeedMobility. <i>ULRank2DenseEqualMcsOffset</i>	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Removed the mutually exclusive relationship between High-speed Railway Superior Experience and "quadruple code division for format-4 resource allocation" from 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Optimized the impact relationships between High-speed Railway Superior Experience and the following functions: RedCap UL skipping and RedCap DRX adaptation. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
 - Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.
 - Revised the processing mechanism of the directional inter-TRP handover function. For details, see [5.2.1 Principles](#).
 - Added descriptions of the mutually exclusive relationships between "NR DU cell resource management between network slices" and the following functions: uplink RB restriction for the low-speed cell and downlink RB restriction for the low-speed cell. For details, see [12.1.3.2.2 Mutually Exclusive Functions](#).
 - Revised descriptions of the principles of "in-stop-train UE identification and experience guarantee". For details, see [11.5.1 Principles](#).
 - Changed the mutually exclusive relationship between High-speed Railway Superior Experience and RF module deep dormancy into an impact relationship. For details, see [4.3.2.3 Function Impacts](#).
 - Changed the mutually exclusive relationship between High-speed Railway Superior Experience and RF module super dormancy into an impact relationship. For details, see [4.3.2.3 Function Impacts](#).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-030207	High-speed Railway Superior Experience	4 High-speed Railway Superior Experience Basic Functions 5 High-Speed Cell Switching Performance Optimization 6 Coverage Improvement for High-Speed Cells 7 High-Speed Cell Capacity Optimization 8 Frequency Offset and Channel Time-Variation Optimization for High-Speed Cells 9 Interference Resistance Optimization for High-Speed Cells 10 High-Speed Cell Scheduling Performance Optimization 11 Differentiated Scheduling Policy Management for High- and Low-Speed UEs 12 Interference Policy Management for Public and Dedicated Networks 13 Mobility Enhancement Policy Management 14 Multi-Carrier Coordination Policy Management

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
High-speed Railway Superior Experience	None	4 High-speed Railway Superior Experience Basic Functions
High-speed cell performance optimization	The following functions take effect only on SA UEs: • Incoming handover protection for high-speed cells • Adaptive handover in high-speed cells For other functions, there is no difference.	5 High-Speed Cell Switching Performance Optimization 6 Coverage Improvement for High-Speed Cells 7 High-Speed Cell Capacity Optimization 8 Frequency Offset and Channel Time-Variation Optimization for High-Speed Cells 9 Interference Resistance Optimization for High-Speed Cells 10 High-Speed Cell Scheduling Performance Optimization
High speed specified policy management	The following functions take effect only on SA UEs: • "Measurement-based outmigration only for SA UEs" in "low-speed UE outmigration" • High-speed UE return For other functions, there is no difference.	11 Differentiated Scheduling Policy Management for High- and Low-Speed UEs 12 Interference Policy Management for Public and Dedicated Networks 13 Mobility Enhancement Policy Management 14 Multi-Carrier Coordination Policy Management

Differences Between High Frequency Bands and Low Frequency Bands

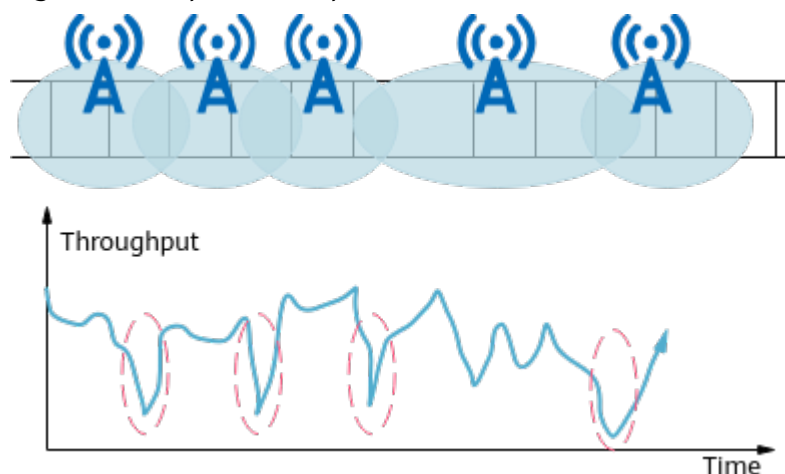
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
High-speed Railway Superior Experience	Supported only in low frequency bands	4 High-speed Railway Superior Experience Basic Functions
High-speed cell performance optimization	Supported only in low frequency bands	5 High-Speed Cell Switching Performance Optimization 6 Coverage Improvement for High-Speed Cells 7 High-Speed Cell Capacity Optimization 8 Frequency Offset and Channel Time-Variation Optimization for High-Speed Cells 9 Interference Resistance Optimization for High-Speed Cells 10 High-Speed Cell Scheduling Performance Optimization
High speed specified policy management	Supported only in low frequency bands	11 Differentiated Scheduling Policy Management for High- and Low-Speed UEs 12 Interference Policy Management for Public and Dedicated Networks 13 Mobility Enhancement Policy Management 14 Multi-Carrier Coordination Policy Management

3 Overview

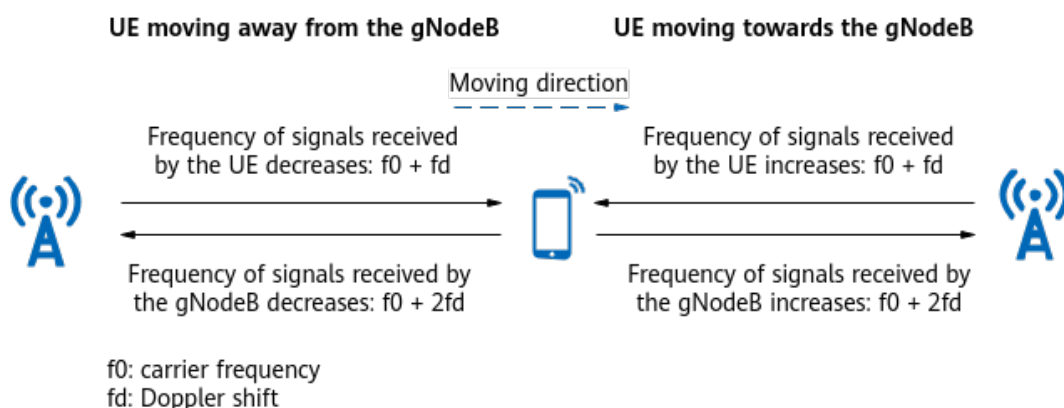
In high-speed transit, such as high-speed railways and maglev trains, UEs are susceptible to Doppler shifts and frequent inter-cell handovers. This may result in performance issues such as high service drop rates, difficult access, and low throughput. [Figure 3-1](#) shows the impact of frequent handovers.

Figure 3-1 Impact of frequent handovers



In high speed mobility scenarios, the received signal frequency at the receiver changes due to the Doppler effect. As a result, data is transmitted and received at different frequencies. This phenomenon is referred to as the Doppler shift. [Figure 3-2](#) shows the uplink and downlink frequency offsets when a UE is moving relative to a gNodeB.

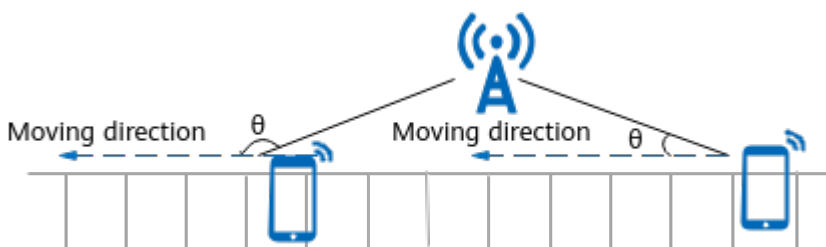
Figure 3-2 Doppler shift



The Doppler shift is calculated as follows: $f_d = f_0 / C \times v \times \cos\theta$

In the preceding formula: C is the electromagnetic wave propagation speed, v is the moving speed of the UE, and θ is the angle between the direction in which the UE is moving relative to the gNodeB and the direction in which the gNodeB transmits signals. **Figure 3-3** shows the angle θ .

Figure 3-3 Diagram of angle θ



As the preceding formula indicates, Doppler shifts increase as the moving speed of a UE relative to the gNodeB increases. A Doppler shift alters the frequency for data receiving, which causes intra- and inter-OFDM symbol interference. In severe cases, receivers fail to demodulate data from transmitters and UEs cannot access the network. Therefore, in high speed mobility scenarios, Doppler shifts must be corrected to mitigate their impacts and meet performance requirements. Given this context, Huawei has introduced the High-speed Railway Superior Experience feature.

If an RF module of 8T8R or lower is used, Hyper Cell or Cell Combination is recommended, to reduce frequent handovers between cells. For details about Hyper Cell, see *Hyper Cell*. For details about Cell Combination, see *Cell Combination*.

High-Speed Cell Performance Optimization

Based on the main optimization effects, all functions in high-speed cell performance optimization are classified into handover performance optimization, coverage improvement optimization, capacity optimization, frequency offset and channel time-varying optimization, anti-interference optimization, and scheduling performance optimization.

Handover performance optimization:

Function Name	Solution Principles	Section
Incoming handover protection for high-speed cells	After this function is enabled, the number of ping-pong handovers of UEs in high-speed cells decreases. This function takes effect only on SA UEs.	5.1 Incoming Handover Protection for High-speed Cells
Directional inter-TRP handover	After this function is enabled, high-speed UEs can be handed over to a specific target TRP, improving the timeliness of intra-cell inter-TRP handovers for high-speed UEs.	5.2 Directional Inter-TRP Handover
Adaptive handover in high-speed cells	After this function is enabled, the base station controls the offset level for adaptive handovers in a high-speed cell to improve the timeliness of inter-cell handovers for high-speed UEs.	5.3 Adaptive Handover in High-Speed Cells

Coverage improvement optimization:

Function Name	Solution Principles	Section
Uplink waveform adaptation in high-speed cells	After this function is enabled, the network side instructs a UE to adaptively select a cyclic prefix-orthogonal frequency division multiplexing (CP-OFDM) or discrete Fourier transform-spread OFDM (DFT-s-OFDM) waveform.	6.1 Uplink Waveform Adaptation in High-Speed Cells
Adaptation of initial access beams in multi-SSB scenarios	After this function is enabled, the cell coverage rate and high-speed UE access reliability can both be ensured when narrow beams are used to improve SSB coverage.	6.2 Adaptation of Initial Access Beams in Multi-SSB Scenarios

Capacity optimization:

Function Name	Solution Principles	Section
SRS period and CDM adaptation	After this function is enabled, the base station adaptively allocates SRS resources of different periods to newly admitted UEs based on the number of UEs in a cell and supports SRS CDM between two UEs.	7.1 SRS Period and CDM Adaptation

Function Name	Solution Principles	Section
Uplink DCI extension-based scheduling in high-speed cells	After this function is enabled, uplink DCI can be transmitted in addition downlink slots in a high-speed cell in heavy-load scenarios. This improves the utilization of uplink RB resources and increases the average uplink cell throughput.	7.2 Uplink DCI Extension-based Scheduling in High-Speed Cells
SRS resource extension in a high-speed cell	After this function takes effect, the base station allocates more SRS resources to UEs and more RedCap UEs can be served.	7.3 SRS Resource Extension in a High-Speed Cell
Uplink and downlink additional DMRS position adaptation	After this function is enabled, when a train has arrived at a cell, the cell instructs neighboring high-speed cells to adjust the number of uplink and downlink additional DMRS positions based on the train speed. In this way, UEs in the train can obtain the optimal network performance after entering areas covered by these neighboring cells.	7.4 Uplink and Downlink Additional DMRS Position Adaptation
MU-MIMO in high-speed railway scenarios	This function enables the collaboration with MU-MIMO in high-speed railway scenarios.	7.5 MU-MIMO in High-Speed Railway Scenarios (Trial)

Frequency offset and channel time-varying optimization:

Function Name	Solution Principles	Section
Uplink frequency offset range estimation	After this function is enabled, the gNodeB can accurately measure the frequency offset of high-speed UEs even when one uplink front-loaded DMRS position and one uplink additional DMRS position are configured.	8.1 Uplink Frequency Offset Range Estimation
Rank restriction	This function improves the average downlink throughput of high-speed UEs by limiting the effective ranks to rank 1 and rank 2.	8.2 Rank Restriction

Function Name	Solution Principles	Section
Multi-TRS measurement (trial)	After this function is enabled, a UE that is capable of measuring multiple sets of TRSs can simultaneously measure multiple sets of TRSs in the serving cell. This reduces the risk of performance deterioration caused by frequency offset adjustment during handovers of high-speed UEs.	8.3 Multi-TRS Measurement (Trial)
Uplink SINR adjustment	This function adjusts the SINR used for MCS index selection to make the MCS index more suitable for the actual channel quality. In this way, the average uplink throughput of high-speed UEs can be increased in scenarios where channel conditions change greatly with time.	8.4 Uplink SINR Adjustment
Uplink MCS index selection optimization based on narrowband SRS	After this function is enabled, filtered SRS SINRs of the frequency hopping subbands are used in MCS index selection for high-speed UEs involved in SRS frequency hopping.	8.5 Uplink MCS Index Selection Optimization based on Narrowband SRS
Uplink frequency offset filtering adaptation	After this function is enabled, the base station adjusts the uplink frequency offset filtering coefficient based on the frequency offset fluctuation, improving the accuracy of uplink frequency offset measurement results.	8.6 Uplink Frequency Offset Filtering Adaptation
Uplink channel densified equalization for high-speed UEs	The number of symbols used for uplink equalization calculation in the time domain is increased to effectively resist time-dependent channel variations and improve uplink performance.	8.7 Uplink Channel Densified Equalization for High-Speed UEs

Anti-interference optimization:

Function Name	Solution Principles	Section
Inter-TRP interference coordinated scheduling for high-speed railways	After this function is enabled, interference on traffic channels between TRPs is reduced.	9.1 Inter-TRP Interference Coordinated Scheduling for High-Speed Railways
Inter-TRP PDCCH interference avoidance for VoNR	After this function is enabled, interference between the PDCCHs of VoNR UEs served by different TRPs is reduced by staggering time-domain and frequency-domain resources.	9.2 Inter-TRP PDCCH Interference Avoidance for VoNR

Scheduling performance optimization:

Function Name	Solution Principles	Section
Downlink robust weight in high-speed cells	This function enhances the energy in the main path direction of a beam to improve the matching with TRP-level channels. This helps to improve beamforming gains and obtain the beamforming gains of massive MIMO.	10.1 Downlink Robust Weights in High-Speed Cells
Downlink Priority enhancement for Voice Service UEs in a High-Speed Cell	After this function is enabled, the downlink scheduling priority of VoNR UEs is increased to prevent voice packets from failing to be scheduled due to signaling impact.	10.2 Downlink Priority Enhancement for Voice Service UEs in a High-Speed Cell

High Speed Specified Policy Management

The high-speed railway dedicated network is referred to as the dedicated network, which is deployed along high-speed railways. Dedicated-network cells cover high-speed railways and alongside areas and they are configured as high-speed cells (by setting `NRDUCell.HighSpeedFlag` to `HIGH_SPEED`). Public-network cells cover areas adjacent to the high-speed railway dedicated network and provide services for non-high-speed railway UEs. Public-network

cells are configured as low-speed cells (by setting **NRDUCell.HighSpeedFlag** to **LOW_SPEED**).

In intra-frequency band networking of high-speed railway dedicated networks and public networks, the following scenarios may exist on high-speed railway dedicated networks:

- Scenario 1: The high-speed railway dedicated network serves some public-network low-speed UEs. As a result, there are both dedicated-network high-speed UEs and public-network low-speed UEs in high-speed cells on the high-speed railway dedicated network. Public-network low-speed UEs occupy resources of the dedicated network, affecting user experience on the dedicated network.
- Scenario 2: The public network covers areas around the high-speed railway dedicated network, and the public network may cause severe interference to the dedicated network.

In the preceding scenarios, the high speed specified policy management function is introduced to ensure user experience of high-speed UEs on the dedicated network. The high speed specified policy management function enables differentiated processing on high-speed and low-speed UEs in high-speed cells of high-speed railway dedicated networks. In addition, when a high-speed train arrives in the areas with public network coverage, coordination with surrounding public networks can be performed to improve the experience of dedicated network users. High speed specified policy management includes differentiated scheduling for high- and low-speed UEs, interference between public and dedicated networks, mobility enhancement, and multi-carrier coordination.

Differentiated scheduling for high- and low-speed UEs:

Function Name	Solution Principles	Section
Differentiated scheduling	This function increases the scheduling priority of high-speed UEs in high-speed cells to ensure that high-speed UEs are preferentially scheduled compared with low-speed UEs.	11.1 Differentiated Scheduling
Differentiated configuration between high- and low-speed UEs	This function provides more flexible differentiated configuration policies for high-speed and low-speed UEs in high-speed cells. The policies include differentiated handover, differentiated DRX, differentiated EPS fallback, and differentiated UE inactivity timer.	11.2 Differentiated Configuration Between High- and Low-speed UEs

Function Name	Solution Principles	Section
Differentiated A2 threshold configuration for high-speed UEs	In scenarios where high-speed cells and adjacent public-network cells use different frequency bands, differentiated A2 thresholds can be configured for high-speed UEs that are newly handed over to high-speed cells. In this way, high-speed UEs that do not stop at railway station platforms are not likely to start gap-assisted measurements, improving user experience.	11.3 Differentiated A2 Threshold Configuration for High-Speed UEs
Low-speed UE performance optimization	This function optimizes the performance of low-speed UEs in high-speed cells.	11.4 Low-Speed UE Performance Optimization
In-stop-train UE identification and experience guarantee	The base station can guarantee user experience of in-stop-train UE through the mechanism consisting of the following procedures in sequence: identifying UEs in the train that stops at a platform, starting scheduling guarantee, and exiting in-stop-train UE experience guarantee.	11.5 In-stop-train UE identification and experience guarantee

Interference between public and dedicated networks:

Function Name	Solution Principles	Section
Interference coordination between public and dedicated networks	The following functions reduce interference from neighboring cells on the public network to the high-speed railway dedicated network. • RB restriction and power restriction for interference coordination between public and dedicated networks • Automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks • Beam avoidance for interference coordination between public and dedicated networks (low-frequency TDD) • TRP-based precise interference coordination for interference coordination between public and dedicated networks • Beam-level power reduction for interference coordination between public and dedicated networks	12.1 Interference Coordination Between Public and Dedicated Networks
Configuration-based scheduling for SSB and RMSI interference avoidance	It is recommended that this function be enabled when spectrum resources overlap between public-network cells and dedicated-network cells, their SSB frequency-domain positions are different, and Xn links between public and dedicated networks are disconnected (for example, the public network and dedicated network use devices from different vendors). After this function is enabled, scheduling avoidance is performed for victim dedicated-network cells based on configurations.	12.2 Configuration-based Scheduling for SSB and RMSI Interference Avoidance

Mobility enhancement:

Function Name	Solution Principles	Section
High-speed UE directional handover	In high-speed railway dedicated networks, after this function is enabled, high-speed UEs can be handed over only to high-speed cells but not low-speed cells.	13.1 High-Speed UE Directional Handover

Function Name	Solution Principles	Section
Inter-frequency directional handover for high-speed UEs	With this function enabled, high-speed UEs in a high-speed cell on a high-speed-railway dedicated network are not allowed to be handed over to neighboring cells working on frequencies for low speed mobility.	13.2 Inter-Frequency Directional Handover for High-Speed UEs
Low-speed UE outmigration	When the high-speed railway does not pass by, the base station migrates a certain number of low-speed UEs from the high-speed railway dedicated network to the public network in each period. Low-speed UE outmigration includes three functions: blind redirection-based outmigration, measurement-based outmigration, and measurement-based outmigration only for SA UEs.	13.3 Low-Speed UE Outmigration
High-speed UE return	This function enables high-speed UEs that access public-network cells to be redirected back to dedicated-network cells.	13.4 High-Speed UE Return
Broadcast of R17-compliant HSDN-related IEs	This function helps the base station broadcast R17-compliant HSDN-related IEs, improving the accuracy of reselection to high-speed cells and high-speed neighboring cells for high-speed UEs.	13.5 Broadcast of R17-compliant HSDN-related IEs
High-speed attribute indication of neighboring E-UTRAN cells	This function helps UEs identify whether neighboring E-UTRAN frequencies are high-speed cell frequencies and optimize camping policies.	13.6 High-Speed Attribute Indication of Neighboring E-UTRAN Cells

Multi-carrier coordination:

Function Name	Solution Principles	Section
Load balancing in fixed proportions for RRC_IDLE UEs	In high-speed railway multi-carrier networking scenarios, load balancing among multiple carriers allows the number of UEs served by each carrier to reach the expected proportion.	14.1 Load Balancing in Fixed Proportions for RRC_IDLE UEs

Function Name	Solution Principles	Section
Uplink-service-based carrier selection in high-speed scenarios	If the base station identifies uplink-preferred UEs, the UEs perform inter-frequency measurements. In this case, the base station performs carrier selection based on the frequency priorities for uplink services to ensure differentiated selection of optimal carriers for UEs with different preferred services.	14.2 Uplink-Service-based Carrier Selection in High-Speed Scenarios
Downlink-service-based carrier selection in high-speed scenarios	If the base station identifies downlink-preferred UEs, the UEs perform inter-frequency measurements. In this case, the base station performs carrier selection based on the frequency priorities for downlink services to ensure differentiated selection of optimal carriers for UEs with different preferred services.	14.3 Downlink-Service-based Carrier Selection in High-Speed Scenarios

It is recommended that the high speed specified policy management function be enabled in high-speed cells where UEs move faster than 120 km/h. Do not enable this function in high-speed cells near high-speed railway platforms.

4 High-speed Railway Superior Experience Basic Functions

4.1 Principles

Basic functions include configuring high-speed cells on the base station, identifying high-speed UEs, and performing uplink frequency correction for high-speed UEs to reduce the impact of the Doppler effect. For details, see [4.1.1 High-Speed Cell](#), [4.1.2 High-speed UE Identification](#), and [4.1.3 Frequency Correction](#).

4.1.1 High-Speed Cell

Because the NR system is sensitive to Doppler shifts due to closely-spaced subcarriers, the effect of Doppler shifts needs to be better controlled. In high-speed railway scenarios, where UEs travel between 120 km/h and 450 km/h, it is recommended that cells be configured as high-speed cells.

A cell can be configured as a high-speed cell using the **NRDUCell.HighSpeedFlag** parameter. [Table 4-1](#) describes suggestions for configuring high-speed and low-speed cells.

Table 4-1 Suggestions for configuring high-speed and low-speed cells

UE Speed in the Cell	Cell Configuration (NRDUCell.HighSpeedFlag)
120 km/h to 450 km/h	HIGH_SPEED
Less than 120 km/h	LOW_SPEED

In high-speed cells, if the **R16_HIGH_SPEED_MEAS_FLAG** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter is selected, the gNodeB broadcasts the high-speed cell measurement flag (indicated by the **highSpeedMeasFlag** field) in the system information. UEs can identify high-speed cells based on this flag and perform corresponding optimization. For details about

the high-speed cell measurement flag, see section 6.3 "RRC information elements" in 3GPP TS 38.331 V17.3.0.

4.1.2 High-speed UE Identification

The mechanism for determining the UE type in a high-speed cell is as follows:

- If the frequency shift of a UE is greater than or equal to the high-speed frequency offset threshold (***NRDUCellHighSpeedMob.HighSpeedFreqOffsetThld***), the UE is identified as a high-speed UE. According to the Doppler shift calculation formula, the frequency offset of a UE is affected by factors such as the moving speed and carrier frequency. Therefore, the high-speed frequency offset threshold must be configured based on the cell frequency band and UE moving speed. To ensure that a UE moving at about 120 km/h can be accurately identified as a high-speed UE, it is recommended that the high-speed frequency offset threshold be set as follows for common frequency bands:
 - When the frequency band is n38 or n41, it is recommended that this parameter be set to **560**.
 - When the frequency band is n77 or n78, it is recommended that this parameter be set to **760**.

NOTE

The configured parameter is a bidirectional frequency offset. Therefore, when calculating the moving speed, you need to divide the configured value by 2 and then use the result in the Doppler shift calculation formula.

- A UE is considered as a low-speed UE if its camping duration in the current TRP is greater than the value of the ***NRDUCellHighSpeedMob.LowSpeedStayTimeThld*** parameter or the frequency offset threshold is less than the value of the ***NRDUCellHighSpeedMob.LowSpeedFreqOffsetThld*** parameter.

NOTE

Do not set the high-speed frequency offset threshold to an excessively large or small value. If this parameter is set to an excessively large value, it is difficult for high-speed UEs to be identified. If this parameter is set to an excessively small value, low-speed UEs may be incorrectly identified as high-speed ones, and the UEs' high-speed attribute may change frequently.

It is recommended that the low-speed camping time threshold be set to a proper value (15s by default) on the live network. If this parameter is set to a large value, low-speed UEs cannot be promptly identified. If this parameter is set to a small value, high-speed UEs may be mistakenly identified as low-speed UEs, and the UEs' high-speed attribute may change frequently.

After a UE is identified as a high-speed UE, the UE's high-speed attribute is transferred to the target cell during an Xn-based handover between high-speed cells (only supported by Huawei base stations).

4.1.3 Frequency Correction

After a cell is configured as a high-speed cell, uplink frequency correction is enabled for high-speed UEs in the cell.

The gNodeB uses paired uplink DMRS symbols to estimate the frequency offset. Given that the impact of a frequency offset on a signal is represented as phase

rotation of the signal in the time domain, a corresponding frequency offset f_d can be calculated by using a phase difference between two adjacent DMRS symbols in a slot. The obtained frequency shift f_d is used as continuous input for UE frequency correction. This reduces the impact of Doppler shifts as much as possible and meets the performance requirement of high speed mobility.

Additional DMRS is required in NR because front-loaded DMRS is insufficient for rapidly changing channel conditions in medium- and high-speed scenarios. It is configured to meet the accuracy requirements of time-variant channel estimation. Generally, a high-speed cell must be configured with one uplink front-loaded DMRS position and two uplink additional DMRS positions (by setting **NRDUCellPusch.UlAdditionalDmrsPos** to **POS2**). Note that in special scenarios, a high-speed cell can be configured with one uplink front-loaded DMRS position and one uplink additional DMRS position (by setting **NRDUCellPusch.UlAdditionalDmrsPos** to **POS1**), as shown in [Table 4-2](#).

Table 4-2 Scenarios supporting one uplink additional DMRS position

RAT	Frequency Band	Maximum Train Speed	Number of Uplink Front-loaded DMRS Positions	Number of Uplink Additional DMRS Positions
NR TDD	2.6 GHz	≤ 300 km/h	1	1
NR TDD	3.5 GHz	≤ 200 km/h	1	1

The impact of frequency shifts on signals is represented as the phase rotation of the signals in the time domain. In the uplink, the gNodeB uses pairs of DMRS symbols to estimate frequency shifts. The fundamental principle is to calculate the frequency shift f_d based on the phase difference between the two adjacent DMRS symbols in a timeslot. The obtained frequency shift f_d is used as continuous input for UE frequency correction. This reduces the impact of Doppler shifts as much as possible and meets the performance requirement of high speed mobility.

4.2 Network Analysis

4.2.1 Benefits

In high speed mobility scenarios, when a UE moves faster than 200 km/h, it is very likely that the UE is incapable of performing services due to Doppler shifts. This may lead to performance issues, such as difficult network access, slow-loading web pages, and reduced UE traffic volume. When an NR TDD cell is configured as a high-speed cell, UEs in this cell can normally perform services when moving at 450 km/h. Configuring a cell as a high-speed cell improves the throughput of UEs moving faster than 120 km/h.

Once frequency correction is enabled, the uplink cell throughput, RRC connection setup success rate that is indicated by **RRC Setup Success Rate (CU)**, and intra-

RAT incoming handover success rate that is indicated by **Intra-RAT Handover In Success Rate (CU)** increase.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Configuring a cell as a high-speed cell (by setting the **NRDUCell.HighSpeedFlag** parameter to **HIGH_SPEED**) brings the following impacts on the network:

- If two uplink additional DMRS positions are configured (with **NRDUCellPusch.UlAdditionalDmrsPos** set to **POS2**), the consumption of uplink pilot resources in the cell will be increased by 4% to 8%.
- To better adapt to high-speed movement of a large number of UEs, uplink measurement and MCS index selection are optimized in high-speed cells. After a low-speed cell is reconstructed into a high-speed cell, the uplink performance of low-speed UEs in the cell may slightly deteriorate by less than 5%.
- To better adapt to high-speed movement of a large number of UEs, a high-speed cell uses a single beam for CSI-RSs and PMI-based weights for downlink PDSCHs by default. After a low-speed cell is reconstructed into a high-speed cell, the service drop rate of low-speed UEs may slightly increase. In non-massive MIMO scenarios, the downlink data rates of low-speed UEs in the cell may decrease by 10% to 20%. In massive MIMO scenarios, the downlink data rate of low-speed UEs in the cell may decrease by 20% to 30%.
- The access capability of high-speed UEs in high-speed cells is enhanced. After low-speed cells along the high-speed railway are reconstructed into high-speed cells, the number of UEs in the cells may increase. As a result, the average uplink and downlink cell throughputs, uplink and downlink user-perceived rates, and uplink and downlink MCS indexes may fluctuate.

Cooperation between high-speed cells and other functions:

- In high-speed cells, DRX can be used only for low-speed UEs. Compared with low-speed cells, high-speed UEs consume more power. For details about DRX for low-speed UEs, see [11.2.1 Principles](#).

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030207	High-speed Railway Superior Experience	NR0S00HSMB00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00HSMB00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the NRDUCell.HighSpeedFlag parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Logical Root Sequence Index	NRDUCell.LogicalRootSequenceIndex	<i>Cell Management</i>	For NR TDD, the NRDUCell.LogicalRootSequenceIndex parameter must be set to a value in the range of 0 to 137 . In high-speed cells, it is recommended that the NRDUCell.LogicalRootSequenceIndex parameter be set to a value in the range of 2 to 78 or 91 to 124 .
Low-frequency TDD	Downlink interference randomization-based scheduling	DL_INTRF_RANDOM_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i>	High-speed-specific interference randomization-based scheduling (controlled by NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw) requires at least one of the following functions to be enabled: • DL_INTRF_RANDOM_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter • NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch
Low-frequency TDD	Uplink interference randomization-based scheduling	NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch	<i>Uplink Scheduling</i>	
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	High-speed-specific interference randomization-based scheduling (controlled by NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw) can be enabled only in high-speed cells in Hyper Cell networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy	<i>Hyper Cell Combination</i>	When high-speed cells are deployed in Hyper Cell or Cell Combination networking, it is recommended that the NRDUCellCsirs.TrsAllocationStrategy parameter be set to MULTI_TRS to prevent performance deterioration caused by frequency offset adjustment during handovers between high-speed cells.
Low-frequency TDD	Cell-specific PUCCH configuration	NRDUCellPucch.StructureType	<i>Channel Management</i>	High-speed cells support only the long PUCCH (with NRDUCellPucch.StructureType set to LONG_STRUCTURE).
Low-frequency TDD	PRACH channel management	NRDUCellPrach.PrachConfigurationIndex	<i>Random Access Control</i>	High-speed NR TDD cells support only PRACH preamble format C2. Format C2 provides a larger subcarrier spacing than format 0, enabling gNodeBs to tolerate larger frequency offsets. This is more suitable for UE access in high speed mobility scenarios. For details about the relationships between the preamble formats used by the gNodeB and PRACH configuration indexes, see the overview of RA preambles in <i>Random Access Control</i> .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI-RS resource management	NRDUCellCsirs.TrsPeriod	<i>Channel Management</i>	High-speed cells support only the TRS period of MS0 , MS10 , or MS20 . This prevents delayed tracking of frequency offsets and time offsets when UEs move at high speeds.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For high-speed NR TDD cells, intra-band CA can be enabled only in Cell Combination, Hyper Cell, or common cell networking. Intra-band CA is not supported in other networking modes. That is, when NRDUCell.HighSpeedFlag is set to HIGH_SPEED for a cell, the INTRA_BAND_CA_SW option of NRDUCellAlgoSwitch.CaAlgoSwitch can be selected for the cell only under any of the following conditions: (1) NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL ; (2) NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode is set to BASIC_MODE or DPS_MODE ; and (3) NRDUCell.NrDuCellNetworkingMode is set to NORMAL_CELL .
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>	High-speed cells do not support UL and DL Decoupling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH MU-MIMO	PDCCH_MU_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	High-speed cells do not support PDCCH MU-MIMO.
Low-frequency TDD	SRS period adaptation	SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithmSwitch parameter	<i>Channel Management</i>	To prevent user experience from deteriorating due to frequent RRC connection reconfigurations in high-speed cells, SRS period adaptation for common cells cannot be enabled in high-speed cells. The SRS period has an immediate impact on the maximum number of UEs supported by the gNodeB. Set NRDUCellSrs.SrsPeriod based on the number of UEs in a cell. SRS period adaptation can be achieved in high-speed cells by enabling the SRS period and CDM adaptation function (by selecting the SRS_PERIOD_AND_CDM_ADAPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgorithmSwitch parameter). In addition, setting NRDUCellSrs.SrsPeriod to SL160 is recommended.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SR period adaptation	SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResoureAlgoSwitch parameter	<i>Channel Management</i>	To prevent user experience from deteriorating due to frequent RRC connection reconfigurations in high-speed cells, SR period adaptation cannot be enabled in high-speed cells.
Low-frequency TDD	CSI reporting period adaptation	CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResoureAlgoSwitch parameter	<i>Channel Management</i>	To prevent user experience from deteriorating due to frequent RRC connection reconfigurations in high-speed cells, CSI reporting period adaptation cannot be enabled in high-speed cells.
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	High-speed cells do not support UE-characteristic-based SRS adaptation.
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	LTE TDD and NR Flash Dynamic Power Sharing and High-speed Railway Superior Experience are mutually exclusive because power estimation may be inaccurate when they are used together.
Low-frequency TDD	Super uplink	SUPER_UL_TDM_SCH_SW or FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink is not supported in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability (trial)	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter	<i>URLLC</i>	High-speed cells do not support the low latency and high reliability function.
Low-frequency TDD	Light Load Time Sequence Opt Switch	LIGHT_LOAD_TIME_SEQ_OPT_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	The light-load time sequence optimization switch cannot be turned on in high-speed cells.
Low-frequency TDD	PUSCH joint power control	PUSCH_JOINT_PC_SW option of the NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	High-speed cells do not support PUSCH joint power control.
Low-frequency TDD	PRACH false alarm detection optimization	PRACH_FALSE_ALARM_DET_ALGO_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	High-speed cells do not support PRACH false alarm detection optimization.
Low-frequency TDD	Extended Cell Range	For TDD: NRDUCell.CellRadius set to a value greater than 14500 and less than or equal to 60000	<i>Extended Cell Range</i>	After Extended Cell Range takes effect, the High-speed Railway Superior Experience feature cannot take effect.
Low-frequency TDD	Interference avoidance for indoor cells	INDOOR_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter	<i>3D Networking Experience Improvement</i>	High-speed cells do not support "interference avoidance for indoor cells".

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink DCI extension-based scheduling (low-frequency TDD)	DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	Uplink DCI extension-based scheduling is not supported in a high-speed cell with the NRDUCell.NrDuCellNetworkingMode parameter set to HYPER_CELL_COMBINE_MODE .
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	NR_IF_HO_MEAS_QTY_PRED_FUNCTION option of the NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter DL_INTRAF_SE_MODEL_BUILD_ALL option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	High-speed cells do not support Virtual Grid-based Multi-Frequency Coordination.
Low-frequency TDD	Closed-loop PUSCH antenna selection	PUSCH_CLOSED_LOOP_ANT_SEL_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	High-speed cells do not support closed-loop PUSCH antenna selection.
Low-frequency TDD	Enhanced interference avoidance for indoor cells	INDOOR_INTRF_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter	<i>3D Networking Experience Improvement</i>	Enhanced interference avoidance for indoor cells cannot be enabled in high-speed cells.
Low-frequency TDD	Optimization of non-contention-based RA beyond cell radius	CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	High-speed cells do not support optimization of non-contention-based RA beyond cell radius.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference resistance for PUSCH demodulation	RIM_PUSCH_DEMOD_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	High-speed cells do not support remote interference resistance for PUSCH demodulation.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_CO_MM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM is not supported in high-speed railway scenarios.
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	High-speed cells do not support inter-FR CA.
Low-frequency TDD	Spatial channel estimation under uplink weak coverage	UL_WEAK_COV_SPACE_CHN_ESTI_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	High-speed cells do not support spatial channel estimation under uplink weak coverage.
Low-frequency TDD	SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	High-speed cells do not support SRS field strength-based positioning.
Low-frequency TDD	U-TDOA-based positioning/TOA fingerprint-based positioning	UTDOA_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	High-speed cells do not support U-TDOA-based positioning/TOA fingerprint-based positioning.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSw. MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	High-speed cells do not support multi-frequency beam coordination.
Low-frequency TDD	Blind SRS interference suppression	SRS_BLIND_IS_SW option of the NRDUCellDlMimo.HighLayerMuMimoSw parameter	<i>MIMO (TDD)</i>	High-speed Railway Superior Experience and blind SRS interference suppression cannot be enabled at the same time.
Low-frequency TDD	DMRS adaptation for voice services	VOICE_DMRS_ADAPT_SW option of the NRDUCellDmrs.UIDmrsSwitch parameter	None	High-speed Railway Superior Experience does not support DMRS adaptation for voice services.
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	High-speed cells do not support CCE-based data split stop for CA UEs.
Low-frequency TDD	CSI-RS resource management	NRDUCellCsirs.TrsPeriod	<i>Channel Management</i>	High-speed cells do not support TRS periods longer than 20 ms. This prevents delayed tracking of frequency offsets and time offsets when UEs move at high speeds.
TDD	Assisted Massive MIMO	UL_ASSISTANCE_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	None	Assisted massive MIMO is not supported in high-speed cells.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Distributed massive MIMO is not supported in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH beam domain enhancement	PUSCH_BEAM_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	None	In massive MIMO scenarios, PUSCH beam domain enhancement is not supported in high-speed cells.
Low-frequency TDD	Extended Cell Range of 90 km (TDD)	NRDUCell.CellRadius set to a value greater than 60000 and less than or equal to 90000	<i>Extended Cell Range</i>	High-speed Railway Superior Experience and Extended Cell Range of 90 km (TDD) cannot be both enabled.
Low-frequency TDD	Blind interference suppression for SRS measurement	SRS_BLIND_IS_MEAS_SW option of the NRDUCellSrsMeas.SrsMeasOptSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	None
Low-frequency TDD	Enhanced uplink waveform adaptation	RANK1_ADAPT_DFT_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>ViNR</i>	Enhanced uplink waveform adaptation (controlled by the RANK1_ADAPT_DFT_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter) cannot be enabled in high-speed cells.
Low-frequency TDD	Inactive RA-SDT	UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter and the RRC_INACTIVE_RA_SDT_SW option of the NRCCellAlgoSwitch.InactiveStrategySwitch parameter	<i>UE Power Saving</i>	If the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED , the RRC_INACTIVE_RA_SDT_SW option of the NRCCellAlgoSwitch.InactiveStrategySwitch parameter cannot be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC coverage expansion	SA networking: SA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter NSA networking: NSA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	High-speed cells do not support SCC coverage expansion.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDlRank.DlRankSelAlgoSw parameter	<i>FWA</i>	High-speed Railway Superior Experience cannot work with downlink 8-layer transmission for a single FWA user.
Low-frequency TDD	Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUlRank.UlRankAlgoSw parameter	<i>FWA</i>	High-speed Railway Superior Experience cannot work with "uplink 4-layer transmission for a single FWA user".
Low-frequency TDD	High-reliability uplink MCS selection	UL_HIGH_RELIAB_MCS_SEL_SW option of the NRDUCellUlAmc.UlAmcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	When High-speed Railway Superior Experience is enabled, high-reliability uplink MCS selection must be disabled.
Low-frequency TDD	Time-domain adaptive filtering	PUSCH_TIME_DOM_ADAPT_FILTER_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	High-speed Railway Superior Experience and time-domain adaptive filtering cannot be enabled at the same time.
Low-frequency TDD	MetaAAU deep energy saving	DEEP_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	MetaAAU deep energy saving does not apply to high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUT DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	High-speed Railway Superior Experience is mutually exclusive with dynamic RF channel shutdown.
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	High-speed Railway Superior Experience is mutually exclusive with 3GPP R15-compliant uplink slot aggregation.
Low-frequency TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	High-speed Railway Superior Experience is mutually exclusive with 3GPP R16-compliant uplink slot aggregation.
Low-frequency TDD	Coherent-codebook-based scheduling	COHERENT_CODEBOOK_SCH_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Channel Management</i>	High-speed Railway Superior Experience cannot work with coherent-codebook-based scheduling.
Low-frequency TDD	Low power high accuracy positioning	LPHAP_REDCAP_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	For an NR TDD cell, when the value of NRDUCell.CellRadius is greater than or equal to 4610 and the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED , LPHAP cannot be enabled.
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_CO MP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	High-speed cells support UL CoMP only in Hyper Cell or Cell Combination networking.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	High-speed cells do not support single DCI.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SUL and Multi-Slot-Config NUL Bind	SUL_WITH_MULTI_SLOT_CFG_NUL_SW option of the NRDUCellCollabCoop.SulAlgoSwitch parameter	<i>Super Uplink</i>	SUL and Multi-Slot-Config NUL Bind is not supported in high-speed cells.
Low-frequency TDD	Flexible spectrum access	FLEX_SPCT_ACCESS_SW option of the NRDUCellCarrMgmt.MultiBandFlexSchSwitch parameter	<i>Uplink Flexible Spectrum Scheduling</i>	Flexible spectrum access is not supported in high-speed cells.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	It is not recommended that this parameter of a cell be set to HIGH_SPEED in Fusion Cell networking.
Low-frequency TDD	Uplink MU grouping and pairing	UL_MU_GRP_PAIR_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink MU grouping and pairing cannot be enabled in high-speed cells.
Low-frequency TDD	UE-number-based BWP1 adaptation	USER_NUM_BASED_BWP_ACTV_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	In a high-speed cell, UE-number-based BWP1 adaptation cannot be enabled.
Low-frequency TDD	PRB-and-CCE-resource-based BWP1 adaptation	RESOURCE_BASED_BWP_ACTV_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	In a high-speed cell, PRB-and-CCE-resource-based BWP1 adaptation cannot be enabled.
Low-frequency TDD	Coverage enhancement	REDCAP_UL_COVERAGE_ENH_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	Coverage enhancement and High-speed Railway Superior Experience cannot be enabled at the same time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inactive RA-SDT	REDCAP_RRC_INACTIVE_RA_SDT_SW option of the NRCellAlgoSwitch.InactiveStrategySwitch parameter	<i>RedCap</i>	If the NRDUCell.CellRadius parameter is set to a value greater than or equal to 4610 and the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED , the inactive RA-SDT function cannot be enabled.
Low-frequency TDD	UE-number-based periodic BWP load balancing	UE_NUM_BASED_PRDC_BLB_SW option of the NRDUCellRedCap.RedCapAlgoExtSwitch parameter	<i>RedCap</i>	UE-number-based periodic BWP load balancing cannot be enabled in high-speed cells.
Low-frequency TDD	DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgorithmSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	DCI-based dynamic waveform switching cannot be enabled in high-speed cells.
Low-frequency TDD	Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	High-speed cells do not support intelligent power control energy saving.
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Inter-cell interference avoidance cannot be enabled in high-speed scenarios.
Low-frequency TDD	DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgorithmSwitch parameter	<i>Uplink Scheduling</i>	DCI-based dynamic waveform switching cannot be enabled in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS IC	SRS_IC_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	High-speed cells do not support SRS IC.
Low-frequency TDD	SRS joint power control	SRS_JOINT_PC_SW option of the NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	SRS joint power control is not supported in high-speed cells.
Low-frequency TDD	Intelligent MCS optimization for downlink SU-MIMO	DL_SU_MCS_INTEL_OPT_SW option of the NRDUCellAiAlgo.AiAimcAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Intelligent MCS optimization for downlink SU-MIMO cannot be enabled in high-speed cells.
Low-frequency TDD	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	DL_MU_NLOS_IR_PRECODING_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios cannot be enabled in high-speed cells.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	High-speed cells do not support experience-based smart carrier selection (controlled by the MULTI_FREQ_SMART_SEL_SW option of the NRCCellAlgoSwitch.MultiFreqAlgoSwitch parameter).
Low-frequency TDD	Uplink Boosting 2.0	UL_LOW_NOISE_PHASE2_SW option of the NRDUCellFeatureSwitch.MimoFeatureSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	High-speed cells do not support Uplink Boosting 2.0.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	- On the LTE side: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter - On the NR side: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	L/NR Inter-band Deep Coordination and High-speed Railway Superior Experience are mutually exclusive because power estimation may be inaccurate when they are used together.
Low-frequency TDD	Blind SRS interference suppression enhancement	SRS_ENH_BLIND_IS_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When blind SRS interference suppression enhancement is enabled for a cell, the cell cannot be configured as a high-speed cell.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS resource management	NRDUCellSrs.SrsPeriod	<i>Channel Management</i>	In high speed mobility scenarios, UEs move at high speeds. If the sounding reference signal (SRS) period (specified by NRDUCellSrs.SrsPeriod) is overly long, UE performance is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink Time Alignment Timer	NRDUCellUPLinkTimeAlignmentTimer	<i>5G Networking and Signaling</i>	UEs are more likely to be out of synchronization in high speed mobility scenarios. The larger the value of the uplink time alignment timer, the higher the probability that UEs are out of synchronization.
Low-frequency TDD	Basic functions of uplink scheduling	NRDUCellPUSCH-ULAdditionalDMRSPositions	<i>Uplink Scheduling</i>	A high-speed cell supports two uplink additional DMRS positions. That is, NRDUCellPUSCH-ULAdditionalDMRSPositions should be set to POS2. For NR TDD, if the maximum train speed is 300 km/h in the 2.6 GHz frequency band or 200 km/h in the 3.5 GHz frequency band, a high-speed cell can be configured with one uplink additional DMRS (by setting NRDUCellPUSCH-ULAdditionalDMRSPositions to POS1).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic functions of downlink scheduling	NRDUCellPdsch.DLAdditionalDmrsPositions NRDUCellPdsch.DLDmrsMaxLength	<i>Downlink Scheduling</i>	The maximum downlink DMRS symbol length can only be one symbol for a high-speed cell. That is, NRDUCellPdsch.DLDmrsMaxLength must be set to 1SYMBOL. A high-speed cell supports one or two downlink additional DMRS positions, that is, NRDUCellPdsch.DLAdditionalDmrsPos must be set to POS1 or POS2. This guarantees the PDSCH demodulation performance of UEs in high speed mobility scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table	NRDUCellAlgoSwitch.Ul256QamSwitch	<i>Modulation Schemes</i>	It is recommended that adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table be disabled for high-speed cells because MCS index adaptation is difficult to perform while UEs move at high speeds. In high-speed railway scenarios, the 256QAM MCS index table is recommended (by setting the NRDUCellAlgoSwitch.Ul256QamSwitch parameter to UL_256QAM_FIXED). In a specific scenario, whether to optimize the configuration depends on the SINR and MCS index distribution.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table	DL_MCS_TABLE_ADAPT_SW option of the NRDUCellPdsch.DLLinkAdaptAlgoSwitch parameter	<i>Modulation Schemes</i>	It is recommended that adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table be disabled for high-speed cells because MCS index adaptation is difficult to perform while UEs move at high speeds. In high-speed railway scenarios, the 256QAM MCS index table is recommended (by setting the NRDUCellAlgoSwitch.DL256QamSwitch parameter to ON). In a specific scenario, whether to optimize the configuration depends on the SINR and MCS index distribution.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RRC-reconfiguration-based dynamic waveform switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAId parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	After RRC-reconfiguration-based dynamic waveform switching is enabled in a high-speed cell, UEs whose uplink capability is rank 1 always use the DFT waveform when they perform full buffer services. It is recommended that this function be enabled for high-speed railways with poor overall coverage (with the average uplink MCS index being less than or equal to 8 if the 256QAM MCS index table is used, or being less than or equal to 14 if the 64QAM MCS index table is used). When the overall coverage along the high-speed railway is poor, enabling this function increases the uplink throughput of UEs performing full buffer services; in other scenarios, enabling this function may decrease the uplink throughput of UEs performing full buffer services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic cellular-based positioning	NR cell ID-based positioning: none NR E-CID-based positioning: ECID_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	Currently, positioning for high-speed moving UEs is not supported. As a result, the positioning accuracy deteriorates in high speed mobility scenarios.
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnlIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When High-speed Railway Superior Experience takes effect, interference avoidance during scheduling is allowed for deep-coverage UEs performing full buffer services.
Low-frequency TDD	UL skipping	UPLINK_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	When High-speed Railway Superior Experience is enabled, the UL skipping function does not take effect.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When the High-speed Railway Superior Experience function is enabled, only intra-base-station joint transmission by TRPs for Hyper Cell or Cell Combination is supported.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	For high-speed cells, coordinated beamforming (CBF) does not take effect.
Low-frequency TDD	SRS noise reduction for low-SNR channels	LOW_SNR_CHN_DENOISE_SW option of the NRDUCellAIgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	SRS noise reduction for low-SNR channels does not take effect for high-speed cells.
Low-frequency TDD	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAIgoExtSwitch parameter	<i>Channel Management</i>	In high speed mobility scenarios, UEs move fast, and the traffic volume in the cell is heavy and burst services dominate. To prevent service drops of a large number of UEs, PDCCH symbol number adaptation is not recommended in high-speed cells.
Low-frequency TDD	Interference rejection precoding	DL_MU_MIMO_IR_PRECODING_SW option of the NRDUCellAIgoSwitch.MuMimoSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Interference rejection precoding does not take effect in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-frequency dedicated-priority-based idle mode MLB	NRCeMlbPolicy set to DEDICATED_PRIORITY and the INTER_FREQ_IDLE_MLB_SW option of the NRCeMlbPolicySwitch parameter selected	<i>Mobility Load Balancing in Idle Mode</i>	High-speed Railway Superior Experience takes effect when it is enabled together with low-frequency dedicated-priority-based idle mode MLB and the NRCeMlbPolicySwitch.IdleMlbUeReleaseRatio parameter is set to 0.
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	After remote interference adaptive avoidance is enabled, only RIM-RS transmission and detection take effect in high-speed cells (so that interference avoidance can be triggered in common cells when necessary), while GP adjustment does not. High-speed cells have high traffic load, and fewer resources can be consumed in such cells if GP adjustment does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAIgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	After enhanced remote interference adaptive avoidance is enabled, only RIM-RS transmission and detection take effect in high-speed cells (so that interference avoidance can be triggered in common cells when necessary), while adjustment in the time domain range for scheduling UEs does not. High-speed cells have high traffic load, and fewer resources can be consumed in such cells if adjustment in the time domain range for scheduling UEs does not take effect.
Low-frequency TDD	Super uplink	FLEX_SUPER_UPLINK_SW option of the NRDUCellAIgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with SUL and NR FDD co-deployment does not take effect in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive suppression	RIM_ADAPT_SUPPRESSION_SW option of the NRDUCellAIgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	In high-speed scenarios, remote interference adaptive suppression yields fewer gains than it would in low-speed scenarios.
Low-frequency TDD	Remote interference resistance for random access	RA_REMOTE_INTRF_RESIST_SW option of the NRDUCellAIgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	In high-speed scenarios, remote interference resistance for random access yields fewer gains than it would in low-speed scenarios.
Low-frequency TDD	RedCap UL skipping	REDCAP_UL_LINK_SKIP_SW or REDCAP_UL_LINK_SKIP_EXT_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>RedCap</i>	When High-speed Railway Superior Experience is enabled, RedCap UL skipping does not take effect.
Low-frequency TDD	RedCap DRX adaptation	REDCAP_DRX_ADAPTATION_SW or REDCAP_DRX_ADAPTATION_EXT_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>RedCap</i>	When High-speed Railway Superior Experience is enabled, RedCap DRX adaptation does not take effect.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SLEEP	<i>Multi-RAT Coordinated Energy Saving</i>	NR/LNR load-based RF module deep dormancy is not recommended in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEPSUPER	<i>Multi-RAT Coordinated Energy Saving</i>	NR/LNR load-based RF module super dormancy is not recommended in high-speed cells.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When the high-speed cell function in high speed specified policy management is enabled, you are advised not to enable NR load-based RF module fast deep dormancy or LNR load-based RF module fast deep dormancy.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

NR TDD:

All NR-capable main control boards support this function. If the number of RF channels is 1T1R, 2T2R, 4T4R or 8T8R, all NR-capable main control boards and NR TDD baseband processing units support this function. If the number of RF channels is 32T32R or 64T64R, BBPs except for the UBBPfw series ones support this function. The BBPs must work in NR only mode. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

NR TDD: Only NR TDD low-frequency RF modules with the number of RF channels being 1T1R, 2T2R, 4T4R, 8T8R, 32T32R, or 64T64R support this function. In massive MIMO scenarios, only NR TDD low-frequency RF modules complying with the eCPRI protocol are supported.

4.3.4 Networking

To prevent frequent handovers in high-speed scenarios, it is recommended that high-speed cells use Hyper Cell or Cell Combination networking. In Cell Combination networking, the TX/RX mode cannot be 32T32R or 64T64R.

In Hyper Cell or Cell Combination networking scenarios, it is recommended that the **NRDUCellTrp.NrDuCellTrpld** parameter values of six physically-adjacent transmission reception points (TRPs) along high-speed railways be planned sequentially based on their TRP-ID-mod-6 results. This prevents performance issues such as service drops and UE throughput decrease caused by TRS interference between TRPs. For details about Hyper Cell, see *Hyper Cell*. For details about Cell Combination, see *Cell Combination*. For example, as shown in [Figure 4-1](#) or [Figure 4-2](#), the **NRDUCellTrp.NrDuCellTrpld** parameter values of the six TRPs are planned as 15 (mod 6 = 3), 16 (mod 6 = 4), 17 (mod 6 = 5), 18 (mod 6 = 0), 19 (mod 6 = 1), and 20 (mod 6 = 2).

Figure 4-1 Planning the TRP IDs of NR DU cells (hyper cell)

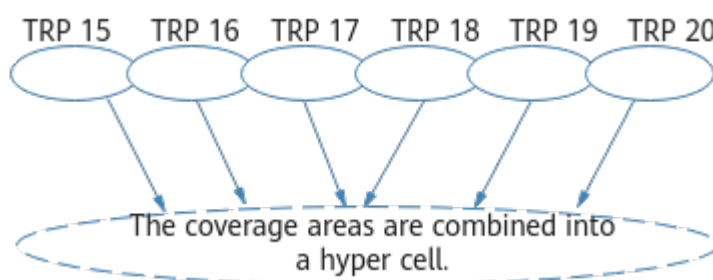
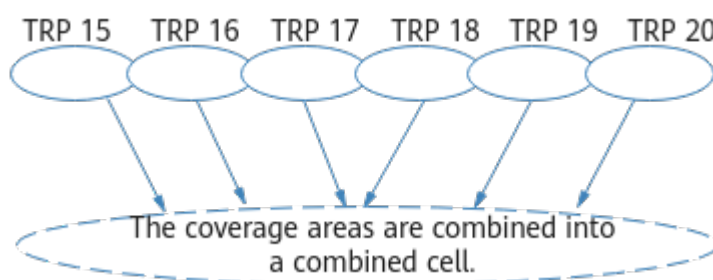


Figure 4-2 Planning the TRP IDs of NR DU cells (combined cell)



4.3.5 Cells

NR TDD:

- When **NRDUCell.NrDuCellNetworkingMode** is set to **HYPER_CELL_COMBINE_MODE** for a cell, **NRDUCellTrp.TxRxMode** of the cell must be set to **1T1R**, **2T2R**, **4T4R**, or **8T8R**.

- In addition, when **NRDUCell.NrDuCellNetworkingMode** is set to **HYPER_CELL**, 32T32R and 64T64R RF modules cannot be configured together with other low-frequency RF modules.
- 32T32R and 64T64R RF modules can be configured only when **NRDUCell.NrDuCellNetworkingMode** is set to **NORMAL_CELL** or **HYPER_CELL**. When 32T32R and 64T64R RF modules are used in high-speed cells, dedicated carriers must be used for high-speed UEs or these RF modules must be used only for high-speed railway coverage to prevent serving a large number of low-speed UEs at the same time.

4.3.6 Others

UEs must support two uplink additional DMRS positions. Otherwise, service drops may occur when UEs move at a high speed.

According to section 6.3 "RRC information elements" in 3GPP TS 38.331 V17.3.0, if the **oneFL-DMRS-TwoAdditionalDMRS-UL** field is **supported** in the UE capability, the UE supports two uplink additional DMRS positions.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

If Hyper Cell networking or Cell Combination networking is used for high-speed cells, configure hyper cells or combined cells by referring to *Hyper Cell* or *Cell Combination*, respectively.

Table 4-3 describes the parameters used for function activation and **Table 4-4** describes other parameters. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	High Speed Flag	NRDUCell.Hi ghSpeedFlag	Set this parameter to HIGH_SPEED for high-speed cells and LOW_SPEED for low-speed cells.
Low-frequency TDD	Duplex Mode	NRDUCell.Du plexMode	For NR TDD, set this parameter to CELL_TDD .

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgorithmOptSw	Select the R16_HIGH_SPEED_MEAS_FLAG option if gNodeBs are required to broadcast the high-speed cell measurement flag in system information.
Low-frequency TDD	Uplink Additional DMRS Position	NRDUCellPusch.UlAdditionalDmrsPos	New high-speed cell: The gNodeB automatically sets this parameter to the default value POS2 for high-speed cells. A low-speed cell reconfigured as a high-speed cell: Set this parameter to POS2 .
Low-frequency TDD	Downlink DMRS Maximum Length	NRDUCellPdsch.DlDmrsMaxLength	New high-speed cell: The gNodeB automatically sets this parameter to the default value 1SYMBOL for high-speed cells. A low-speed cell reconfigured as a high-speed cell: Set this parameter to 1SYMBOL .
Low-frequency TDD	Uplink Time Alignment Timer	NRDUCellUplinkConfig.UplinkTimeAlignmentTimer	New high-speed cell: The gNodeB automatically sets this parameter to the default value 1280SF for high-speed cells. A low-speed cell reconfigured as a high-speed cell: Setting this parameter is required. The value 1280SF is recommended.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	High Speed Intrf Random Schedule Sw	NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw	High-speed-specific interference randomization-based scheduling can be enabled only in high-speed cells in Hyper Cell networking. After high-speed-specific interference randomization-based scheduling is enabled, the scheduling positions for TRPs can be staggered based on the TRP number modulo 2, reducing mutual interference and improving user-perceived rates in the uplink and downlink. To enable high-speed-specific interference randomization-based scheduling for high-speed cells, set this parameter to ON.
Low-frequency TDD	High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgorithmOptSw	It is recommended that the UL_CHANNEL_ESTIMATE_OPT_SW option be selected to optimize uplink channel estimation for UEs in high-speed scenarios and increase the uplink user-perceived rate in non-line-of-sight (NLOS) scenarios. This function takes effect only when NRDUCellPusch.UlAdditionalDmrsPos is set to POS2.
Low-frequency TDD	High Speed Frequency Offset Threshold	NRDUCellHighSpeedMob.HighSpeedFreqOffsetThld	This parameter is used to identify high- and low-speed UEs. Set this parameter to its recommended value so that UEs moving faster than 120 km/h can be identified as high-speed UEs. If this parameter is set to an excessively small value, the high-speed and low-speed attributes of UEs change continuously, affecting user experience.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Low Speed Stay Time Threshold	NRDUCellHighSpeedMob.LowSpeedStayTimeThld	This parameter is used to identify high- and low-speed UEs. Use the recommended value.
Low-frequency TDD	Low Speed Frequency Offset Threshold	NRDUCellHighSpeedMob.LowSpeedFreqOffsetThld	This parameter is used to identify high- and low-speed UEs. Use the recommended value.

Table 4-4 Parameters used for optimization

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	SRS Period	NRDUCellSrs.SrsPeriod	High-speed cells do not support SRS period adaptation. Therefore, the SRS period has an immediate impact on the maximum number of UEs supported by the gNodeB. Set NRDUCellSrs.SrsPeriod based on its recommended values for different load scenarios. Set this parameter to the same value for high-speed cells served by the same baseband processing unit. To allow more high-speed UEs to access a high-speed NR TDD cell, it is recommended that the SRS period be set to SL160 and SRS period and CDM adaptation be enabled (by selecting the SRS_PERIOD_AND_CDM_ADAPT_SW option of the NRDUCellHighSpeed-Mob.HighSpeedAlgoOptSw parameter).
Low-frequency TDD	TRS Period	NRDUCellCsiRs.TrsPeriod	System performance can be improved by adjusting this parameter. Retain the default value.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	TRP Handover Delta RSRP	NRDUCellMultiTrp.TrpHoDeltaRsrp	Use the recommended value.
Low-frequency TDD	PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgorithmSwitch	It is recommended that the following options be selected to allow more high-speed UEs to access high-speed cells: • FORMAT1_SR_CHN_ALLOC_OPT_SW • PUCCH_RBRES_ADAPTIVE_SWITCH
Low-frequency TDD	SR Period	NRDUCellPucch.SrPeriod	To enable more high-speed UEs to access a high-speed NR TDD cell, it is recommended that this parameter be set to SLOT80 . If the number of high-speed UEs in a cell continuously increases and exceeds, for example, 600, you can set this parameter to SLOT160 .
Low-frequency TDD	CSI Report Period	NRDUCellPucch.CsiReportPeriod	To enable more high-speed UEs to access a high-speed NR TDD cell, it is recommended that this parameter be set to SLOT160 . If the number of high-speed UEs in a cell continuously increases and exceeds, for example, 600, you can set this parameter to SLOT320 .

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	SRS Algorithm Extension Switch	NRDUCellSrs.SrsAlgoExtSwitch	To allow more high-speed UEs to access high-speed cells, it is recommended that the SPEC_INCREASE_SW option be selected. If the preceding option is selected, all high-speed cells served by the same baseband processing unit must have the same values regarding the following parameters and you are advised to set these parameters to their recommended values. • NrDUCellSrs.SrsNarrowBandIndexCsrs • NrDUCellSrs.SrsNarrowBandIndexBsrs • NrDUCellSrs.SrsWideBandIndexCsrs • NrDUCellSrs.SrsWideBandIndexBsrs
Low-frequency TDD	Downlink Additional DMRS Position	NRDUCELLPDSCH.DLAdditionalDmrsPositions	If Downlink Additional DMRS Position is set to POS2, the downlink channel estimation performance improves when UEs moves at a high speed, but DMRS overhead increases. Therefore, it is recommended that this parameter be set to POS2 in cells where the trains on high-speed railways move faster than 300 km/h to improve the performance of high-speed UEs.
Low-frequency TDD	SRS Resource Multiplexing Policy	NRDUCellHighSpeedMob.SrsResourceMultiplexPolicy	It is recommended that this parameter be set to NO_CDM to reduce the service drop rate in high-speed railway overshoot coverage scenarios.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Uplink PUSCH Algorithm Switch	NRDUCCELLPUSCH.UlPuschAlgoSwitch	If the NRDUCell.NrDuCellNetworkingMode parameter is set to a value other than HYPER_CELL_COMBINE_MODE for a high-speed cell and uplink CCEs are limited (for example, the uplink CCE allocation success rate is lower than 90%), it is recommended that the DCI_EXTEND_UL_CONGEST_SW option be selected, the UL_SCH_DCI_EXT_SW option of the NRDUCellUISch.UlSchAlgoSwitch parameter be selected, and the NRDUCellUISch.UlSchDciExtUsrNumThld and NRDUCellUISch.UlSchDciExtExitThld parameters be set to 200 and 80 , respectively, to expand uplink CCEs and increase uplink capacity for the cell.
Low-frequency TDD	Multi-Symbol CORESET Switch	MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	It is recommended that the MULTI_SYM_CORESET_SW option be selected to improve the user-perceived rates of high-speed UEs. After this function is enabled, the average uplink and downlink PDCCH CCE aggregation levels and the CCE usage may decrease, and the uplink and downlink PDCCH CCE allocation success rates may increase ^a . For details, see <i>Channel Management</i> .

 NOTE

a:

- Average uplink PDCCH CCE aggregation level = $(N.CCE.UL.AggLvl16Num \times 16 + N.CCE.UL.AggLvl8Num \times 8 + N.CCE.UL.AggLvl4Num \times 4 + N.CCE.UL.AggLvl2Num \times 2 + N.CCE.UL.AggLvl1Num) / (N.CCE.UL.AggLvl16Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl1Num)$
- Average downlink PDCCH CCE aggregation level = $(N.CCE.DL.AggLvl16Num \times 16 + N.CCE.DL.AggLvl8Num \times 8 + N.CCE.DL.AggLvl4Num \times 4 + N.CCE.DL.AggLvl2Num \times 2 + N.CCE.DL.AggLvl1Num) / (N.CCE.DL.AggLvl16Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl1Num)$
- CCE usage = $N.CCE.Used.Avg / N.CCE.Avail.Avg$
- Uplink PDCCH CCE allocation success rate = $(N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num) / (N.CCE.UL.AllocReq.Num)$
- Downlink PDCCH CCE allocation success rate = $(N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num) / (N.CCE.DL.AllocReq.Num)$

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Adding a high-speed NR TDD cell: For details about the recommended settings of other parameters in the **NRDUCell MO**, see *Cell Management*.
//Adding a high-speed cell with HighSpeedFlag set to HIGH_SPEED and DuplexMode set to CELL_TDD
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="0_TDD", DuplexMode=CELL_TDD, CellId=255, PhysicalCellId=7, FrequencyBand=N77, DInArfcn=646666, ULBandwidth=CELL_BW_100M, DLBandwidth=CELL_BW_100M, SlotAssignment=4_1_DDSU, SlotStructure=SS2, TrackingAreaId=0, SsbFreqPos=8012, LogicalRootSequenceIndex=2, HighSpeedFlag=HIGH_SPEED;
//Enabling high-speed-specific interference randomization-based scheduling in Hyper Cell networking
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighSpeedIntrfRandomSchSw=ON;
//Turning on the UL Channel Estimation Optimization Sw
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=UL_CHANNEL_ESTIMATE_OPT_SW-1;
//Enabling the gNodeB to broadcast the 3GPP Release 16-compliant high-speed cell measurement flag in system information if required
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=R16_HIGH_SPEED_MEAS_FLAG-1;
//If high-speed cells are deployed in Hyper Cell or Cell Combination networking, it is recommended that the TRS allocation strategy be set to MULTI_TRS to prevent performance deterioration caused by frequency offset adjustment during handovers between high-speed cells.
MOD NRDUCELLCSIRS: NrDuCellId=0, TrsAllocationStrategy=MULTI_TRS;
- Changing a low-speed cell to a high-speed cell (This will cause the cell to automatically restart for the configuration to take effect.)
Before changing cell configurations, ensure that the license for the High-speed Railway Superior Experience feature has been installed on the gNodeB. Otherwise, the cell cannot be activated after automatic reset.

If an activated TDD massive MIMO cell complies with the CPRI protocol, changing the cell type to high-speed (setting the **NRDUCell.HighSpeedFlag** parameter to **HIGH_SPEED**) will fail.

- Changing some default configurations of the low-speed cell to those for high-speed cells:

```
//Changing the uplink additional DMRS position configuration to POS2
MOD NRDUCELLPUSCH: NrDuCellId=0, UAdditionalDmrsPos=POS2;
//Changing the downlink DMRS maximum length to 1SYMBOL
MOD NRDUCELLPDSCH: NrDuCellId=0, DLDmrsMaxLength=1SYMBOL;
//Disabling SR period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_ADAPT_SWITCH-0;
//Disabling SRS period adaptation
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-0;
//Disabling CSI reporting period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0,
CsiResoureAlgoSwitch=CSI_REPORT_PERIOD_ADAPT_SWITCH-0;
//Disabling quadruple code division for format-4 resource allocation
MOD NRDUCELLPUCCH: NrDuCellId=0,
PucchAlgoSwitch=FORMAT4_OCC_LENGTH_N4_SWITCH-0;
//Setting the uplink time alignment timer
MOD NRDUCELLULTACONFIG: NrDuCellId=0, ULTimeAlignmentTimer=1280SF;
```

- For NR TDD:

```
//(NR TDD) Changing the high speed flag of the cell to HIGH_SPEED. Before the modification,
check that the UE-characteristic-based SRS period adaptation switch is turned off.
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=USER_CHARACTER_SRS_ADAPT_SW-0;
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, HighSpeedFlag=HIGH_SPEED;
//If high-speed cells are deployed in Hyper Cell or Cell Combination networking, it is
recommended that the TRS allocation strategy be set to MULTI_TRS to prevent performance
deterioration caused by frequency offset adjustment during handovers between high-speed cells.
MOD NRDUCELLCSIRS: NrDuCellId=0, TrsAllocationStrategy=MULTI_TRS;
```

- The following are examples of parameter settings for optimization in high-speed NR TDD cells:

- After a high-speed cell with the bandwidth lower than 100 MHz is set up:

```
//Setting the SRS period to SL160 for a cell with a bandwidth lower than 100 MHz. This
parameter must be set to the same value for high-speed cells served by the same baseband
processing unit.
MOD NRDUCELLSRS: NrDuCellId=0, SrsPeriod=SL160;
//Setting the TRS period
MOD NRDUCELLCSIRS: NrDuCellId=0, TrsPeriod=MS20;
```

- The following operations are required to allow more high-speed UEs to access high-speed cells:

```
//Turning on the switches for PUCCH RB adaptation and format-1 SR code channel allocation
optimization
MOD NRDUCELLPUCCH: NrDuCellId=0,
PucchAlgoSwitch=FORMAT1_SR_CHN_ALLOC_OPT_SW-1&PUCCH_RBRES_ADAPTIVE_SWITCH-1;
//Setting the SR period to SLOT80 as recommended
MOD NRDUCELLPUCCH: NrDuCellId=0, SrPeriod=SLOT80;
//Setting the CSI reporting period to SLOT160 as recommended
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiReportPeriod=SLOT160;
//Turning on the SRS specification improvement switch. If this option is selected, all high-speed
cells served by the same baseband processing unit must have the same values regarding the
NrDUCellSrs.SrsNarrowBandIndexCsrs, NrDUCellSrs.SrsNarrowBandIndexBsrs,
NrDUCellSrs.SrsWideBandIndexCsrs and NrDUCellSrs.SrsWideBandIndexBsrs parameters and you
are advised to set these parameters to their recommended values.
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SPEC_INCREASE_SW-1;
```

- If the number of UEs in a cell exceeds 600, modify the settings of the following parameters:

```
//Prolonging the SRS period, SR period, and CSI period as the number of UEs keeps increasing
MOD NRDUCELLSRS: NrDuCellId=0, SrsPeriod=SL320;
MOD NRDUCELLPUCCH: NrDuCellId=0, SrPeriod=SLOT160;
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiReportPeriod=SLOT320;
```

- Activation command examples for high-speed and low-speed UE identification:
 - The following is an example of parameter settings for NR TDD:
//Configuring the high-speed frequency offset threshold and low-speed camping time threshold (using NRDUCell.FrequencyBand being set to n38 as an example)
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedFreqOffsetThld=560, LowSpeedStayTimeThld=15, LowSpeedFreqOffsetThld=250;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

The following provides command examples for changing a high-speed cell to a low-speed one. This operation will cause the cell to automatically reset so that the configuration takes effect.

- For NR TDD:
//(NR TDD) Changing the high speed flag of the cell to LOW_SPEED.
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, HighSpeedFlag=LOW_SPEED;
//Restoring the uplink additional DMRS position, downlink DMRS maximum length, and SR period adaptation to default settings for low-speed cells
MOD NRDUCELLPUSCH: NrDuCellId=0, ULAdditionalDmrsPos=POS1;
MOD NRDUCELLPDSCH: NrDuCellId=0, DLDmrsMaxLength=2SYMBOL;
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_ADAPT_SWITCH-1;
//Disabling high-speed-specific interference randomization-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighSpeedIntrfRandomSchSw=OFF;
//Turning off the UL Channel Estimation Optimization Sw
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=UL_CHANNEL_ESTIMATE_OPT_SW-0;
//Enabling SRS period adaptation for the low-speed cell as recommended
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-1;
//Changing the uplink time alignment timer to its recommended value INFINITY for the low-speed cell
MOD NRDUCELLULTACONFIG: NrDuCellId=0, ULTimeAlignmentTimer=INFINITY;
//Changing the switch for CSI reporting period adaptation to the recommended value for the low-speed cell (that is, turning on the related switch)
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_REPORT_PERIOD_ADAPT_SWITCH-1;
//Using the recommended quadruple code division for format-4 resource allocation configuration for a low-speed cell (that is, turning on the related switch)
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT4_OCC_LENGTH_N4_SWITCH-1;
//Turning off the switch for delivering 3GPP Release 16-compliant high-speed measurement flag
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=R16_HIGH_SPEED_MEAS_FLAG-0;

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Step 1** Run the **LST NRDUCELL** command and check the value of **High Speed Flag** in the command output. Run the **DSP NRCELL** command and check the value of **Cell Available State** in the command output.

If the value of **High Speed Flag** is **High Speed Cell** and the value of **Cell Available State** is **Available**, the high-speed cell has been successfully set up.

Step 2 Verify cell functions.

If a UE can access the high-speed cell, the high-speed cell is working properly, that is, the High-speed Railway Superior Experience feature has taken effect.

----End

4.4.3 Network Monitoring

In high-speed cells, you can observe the following counters and KPIs to evaluate network performance.

The following KPIs can be observed:

- RRC connection setup success rate (**RRC Setup Success Rate (CU)**)
- Intra-RAT incoming handover success rate (**Intra-RAT Handover In Success Rate (CU)**)

The following counters can be observed.

Counter ID	Counter Name
1911816696	N.ThpVol.UL.Cell
1911816550 to 1911816581	N.ChMeas.PUSCH.MCS.0 to N.ChMeas.PUSCH.MCS.31
1911829336	N.ThpVol.UL.Hs
1911829331	N.ThpTime.UE.UL.RmvSmallPkt.Hs
1911829333	N.ThpVol.UE.UL.SmallPkt.Hs
1911829330	N.ThpVol.DL.Hs
1911829328	N.ThpVol.DL.LastSlot.Hs
1911829334	N.ThpTime.DL.RmvLastSlot.Hs
1911829339	N.User.RRCConn.Hs.Avg
1911829338	N.User.RRCConn.Hs.Max

5 High-Speed Cell Switching Performance Optimization

5.1 Incoming Handover Protection for High-speed Cells

5.1.1 Principles

If an unnecessary handover function is enabled for multiple cells, a UE will be handed over back to the original source cell, due to inappropriate network planning, through unnecessary handovers of the same or different types. That is, the UE experiences ping-pong handovers. The "incoming handover protection for high-speed cells" function is used to prevent ping-pong handovers. This function takes effect only on SA UEs.

When a UE is handed over to a high-speed cell and the cause value carried in the handover request is **Handover Desirable for Radio Reasons**, the following operations are performed:

- The target cell records the source cell of this incoming handover.
- After the handover is complete, the target cell starts the incoming handover protection timer for high-speed cells. This timer is controlled by the **NRCellMobilityConfig.HsCellHoProtectTimer** parameter. If this parameter is set to a value other than 0, this function is enabled.

After receiving an intra-frequency or inter-frequency measurement report from the UE, the target cell performs the following operations before the next handover:

- If the cell in the measurement report is the source cell of the last incoming handover, the target cell checks whether the incoming handover protection timer expires.
 - If the timer has not expired, the source cell is filtered out from the HO Candidate List and no handover to the source cell is triggered.
 - If the timer has expired, a handover to the source cell can be triggered.
- If the cell in the measurement report is not the source cell of the last incoming handover, the handover is directly performed.

5.1.2 Network Analysis

5.1.2.1 Benefits

After this function is enabled, ping-pong handovers in high-speed scenarios are reduced and handover success rates (**Intra-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)**) increase.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

No impact

5.1.3 Requirements

5.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeed Flag	<i>High Speed Mobility</i>	The "incoming handover protection for high-speed cells" function requires that NRDUCell.HighSpeed Flag be set to HIGH_SPEED .

5.1.3.2.2 Mutually Exclusive Functions

None

5.1.3.2.3 Function Impacts

None

5.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

[Table 5-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Cell Handover Protect Timer	NRCellMobilityConfig.HsCellHoProtectTimer	Set this parameter to a value other than 0 to enable incoming handover protection for high-speed cells. A smaller value of this parameter results in a higher probability that the UE is handed over back to the source cell due to signal fluctuation. A larger value of this parameter results in a lower probability that the UE is handed over back to the source cell.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling incoming handover protection for high-speed cells (using  
NRCellMobilityConfig.HsCellHoProtectTimer being set to 2 as an example)  
MOD NRCELLMOBILITYCONFIG: NrcellId=0, HsCellHoProtectTimer=2;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling incoming handover protection for high-speed cells  
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HsCellHoProtectTimer=0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

Run the **LST NRCELLMOBILITYCONFIG** command and check the value of **High Speed Cell Handover Protect Timer(0.5s)** in the command output. If the value is not **0**, incoming handover protection for high-speed cells has been enabled.

5.1.4.3 Network Monitoring

The indicators listed in [5.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.2 Directional Inter-TRP Handover

5.2.1 Principles

In high-speed railway scenarios, the directional inter-TRP handover function is introduced to enable directional inter-TRP handovers to target TRPs with better signal quality (referred to as the target TRP in this document) for high-speed UEs, improving the timeliness of intra-cell inter-TRP handovers for these UEs. The directional inter-TRP handover function takes effect only in a hyper cell or combined cell. This function is controlled by the admission RSRP for directional inter-TRP handovers (specified by the **NRDUCellMultiTrp.DirectTrpHoAdmitRsrp** parameter). The delta RSRP for directional inter-TRP handovers (specified by the **NRDUCellMultiTrp.DirectTrpHoDeltaRsrp** parameter) is used to adjust the timeliness of inter-TRP handovers. This function takes effect only when the admission RSRP for directional inter-TRP handovers is set to a value other than **0**.

The processing mechanism of this function is as follows:

1. Identifying candidate TRPs: If the difference in the SRS RSRP between a non-serving TRP and the serving TRP is greater than or equal to the value of the **NRDUCellMultiTrp.DirectTrpHoAdmitRsrp** parameter, the gNodeB considers the non-serving TRP as a candidate TRP for a directional inter-TRP handover.
2. Determining target TRPs: Based on the trend in SRS RSRP changes among candidate TRPs, the gNodeB selects a candidate TRP that meets the following conditions as a target TRP for directional handover:

- The difference in the SRS RSRP between the candidate TRP and the serving TRP is greater than or equal to the delta RSRP for directional inter-TRP handovers (specified by the **NRDUCellMultiTrp.DirectTrpHoDeltaRsrp** parameter).
- The difference in the SRS RSRP between the candidate TRP and the serving TRP continuously decreases, and the number of consecutive decrease times is greater than the number of RSRP evaluations for inter-TRP handovers (specified by the **NRDUCellMultiTrp.InterTrpHoRsrpEvalTimes** parameter).

 NOTE

- For details about Cell Combination networking, see *Cell Combination*.
- For details about Hyper Cell, see *Hyper Cell*.

5.2.2 Network Analysis

5.2.2.1 Benefits

In high-speed cells, it is recommended that directional inter-TRP handover and the function of adaptation of the number of times TRS MAC CEs are transmitted in an inter-TRP handover be both enabled, to improve the timeliness of handovers and decrease the service drop rate in high-speed cells. The function of adaptation of the number of times TRS MAC CEs are transmitted in an inter-TRP handover is controlled by the **NRDUCellMultiTrp.TrpHoMacCeAdpTransNumOfs** and **NRDUCellMultiTrp.TrpHoMacCeAdpTransBlerThld** parameters. For details, see *Cell Combination* or *Hyper Cell*.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

- The smaller the admission RSRP for directional inter-TRP handovers, the delta RSRP for directional inter-TRP handovers, or the number of RSRP evaluations for inter-TRP handovers, the higher the probability of directional inter-TRP handovers, and the higher the probability of ping-pong handovers in high-speed scenarios.
- The larger the admission RSRP for directional inter-TRP handovers, the delta RSRP for directional inter-TRP handovers, or the number of RSRP evaluations for inter-TRP handovers, the less timely the handovers for UEs, and the lower the probability of ping-pong inter-TRP handovers.
- If the **NRDUCellMultiTrp.HyperCellTrpHoRsrpPolicy** parameter is set to **DL_EQ_RSRP** in hyper cells and the **NRDUCellTrp.MaxTransmitPower** parameter value across TRPs differs greatly, UEs may be handed over prematurely to high-power TRPs. As a result, the uplink throughput of the high-speed cell and the uplink user-perceived rate of high-speed UEs decrease, and the downlink throughput of the high-speed cell and the downlink user-perceived rate of high-speed UEs may fluctuate.
- When directional inter-TRP handover takes effect, it is recommended that the function of preventing handovers to overshoot-coverage TRPs (by configuring the **NRDUCellMultiTrp.TrpTrpOverCovTaThld** and

NRDUCellMultiTrp.TrpTrpOverCovHoRsrpThldOfs parameters) be enabled at the same time. It prevents incorrect target TRP determination and incorrect inter-TRP handovers due to overshoot coverage. Incorrect inter-TRP handovers may increase the uplink and downlink bit errors of the cell and lead to user experience deterioration. In severe cases, accessibility and retainability KPIs may fluctuate.

5.2.3 Requirements

5.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

None

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

None

5.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

5.2.3.4 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

[Table 5-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Directional TRP HO Admission RSRP	NRDUCellMultiTrp.DirectTrpHoAdmitRsrp	In high-speed cells, it is recommended that this parameter be set to -24 to improve the timeliness of inter-TRP handovers for high-speed UEs.
Directional TRP HO Delta RSRP	NRDUCellMultiTrp.DirectTrpHoDeltaRsrp	Set this parameter to its recommended value if directional inter-TRP handover is enabled.
Inter-TRP Handover RSRP Evaluation Times	NRDUCellMultiTrp.InterTrpHoRsrpEvalTimes	Set this parameter to its recommended value if directional inter-TRP handover is enabled.
Target TRP Overshoot Coverage TA Threshold	NRDUCellMultiTrp.TrpTrpOverCovTaThld	It is recommended that this parameter be set to 56 if directional inter-TRP handover is enabled.
Target TRP Overshoot Cov HO RSRP Thld Offset	NRDUCellMultiTrp.TrpTrpOverCovHoRsrpThldOfs	It is recommended that this parameter be set to 30 if directional inter-TRP handover is enabled.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling directional inter-TRP handover (It is recommended that the function of preventing handovers to overshoot-coverage TRPs be enabled at the same time.)  
MOD NRDUCELLMULTITRP: NrDuCellId=0, DirectTrpHoAdmitRsrp=-24, DirectTrpHoDeltaRsrp=0,  
InterTrpHoRsrpEvalTimes=2, TrpTrpOverCovTaThld=56, TrpTrpOverCovHoRsrpThldOfs=30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.


```
//Disabling directional inter-TRP handover  
MOD NRDUCELLMULTITRP: NrDuCellId=0, DirectTrpHoAdmitRsrp=0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Compare the service drop rates before and after the function of directional inter-TRP handover and the function of adaptation of the number of times TRS MAC CEs are transmitted in an inter-TRP handover are enabled, in a high-speed cell with a high service drop rate. If the service drop rate decreases after these functions are enabled, it indicates that the two functions have taken effect. For details about how to observe the function of adaptation of the number of times TRS MAC CEs are transmitted in an inter-TRP handover, see *Cell Combination or Hyper Cell*.

5.2.4.3 Network Monitoring

The indicators listed in [5.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

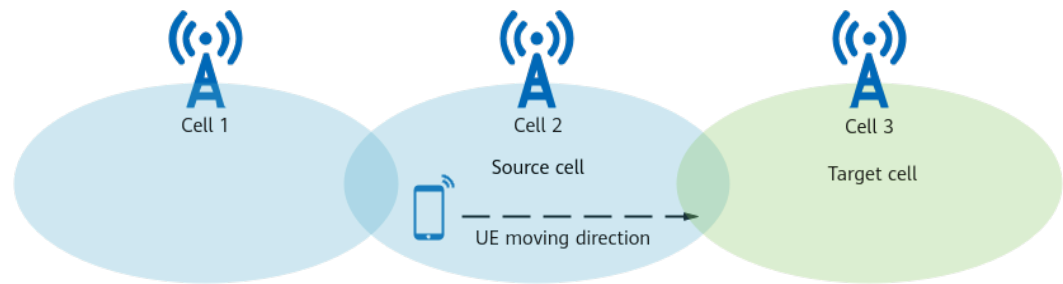
5.3 Adaptive Handover in High-Speed Cells

5.3.1 Principles

In high-speed railway scenarios, UEs move fast and signals change rapidly. As a result, inter-cell handovers may not be performed in a timely manner, affecting user experience of high-speed UEs. The function of adaptive handover in high-speed cells is introduced to ensure that high-speed UEs can be handed over between cells in a timely manner.

This function takes effect in high-speed cells and requires that each high-speed cell has only two high-speed intra-frequency neighboring cells. [Figure 5-1](#) shows an example, in which this function can take effect in cell 2 only if cells 1 and 3 are high-speed intra-frequency neighboring cells of cell 2. When a UE moves from cell 2 to cell 3, cell 2 is the source cell and cell 3 is the target cell for the handover. This function takes effect only on SA UEs with historical NR CGIs and high-speed flags, that is, high-speed UEs that access the high-speed cells through incoming RRC connection reestablishments or handovers.

Figure 5-1 Adaptive handover in high-speed cells



This function is controlled by the **NRCellMobilityConfig.HsCellAdaptHoOffsetLevel** parameter. If this parameter is set to a non-zero value, this function takes effect.

With this function, the base station identifies the moving direction of a high-speed UE based on the NR CGI of the source cell, and adjusts the CIOs of the source and target cells based on the value of the **NRCellRelation.CellIndividualOffset** parameter. Specifically, the base station adds a CIO offset to the CIO of the target cell to improve the timeliness of handovers of high-speed UEs between cells. The base station subtracts the CIO offset from the CIO of the source cell to prevent ping-pong handovers of high-speed UEs back to the source cell. The CIO indicates the difficulty level of an inter-cell handover for a UE.

A high-speed UE reports the measurement result based on the adjusted CIO, and the base station determines whether to hand over the high-speed UE based on the measurement report.

5.3.2 Network Analysis

5.3.2.1 Benefits

If the SSB RSRP difference of a high-speed UE before and after an inter-cell handover is large, the inter-cell handover is not performed in time. After this function is enabled, high-speed UEs can be handed over between cells in a timely manner, reducing the number of service drops caused by delayed handovers. In addition, the non-contention-based access success rate (Non Contention RA Success Rate = $\frac{\text{N.RA.Dedicated.Msg3}}{\text{N.RA.Dedicated.Att}} \times 100\%$) increases, the handover success rate (indicated by the **Intra-RAT Handover In Success Rate (CU)**) may increase, and the sum of all non-operator-based handover durations (indicated by HO Duration Avg Summary) may decrease.

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

A larger value of the **NRCellMobilityConfig.HsCellAdaptHoOffsetLevel** parameter results in premature handovers of high-speed UEs to target cells. As a result, the user-perceived rate of high-speed UEs may decrease. In overshoot coverage scenarios, this function makes it more difficult for UEs to be handed over to the source cell. As a result, the user-perceived rate of high-speed UEs may fluctuate. Overshooting refers to the process in which a UE is handed over from one cell to another.

5.3.3 Requirements

5.3.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

None

5.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level intra-frequency MRO	INTRA_FREQ_MRO_SW option of the NRCellMro.MroSwitch parameter	<i>MRO</i>	Adaptive handover in high-speed cells is mutually exclusive with cell-level intra-frequency MRO.

5.3.3.2.3 Function Impacts

None

5.3.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

5.3.3.4 Others

None

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

Table 5-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High-Speed Cell Adaptive HO Offset Level	NRCeIlMobIliTyConfiG.HsCellAdaptHoOffsetLevel	When handovers are not performed in time in high-speed scenarios, you are advised to set the NRCeIlMobIliTyConfiG.HsCellAdaptHoOffsetLevel parameter to a value other than 0 , to improve the handover timeliness of high-speed UEs.

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling adaptive handover in high-speed cells  
MOD NRCELLMOBILITYCONFIG: NrcellId=0, HsCellAdaptHoOffsetLevel=1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling adaptive handover in high-speed cells  
MOD NRCELLMOBILITYCONFIG: NrcellId=0, HsCellAdaptHoOffsetLevel=0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

Compare the values of **Q-OffsetRange** in **CellIndividualOffset** in the RRC signaling message delivered by the base station before and after this function is enabled. If the values are different, this function has taken effect.

5.3.4.3 Network Monitoring

The indicators listed in [5.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6 Coverage Improvement for High-Speed Cells

6.1 Uplink Waveform Adaptation in High-Speed Cells

6.1.1 Principles

NR supports two waveforms: CP-OFDM waveform and DFT-s-OFDM waveform.

- CP-OFDM is characterized by flexible resource allocation and large frequency diversity gains due to the use of non-contiguous frequency-domain resources. However, this allocation of resources can produce a high peak to average power ratio (PAPR).
- DFT-s-OFDM features a low PAPR (close to that of a single carrier) and high transmit power. However, it allows for the use of only contiguous frequency-domain resources.

In high-speed scenarios, the network side instructs UEs to always use the CP-OFDM waveform. To improve uplink coverage in high-speed railway scenarios, the function of uplink waveform adaptation in high-speed cells is introduced.

After this function is enabled, the network side instructs UEs to adaptively select the CP-OFDM or DFT-s-OFDM waveform. In high-speed scenarios:

- In high-speed scenarios, the network side instructs the following UEs to select the DFT-s-OFDM waveform if they support rank 1: full-buffer UEs, high-speed UEs that newly access a cell (through initial access, incoming handovers, or RRC connection reestablishments) when the train passes by, and unidentified UEs. In addition, full-buffer UEs and VoNR UEs are preferentially guaranteed to use the DFT-s-OFDM waveform. UEs using the DFT-s-OFDM waveform can transmit signals at higher power, improving user experience of high-speed UEs with limited uplink coverage. For details about the transmit power, see 3GPP TS 38.101.
- For non-full-buffer UEs, UEs before the train passes by, low-speed UEs when the train passes by, and UEs that does not support rank 1 in the uplink, the network side instructs them to use the CP-OFDM waveform.

This function is controlled by the **HS_UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. The number of cells with the **HS_UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter selected cannot exceed eight within a base station.

6.1.2 Network Analysis

6.1.2.1 Benefits

After this function is enabled, the average uplink throughput of high-speed UEs increases in high-speed cells with limited uplink coverage (the average uplink MCS index is less than 20).

NOTE

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

Average uplink MCS index = $\sum(n \times N.ChMeas.PUSCH.MCS.n) / \sum(N.ChMeas.PUSCH.MCS.n)$

6.1.2.2 Impacts

The impacts between this function and other functions are described in [6.1.3.2.3 Function Impacts](#).

- CP-OFDM can use non-contiguous frequency-domain resources while DFT-s-OFDM can use only contiguous frequency-domain resources. If UEs select DFT-s-OFDM rather than CP-OFDM in non-contiguous frequency-domain scenarios, the scheduling bandwidth is smaller, and the single-UE throughput may not increase or even decrease. The specific negative impacts depend on the changes in scheduling bandwidth and power spectral density as well as the increase in UE transmit power after switching from CP-OFDM to DFT-s-OFDM.
- After the function of uplink waveform adaptation in high-speed cells is enabled, compatibility issues may occur on UEs not supporting DFT-s-OFDM, causing service drops.
- The number of DFT scrambling code IDs defined in 3GPP specifications is limited. When different cells allocate the same DFT scrambling code ID to UEs, inter-UE interference may occur. As a result, the average uplink user-perceived rate fluctuates, and even access- or retainability-related counters of the cells are affected.
- If the DFT waveform is used, higher transmit power is obtained. After the power is increased, the uplink bit error rate may decrease and the uplink MCS index may increase. The proportion of uplink 256QAM may increase, the proportion of uplink QPSK may decrease, and the proportions of other modulation schemes may fluctuate.
- The increase in uplink throughput may increase the downlink traffic volume of TCP services. As a result, the downlink PDCP-layer and RLC-layer traffic volumes of the cell increase, the downlink transmission duration may increase, the downlink PDSCH RB usage increases, and the average downlink user-perceived rate, average downlink cell throughput, and average MCS index for downlink scheduling may fluctuate.

6.1.3 Requirements

6.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink waveform adaptation in high-speed cells requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .
Low-frequency TDD	Uplink waveform adaptation	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Uplink waveform adaptation in high-speed cells requires that the UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter be selected. In low-frequency TDD scenarios, uplink waveform adaptation in high-speed cells requires that the TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter be selected.

6.1.3.2.2 Mutually Exclusive Functions

None

6.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VoNR	VONR_SW option of the NRCellAlgoSwitch . <i>VonrSwitch</i> parameter	<i>VoNR</i>	After uplink DFT waveform adaptation is enabled in a high-speed cell, full buffer UEs and VoNR UEs preferentially share reserved DFT-s-OFDM resources. Due to limited DFT-S-OFDM resources, there is a low probability that VoNR UEs cannot use the DFT-OFDM waveform when there are a large number of full-buffer UEs and high-speed UEs in a cell.

6.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

6.1.3.4 Others

None

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

[Table 6-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	In high-speed cells, it is recommended that the HS_UL_WAVEFORM_ADAPT_SW option be selected to increase the transmit power of high-speed UEs and improve user experience at the coverage edge.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to [6.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

For low-frequency TDD:

```
//Enabling uplink waveform adaptation in high-speed cells
MOD NRDUCELHIGHSPEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=HS_UL_WAVEFORM_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

For low-frequency TDD:

```
//Disabling uplink waveform adaptation in high-speed cells
MOD NRDUCELHIGHSPEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=HS_UL_WAVEFORM_ADAPT_SW-0;
```

6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

Compare the average uplink throughputs of high-speed UEs before and after the function of uplink waveform adaptation in high-speed cells is enabled in a high-

speed cell with limited uplink coverage (the average uplink MCS index is less than 20). If the average uplink throughput of high-speed UEs increases after this function is enabled, it indicates that this function has taken effect.

NOTE

Average downlink throughput of high-speed UEs = $(N.ThpVol.DL.Hs - N.ThpVol.DL.LastSlot.Hs) / N.ThpTime.DL.RmvLastSlot.Hs$

Average uplink MCS index = $\sum(n \times N.ChMeas.PUSCH.MCS.n) / \sum(N.ChMeas.PUSCH.MCS.n)$

6.1.4.3 Network Monitoring

The indicators listed in [6.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.2 Adaptation of Initial Access Beams in Multi-SSB Scenarios

6.2.1 Principles

When channels and signals of a high-speed cell are configured to be sent in wide beam mode (see [Table 6-2](#) for parameter setting), both the downlink SSB beam and downlink initial access beams (including the Msg2, Msg4, PMI, and downlink data transmission before SRS reporting) are sent in wide beam mode, improving access reliability for UEs moving at high speeds.

When channels and signals of a high-speed cell are configured to be sent in narrow beam mode (see [Table 6-2](#) for parameter setting), both the downlink SSB beams and downlink initial access beams (including the Msg2, Msg4, PMI, and downlink data transmission before SRS reporting) are sent in narrow beam mode. In this case, the SSB RSRP of the cell is better than when wide beam mode is used. However, narrow beams need to be updated in a timely manner when UEs moving at high speeds access the cell, and the indicators related to RRC connection setups are worse than when wide beam mode is used.

Therefore, it is recommended that the SSB beam be sent in wide beam mode to ensure UE access in high-speed cells. However, if narrow beam mode is used to improve SSB coverage, it is recommended that adaptation of initial access beams in multi-SSB scenarios be enabled to ensure both the cell coverage rate and high-speed UE access reliability.

This function is controlled by the **DL_INITIAL_BEAM_ADAPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. This function takes effect only when the **DL_INITIAL_BEAM_SELECT_SW** option of the **NRDUCellAlgoSwitch.BeamOptSwitch** parameter is also selected. After this function is enabled, when a train arrives in a high-speed cell, the downlink SSB beams are sent in narrow beam mode and the downlink initial access beams are sent in wide beam mode. When there is no train in the high-speed cell, both the downlink SSB beams and initial access beam are sent in narrow beam mode.

Table 6-2 Parameter settings for wide and narrow beams

Beam Type	Parameter Setting
Wide beam	• The DL_INITIAL_BEAM_SELECT_SW option of the NRDUCELLALGOSWITCH.BeamOptSwitch parameter is selected. • The NRDUCELLTRPBEAM.CoverageScenario parameter is set to EXPAND_SCENARIO_1.
Narrow beam	• The DL_INITIAL_BEAM_SELECT_SW option of the NRDUCELLALGOSWITCH.BeamOptSwitch parameter is selected. • The NRDUCELLTRPBEAM.CoverageScenario parameter is set to DEFAULT (default scenario) or other values.

6.2.2 Network Analysis

6.2.2.1 Benefits

If a high-speed cell is configured with a wide SSB beam, the SSB coverage needs to be improved, and the deterioration of indicators such as the RRC setup success rate needs to be alleviated, the cell can be configured with narrow SSB beams and this function can be enabled. After this function is enabled, the SSB coverage of the cell improves. If narrow SSB beams have been configured for the high-speed cell, enabling this function will increase the values of indicators related to accessibility, such as the RRC setup success rate.

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

After this function is enabled:

- Assume that the SSB beams of the high-speed cell are currently sent in wide beam mode. After the beam mode is changed to narrow beam, values of accessibility-related indicators such as the RRC setup success rate and random access success rate decrease.
- If the SSB beams of the high-speed cell are currently sent in narrow beam mode, values of accessibility-related indicators such as the contention-based access success rate and random access success rate decrease. In addition, if there are a large number of CEUs in the high-speed cell, the RRC setup success rate in the high-speed cell may still decrease. However, the decrease is less than that caused by change from wide beam mode to narrow beam mode for the SSB beam.
- After this function takes effect, the number of UEs that have accessed the network at the cell edge decreases, and the values of the following indicators (including but not limited to) change:

Indicators related to the number of UEs	The values of the following indicators decrease: • Average number of UEs in a cell • Maximum number of UEs in a cell • Average number of UEs scheduled in the uplink in a cell • Average number of UEs scheduled in the downlink in a cell
Indicators related to channel quality	The values of the following indicators may increase: • PUSCH RSRP distribution • PUSCH SINR distribution • Average MCS index selected for uplink 256QAM scheduling in a cell The values of the following indicators may fluctuate or decrease: • Proportion of 16QAM modulation on the downlink SCH • Proportion of QPSK modulation on the downlink SCH The values of the following indicators decrease: • Number or proportion of downlink HARQ retransmissions • Downlink RBLER After energy saving takes effect, the values of the following indicators may decrease: • Average transmit power of RF units • Average downlink cell power • Maximum transmit power of RF units
Indicators related to load and usage	The values of the following indicators may decrease: • Uplink PUSCH RB usage of a cell • Downlink PDSCH RB usage of a cell • Average number of PDCCH CCEs used for uplink DCI • Average number of PDCCH CCEs used for downlink DCI • PDCCH CCE usage • Uplink PRB usage distribution • Downlink PRB usage distribution • Average user-plane CPU usage of a board in a gNodeB
Indicators related to the data rate and traffic volume	The values of the following indicators may decrease: • Uplink RLC-layer traffic volume • Downlink RLC-layer traffic volume • Uplink PDCCP-layer traffic volume in a cell • Downlink PDCCP-layer traffic volume in a cell • Uplink/Downlink RLC-layer throughput distribution

6.2.3 Requirements

6.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Adaptation of initial access beams in multi-SSB scenarios requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .
Low-frequency TDD	Downlink initial beam selection	DL_INITIAL_BEAM_SELECT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	<i>Beam Management</i>	Adaptation of initial access beams in multi-SSB scenarios can take effect only when the DL_INITIAL_BEAM_SELECT_SW option be selected for the cell.

6.2.3.2.2 Mutually Exclusive Functions

None

6.2.3.2.3 Function Impacts

None

6.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

6.2.3.4 Cells

It is recommended that this function be enabled in 8T8R, 32T32R, or 64T64R cells.

6.2.3.5 Others

None

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

Table 6-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	It is recommended that the DL_INITIAL_BEAM_ADAPT_SW option be selected only when both SSB coverage needs to be improved and the RRC setup success rate of the high-speed cell needs to be improved to some extent. Compared with when the wide SSB beam is configured for a cell, when narrow SSB beams are configured for the cell and the DL_INITIAL_BEAM_ADAPT_SW option is selected, the coverage range increases and the deterioration of the RRC setup success rate is alleviated.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to **6.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on DL_INITIAL_BEAM_SELECT_SW  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, BeamOptSwitch=DL_INITIAL_BEAM_SELECT_SW-1;  
//Enabling adaptation of initial access beams in multi-SSB scenarios  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=DL_INITIAL_BEAM_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling adaptation of initial access beams in multi-SSB scenarios
```

```
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=DL_INITIAL_BEAM_ADAPT_SW-0;
```

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

Compare the RRC setup success rate before and after this function is enabled. If SSB narrow beams are configured for the cell before this function is enabled, the RRC setup success rate increases after this function is enabled.

6.2.4.3 Network Monitoring

The indicators listed in [6.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7 High-Speed Cell Capacity Optimization

7.1 SRS Period and CDM Adaptation

7.1.1 Principles

In medium-or heavy-load scenarios, it is recommended that SRS period and CDM adaptation be enabled in high-speed cells to increase available SRS resources over the air interface. After this function is enabled, the base station adaptively allocates SRS resources of different periods to newly admitted UEs based on the number of UEs in a cell and supports SRS CDM between two UEs.

- When the SRS period is set to 80 ms (that is, the **NRDUCellSrs.SrsPeriod** parameter is set to **SL160**), the system adaptively allocates SRS resources with the period being 80 ms and 160 ms to UEs and supports SRS CDM between two UEs.
- When the SRS period is set to 160 ms (that is, the **NRDUCellSrs.SrsPeriod** parameter is set to **SL320**), the system adaptively allocates SRS resources with the period being 160 ms to UEs and supports SRS CDM between two UEs.

For details about SRS code division, see "CDM" in section "SRS Multiplexing" of *Channel Management*.

SRS period and CDM adaptation is controlled by the **SRS_PERIOD_AND_CDM_ADAPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

If the cell bandwidth is 80 MHz, it is recommended that the **NRDUCellSrs.SrsNarrowBandIndexCsrs** parameter be set to **48** to increase available SRS resources.

7.1.2 Network Analysis

7.1.2.1 Benefits

If the maximum number of UEs in a high-speed cell exceeds 600, it is recommended that SRS period and CDM adaptation be enabled. After this

function is enabled, the service drop rate does not increase due to insufficient SRS resources over the air interface.

7.1.2.2 Impacts

The impacts between this function and other functions are described in [7.1.3.2.3 Function Impacts](#).

After SRS period and CDM adaptation is enabled, the proportion of 2-port resource allocation decreases. If the proportion of UEs supporting 2-port resources in the uplink is high, the uplink UE throughput may slightly decrease. CDM is performed on SRS resources between two UEs. In this case, if overshoot coverage between TRPs in a cell is severe and two UEs are located in different TRPs with a distance over 1 km, the two TRPs can receive uplink SRS signals from the two UEs at the same time and SRS resources may interfere with each other, slightly increasing the service drop rate. In addition, SRS interference may increase between cells with PCI mod 3 values of 0 and 1.

7.1.3 Requirements

7.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeed Flag	<i>High Speed Mobility</i>	SRS period and CDM adaptation requires that the NRDUCell.HighSpeed Flag parameter be set to HIGH_SPEED .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Specification Improvement Switch	SPEC_INCREASE_SW option of the NRDUCELLSRS.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	SRS period and CDM adaptation requires the SPEC_INCREASE_SW option of the NRDUCELLSRS.SrsAlgoExtSwitch parameter to be selected.

7.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPre-code.SuMimoSrsPeriodShortSw parameter	None	"SRS period and CDM adaptation" is mutually exclusive with "SRS period shortening for downlink 1T4R UEs".
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	DL_2T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPre-code.SuMimoSrsPeriodShortSw parameter	None	"SRS period and CDM adaptation" is mutually exclusive with "SRS period shortening for downlink 2T4R UEs".
Low-frequency TDD	Tight SRS multiplexing	SRS_TIGHT_MULTIPLEXING_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	For high-speed cells, tight SRS multiplexing can be enabled only when "SRS period and CDM adaptation" is enabled.

7.1.3.2.3 Function Impacts

None

7.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

7.1.3.4 Others

None

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

Table 7-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	In high-speed cells, it is recommended that the SRS_PERIOD_AND_CDM_ADAPT_SW option be selected to increase available SRS resources over the air interface.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to **7.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SRS period and CDM adaptation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=SRS_PERIOD_AND_CDM_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling SRS period and CDM adaptation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=SRS_PERIOD_AND_CDM_ADAPT_SW-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

When SRS resources over the air interface are insufficient, compare the service drop rate before and after the SRS period and CDM adaptation function is enabled. If the service drop rate issue is optimized after this function is enabled, it indicates that this function has taken effect.

7.1.4.3 Network Monitoring

The indicators listed in [7.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.2 Uplink DCI Extension-based Scheduling in High-Speed Cells

7.2.1 Principles

As more UEs are scheduled in a heavily loaded cell, not all RBs in the cell can be allocated due to limited uplink control channel element (CCE) resources. In this case, the downlink data volume decreases due to delayed uplink TCP ACK feedback. The "uplink DCI extension-based scheduling in high-speed cells" function is introduced to increase the uplink RB usage in a cell. This function uses additional downlink slots to transmit DCI, improving the utilization of uplink RB resources and average uplink cell throughput. This function also prevents the decrease in the downlink data volume caused by delayed uplink TCP ACK feedback, increasing the average downlink cell throughput.

The "uplink DCI extension-based scheduling in high-speed cells" function is jointly controlled by the **UL_SCH_DCI_EXT_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter and the **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch.UIPuschAlgoSwitch** parameter.

After uplink DCI extension-based scheduling in high-speed cells is enabled, it takes effect only when all of the following conditions are met:

- The cell is in Hyper Cell networking.
- The number of UEs in the cell is greater than the UE number threshold for uplink DCI extension-based scheduling (specified by the **NRDUCellUISch.UISchDciExtUsrNumThld** parameter).
- The ratio of the number of UEs to the UE number threshold (specified by the **NRDUCellUISch.UISchDciExtUsrNumThld** parameter) must be greater than the value of the **NRDUCellUISch.UISchDciExtExitThld** parameter.

This function is recommended when uplink CCEs are insufficient (for example, when the uplink CCE allocation success rate is lower than 90%).

For details about uplink DCI extension-based scheduling in low-speed scenarios, see section "Uplink DCI Extension-based Scheduling (Low-Frequency TDD)" in *Uplink Scheduling*.

7.2.2 Network Analysis

7.2.2.1 Benefits

In heavy-load scenarios, if uplink DCI extension-based scheduling is enabled, the number of uplink scheduled UEs, uplink PRB usage (indicated by **Uplink Resource Block Utilizing Rate (DU)**), and average uplink cell throughput (indicated by **Cell Uplink Average Throughput (DU)**) increase, and the CCE allocation failure rate decreases. For details, see the benefits-related description in section "Uplink DCI Extension-based Scheduling (Low-Frequency TDD)" of *Uplink Scheduling*.

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

- The average uplink cell throughput increases, the number of service drops due to uplink CCE congestion decreases, the abnormal QoS flow release rate may decrease, and the number of RRC_CONNECTED UEs may increase.
- The downlink CCE allocation failure rate may increase, and the downlink cell throughput may decrease. In addition, as K1 (interval between downlink scheduling and uplink ACK/NACK feedback) and K2 (interval between uplink grant DCI reception and corresponding PUSCH transmission) increase, the downlink or uplink UE scheduling delay prolongs, which may slightly affect user experience in light-load scenarios.
- As the number of UEs scheduled in the uplink per slot increases, the UE model and uplink and downlink traffic models may change. Consequently, the values of the indicators related to uplink and downlink links, user experience, and delay may fluctuate. The indicators include the user-perceived rate, MCS index, BLER, CCE aggregation level, CCE usage, and packet delay.
- It is recommended that the **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter be selected when uplink CCEs are insufficient (for example, when the uplink CCE allocation success rate is lower than 90%). It is recommended that **NRDUCellULSch.UlSchDciExtUsrNumThld** (specifying the UE number threshold for entering the uplink DCI extension-based scheduling function) and **NRDUCellULSch.UlSchDciExtExitThld** (specifying the UE number threshold for exiting the uplink DCI extension-based scheduling function) be set based on network conditions. If the parameters are not set properly (for example, the number of uplink CCEs in a cell is sufficient but the parameters are set to excessively small values), uplink and downlink user experience in the cell deteriorates. When uplink CCEs are insufficient, it is recommended that the parameters be set to **200** and **80**, respectively, to expand uplink CCEs and improve uplink capacity in a high-speed cell.

- For details about the network impacts, see the network impacts of uplink DCI extension-based scheduling in *Uplink Scheduling*.

7.2.3 Requirements

7.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink DCI extension-based scheduling in high-speed cells requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation	UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	Uplink DCI extension-based scheduling in high-speed cells requires that PDCCH uplink-to-downlink CCE ratio adaptation be enabled.

7.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Uplink DCI extension-based scheduling is not supported in a high-speed cell with the NRDUCell.NrDuCellNetworkingMode parameter set to HYPER_CELL_COMBINE_MODE .
Low-frequency TDD	Pattern-free PDCCH rate matching activation rate increase	PDCCH_NO_PAT_RM_RATIO_INC_SWITCH option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	<i>Downlink Scheduling</i>	Pattern-free PDCCH rate matching activation rate increase cannot be enabled together with uplink DCI extension-based scheduling.
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If the TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected, the DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter must be deselected.
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If the TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected, the DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter must be deselected.

7.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-TRP interference coordinated scheduling for high-speed railways	NRDUCellAlgoSwitch . HighSpeedIntrfRandomSchSw	<i>High Speed Mobility</i>	When uplink DCI extension-based scheduling takes effect in high-speed cells, the estimated number of RBs in some slots may be inaccurate because the scheduling time is earlier than expected. This affects the proportion of UEs in the overlapping area of the center area for which staggered scheduling based on mod 2 result takes effect.
Low-frequency TDD	Dynamic uplink slot aggregation	REDCAP_UL_COVERAGE_ENH_SW option of the NRDUCellFeatureSw . UeCapbBasedFunSw parameter	<i>RedCap</i>	For an NR TDD cell, uplink DCI extension-based scheduling does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect.

7.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

7.2.3.4 Others

Uplink DCI extension-based scheduling in high-speed cells can be enabled only when all of the following conditions are met:

- The **NRDUCell.DlBandwidth** parameter is set to **CELL_BW_40M**, **CELL_BW_50M**, **CELL_BW_60M**, **CELL_BW_70M**, **CELL_BW_80M**, **CELL_BW_90M**, and **CELL_BW_100M**.
- The **NRCell.FrequencyBand** and **NRDUCell.FrequencyBand** parameters are set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**, and the **NRDUCell.SlotAssignment** parameter is set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.

- The **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL** or **NORMAL_CELL**.

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

Table 7-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
UL Scheduling Algorithm Switch	NRDUCellUISch.UISchAlgoSwitch	It is recommended that the UL_SCH_DCI_EXT_SW option be selected when uplink DCI extension-based scheduling in high-speed cells is enabled.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UIPuschAlgoSwitch	It is recommended that the DCI_EXTEND_UL_CONGEST_SW option be selected when uplink DCI extension-based scheduling in high-speed cells is enabled.
UL Schedule DCI Ext User Number Threshold	NRDUCellUISch.UISchDciExtUsrNumThld	This function takes effect only when the number of UEs in the cell exceeds the UE number threshold for entering the function of uplink DCI extension-based scheduling. If the parameters are not set properly (for example, the number of uplink CCEs in a cell is sufficient but the parameters are set to excessively small values), uplink and downlink user experience in the cell deteriorates. When uplink CCEs are insufficient, it is recommended that the parameter be set to 200 to expand uplink CCEs and improve uplink capacity in a high-speed cell.

Parameter Name	Parameter ID	Setting Notes
UL Schedule DCI Extension Exit Threshold	NRDUCellULSch.UISchDciExtExitThld	This function exits only when the ratio of the number of UEs to the UE number threshold is less than or equal to the threshold for exiting the function of uplink DCI extension-based scheduling. If the parameters are not set properly (for example, the number of uplink CCEs in a cell is sufficient but the parameters are set to excessively small values), uplink and downlink user experience in the cell deteriorates. When uplink CCEs are insufficient, it is recommended that the parameter be set to 80 to expand uplink CCEs and improve uplink capacity in a high-speed cell.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to [7.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on UL_SCH_DCI_EXT_SW for a high-speed cell
MOD NRDUCCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=UL_SCH_DCI_EXT_SW-1;
MOD NRDUCCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-1;
//Setting the UE number thresholds for entering and exiting the function of uplink DCI extension-based scheduling
MOD NRDUCCELLULSCH: NrDuCellId=0, UISchDciExtUsrNumThld=200;
MOD NRDUCCELLULSCH: NrDuCellId=0, UISchDciExtExitThld=80;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink DCI extension-based scheduling for a high-speed cell
MOD NRDUCCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-0;
MOD NRDUCCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=UL_SCH_DCI_EXT_SW-0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

For details, see the activation verification of uplink DCI extension-based scheduling in *Uplink Scheduling*.

7.2.4.3 Network Monitoring

The indicators listed in [7.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.3 SRS Resource Extension in a High-Speed Cell

7.3.1 Principles

In high-speed railway scenarios, channel time variation is severe due to the high-speed movement of UEs. As a result, SRS resource demand increases significantly, which may lead to SRS resource insufficiency. SRS resource extension in a high-speed cell is introduced to increase UE capacity. After this function is enabled, the base station can allocate SRS resources to more UEs through optimized SRS resource allocation.

This function is controlled by the **SRS_RES_EXT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

7.3.2 Network Analysis

7.3.2.1 Benefits

In heavy-load scenarios, enabling SRS resource extension in a high-speed cell increases the available SRS resources for UEs and allows the high-speed cell to support more UEs.

7.3.2.2 Impacts

The impacts between this function and other functions are described in [7.3.3.2.3 Function Impacts](#).

In heavy-load scenarios, enabling SRS resource extension in a high-speed cell may reduce the average uplink and downlink throughputs of UEs in neighboring cells.

7.3.3 Requirements

7.3.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

7.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	SRS resource extension in a high-speed cell takes effect only when the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED .
Low-frequency TDD	SRS period and CDM adaptation	SRS_PERIOD_AND_CDM_ADAPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	SRS resource extension in a high-speed cell requires that "SRS period and CDM adaptation" be enabled.

7.3.3.2.2 Mutually Exclusive Functions

None

7.3.3.2.3 Function Impacts

None

7.3.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

7.3.3.4 Others

None

7.3.4 Operation and Maintenance

7.3.4.1 Data Configuration

7.3.4.1.1 Data Preparation

Table 7-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	To enable SRS resource extension in a high-speed cell, select the SRS_RES_EXT_SW option.

7.3.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SRS resource extension in a high-speed cell  
MOD NRDUCELHIGHSPEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=SRS_RES_EXT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling SRS resource extension in a high-speed cell  
MOD NRDUCELHIGHSPEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=SRS_RES_EXT_SW-0;
```

7.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.3.4.2 Activation Verification

In high-speed heavy-load scenarios, compare the number of UEs in a cell before and after the function is enabled. If the number of UEs in the cell increases, SRS resource extension in a high-speed cell has taken effect.

Cell User Quantity Measurement

Counter ID	Counter Name
1911837919	N.User.RRCCConn.RedCap.Avg
1911840354	N.User.RRCCConn.RedCap.Max

7.3.4.3 Network Monitoring

The indicators listed in [7.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.4 Uplink and Downlink Additional DMRS Position Adaptation

7.4.1 Principles

When high-speed trains are in operation, it is recommended that the **NRDUCellPusch.UlAdditionalDmrsPos** and **NRDUCellPdsch.DlAdditionalDmrsPos** parameters be set to **POS2** to improve performance aspects such as channel estimation and frequency correction for high-speed UEs. The POS2 setting decreases the overall performance of UEs on low-speed trains due to increased pilot overhead.

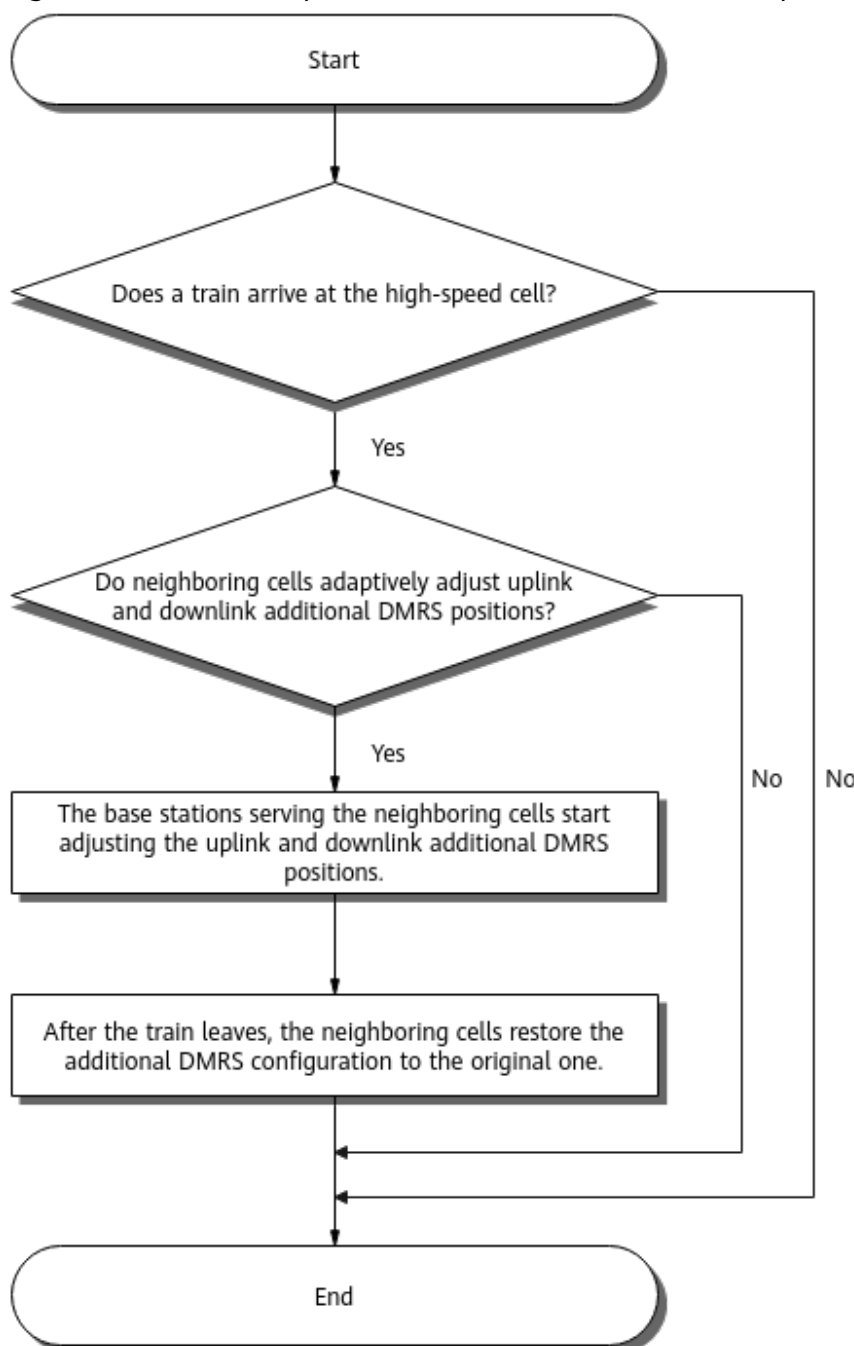
To optimize performance of UEs on low-speed trains, the function of uplink and downlink additional DMRS position adaptation is introduced. It is recommended that this function be enabled when the number of additional DMRS positions is set to POS2. After this function is enabled, when a train arrives at a cell, the cell instructs neighboring high-speed cells to adjust the number of uplink and downlink additional DMRS positions based on the train speed. This ensures UEs onboard obtain the optimal network performance after entering areas covered by these neighboring cells.

- Uplink additional DMRS position adaptation
This function is controlled by the **UL_ADDITIONAL_DMRS_ADAPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. If this switch is turned on for high-speed cells, the number of uplink additional DMRS positions (POS1 or POS2) can be adaptively adjusted for neighboring cells when high-speed trains arrive.
- Downlink additional DMRS position adaptation
This function is controlled by the **DL_POS_ADAPT_SCENARIO_SW** option of the **NRDUCellDmrs.DlPosAdaptOptAlgoSw** parameter. If this switch is

turned on for high-speed cells, the number of downlink additional DMRS positions (POS1 or POS2) can be adaptively adjusted for neighboring cells when high-speed trains arrive.

Figure 7-1 illustrates the processing mechanism of this function.

Figure 7-1 Process of uplink and downlink additional DMRS position adaptation



1. The base station determines whether a train arrives at the high-speed cell. A high-speed cell determines whether a train arrives based on the number of high-speed UEs in the cell. If this number is greater than or equal to the value of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter, it is considered that the train arrives.

2. Neighboring cells adaptively adjust uplink and downlink additional DMRS positions.

Based on the frequency offsets of identified high-speed UEs and the frequency offset threshold, the high-speed cell makes further evaluation. When the number of high-speed UEs whose frequency offset meets the requirement is greater than a specified number, the neighboring cell is instructed to use POS2 as the number of additional DMRS positions. Otherwise, POS1 is used.

The frequency offset threshold used for POS switching of downlink additional DMRS position adaptation is specified by the **NRDUCellDmrs.DlAdpAddDmrsFreqOfsUpThld** parameter. The frequency offset threshold in the uplink is set to a fixed value by the gNodeB.

3. Neighboring cells restore the additional DMRS configuration to the original one.

When a neighboring cell receives the train leaving notification from the serving cell and the train does not arrive at this neighboring cell, the neighboring cell restores the DMRS configuration of POS1.

7.4.2 Network Analysis

7.4.2.1 Benefits

After this function is enabled, if the number of additional DMRS positions for high-speed cells is set to POS2 and the train moves at a low speed, the average uplink and downlink throughput of high-speed UEs increases. If the number of additional DMRS positions for high-speed cells is set to POS1, this function is not recommended.

NOTE

Average downlink throughput of high-speed UEs = $(N.ThpVol.DL.Hs - N.ThpVol.DL.LastSlot.Hs) / N.ThpTime.DL.RmvLastSlot.Hs$

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

7.4.2.2 Impacts

The impacts between this function and other functions are described in [7.4.3.2.3 Function Impacts](#).

This function has no impact.

7.4.3 Requirements

7.4.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

7.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink and downlink additional DMRS position adaptation requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .
Low-frequency TDD	Frequency Band	NRDUCell.FrequencyBand	None	The DL_POS_ADAPT_SCENARIO_SW option of the NRDUCellDmrs.DlPosAdaptOptAlgoSw parameter in the uplink and downlink additional DMRS position adaptation function requires that the NRDUCell.FrequencyBand parameter be set to N38, N41, N77, N78, N1, N66 , or N28 .
Low-frequency TDD	Cell Combination Hyper Cell	Cell Combination: NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE Hyper Cell: NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Cell Combination Hyper Cell</i>	Uplink and downlink additional DMRS position adaptation requires that NRDUCell.NrDuCellNetworkingMode be set to HYPER_CELL_COMBINE_MODE or HYPER_CELL .

7.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink additional pilot adaptation	DL_POS_ADAPT_HIGH_SPEED_SW option of the NRDUCellDmrs.DlPosAdaptOptAlgoSw parameter	None	Downlink additional DMRS position adaptation is mutually exclusive with downlink additional pilot adaptation.

7.4.3.2.3 Function Impacts

None

7.4.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

7.4.3.4 Others

If the UE supports **oneFL-DMRS-TwoAdditionalDMRS-DL** and **oneFL-DMRS-TwoAdditionalDMRS-UL**, it supports two additional DMRS positions in the downlink and uplink, respectively.

7.4.4 Operation and Maintenance

7.4.4.1 Data Configuration

7.4.4.1.1 Data Preparation

[Table 7-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	Select the UL_ADDITIONAL_DMRS_ADAPT_SW option to enable uplink additional DMRS position adaptation in high-speed cells.

Parameter Name	Parameter ID	Setting Notes
DL Pos Adaptation Optimization Algo Switch	NRDUCellDmrs.DlPosAdaptOptAlgoSw	Select the DL_POS_ADAPT_SCENARIO_SW option to enable downlink additional DMRS position adaptation in high-speed cells.
DL Adaptive AddDMRS Freq Ofs Upper Thld	NRDUCellDmrs.DlAdpAddDmrsFreqOfsUpThld	Set this parameter to its recommended value if downlink additional DMRS position adaptation is enabled.
High Speed User Num Thld	NRDUCellHighSpeedMob.HighSpeedUserNumThld	Set this parameter to its recommended value if uplink and downlink additional DMRS position adaptation is enabled.

7.4.4.1.2 Using MML Commands

Before using MML commands, refer to [7.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the threshold for the number of high-speed UEs
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedUserNumThld=10;
//Enabling uplink additional DMRS position adaptation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_ADDITIONAL_DMRS_ADAPT_SW-1;
//Configuring the frequency offset threshold for POS switching of downlink additional DMRS position adaptation
MOD NRDUCELLDMRS: NrDuCellId=0, DlAdpAddDmrsFreqOfsUpThld=1170;
//Enabling downlink additional DMRS position adaptation
MOD NRDUCELLDMRS: NrDuCellId=0, DlPosAdaptOptAlgoSw=DL_POS_ADAPT_SCENARIO_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink additional DMRS position adaptation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_ADDITIONAL_DMRS_ADAPT_SW-0;
//Disabling downlink additional DMRS position adaptation
MOD NRDUCELLDMRS: NrDuCellId=0, DlPosAdaptOptAlgoSw=DL_POS_ADAPT_SCENARIO_SW-0;
```

7.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.4.2 Activation Verification

- Downlink: Check the **RRC_CONN_RECFG** message traced over the Uu interface. If *DMRS-DownlinkConfig* in *dmrs-DownlinkForPDSCH-MappingTypeA* contains the *dmrs-AdditionalPosition* IE with the value of **POS1**, the number of additional DMRS positions for the UE is **POS1**. If *DMRS-DownlinkConfig* in *dmrs-DownlinkForPDSCH-MappingTypeA* does not contain the *dmrs-AdditionalPosition* IE, the number of additional DMRS positions for the UE is **POS2**. Compare the query result with the configuration on the base station side. If they are inconsistent, it indicates that this function has taken effect.
- Uplink: Check the **RRC_CONN_RECFG** message traced over the Uu interface. If *DMRS-UplinkConfig* in *dmrs-UplinkForPUSCH-MappingTypeA* contains the *dmrs-AdditionalPosition* IE with the value of **POS1**, the number of additional DMRS positions for the UE is **POS1**. If *DMRS-UplinkConfig* in *dmrs-UplinkForPUSCH-MappingTypeA* does not contain the *dmrs-AdditionalPosition* IE, the number of additional DMRS positions for the UE is **POS2**. Compare the query result with the configuration on the base station side. If they are inconsistent, it indicates that this function has taken effect.

For details about the messages, see 3GPP TS 38.331 V16.7.0.

7.4.4.3 Network Monitoring

The indicators listed in [7.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.5 MU-MIMO in High-Speed Railway Scenarios (Trial)

PDSCH MU-MIMO and PUSCH MU-MIMO in high-speed railway scenarios are trial functions.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial

functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

7.5.1 Principles

PDSCH MU-MIMO

In high-speed railway scenarios, PDSCH MU-MIMO is supported only in common or hyper cells. PDSCH MU-MIMO in high-speed railway scenarios is a trial function and is supported only by 8T8R RRUs and massive MIMO modules.

This function is controlled by the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter.

For details, see *MIMO (TDD)*.

PUSCH MU-MIMO

In high-speed railway scenarios, PUSCH MU-MIMO is supported only in common or hyper cells. In addition, when MU-MIMO takes effect, **NRDUCellPusch.UIDmrsType** can only be set to **TYPE1**. PUSCH MU-MIMO in high-speed railway scenarios is a trial function and is supported only by 8T8R RRUs and massive MIMO RF modules.

This function is controlled by the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter.

For details, see *MIMO (TDD)*.

8 Frequency Offset and Channel Time-Variation Optimization for High-Speed Cells

8.1 Uplink Frequency Offset Range Estimation

8.1.1 Principles

The impact of frequency shifts on signals is represented as the phase rotation of the signals in the time domain. The gNodeB estimates frequency offsets based on the phase difference between two adjacent uplink DMRS symbols in a timeslot. A smaller interval between two adjacent uplink DMRS symbols indicates a larger range of frequency offsets that can be measured and a higher moving speed of a UE that can be identified by the base station as a high-speed UE.

In high-speed scenarios, if high-speed UEs move excessively fast, the high-speed cell must be configured with one uplink front-loaded DMRS position and two uplink additional DMRS positions. To reduce the pilot resource consumption and ensure that the actual frequency offset is within the measurable range, the uplink frequency offset range estimation function is introduced. After this function is enabled, the gNodeB can accurately measure the frequency offset of high-speed UEs even when one uplink front-loaded DMRS position and one uplink additional DMRS position are configured.

This function is jointly controlled by the **UL_FREQ_OFS_INTVL_ESTIMATE_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter and the **UL_FREQ_OFS_INTVL_EST_OPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. The gNodeB estimates frequency offsets based on the **NRDUCellPusch.UlAdditionalDmrsPos** parameter set to **POS1**.

8.1.2 Network Analysis

8.1.2.1 Benefits

This function increases the uplink user-perceived rate in high-speed cells.

8.1.2.2 Impacts

The impacts between this function and other functions are described in [8.1.3.2.3 Function Impacts](#).

- The increase in uplink throughput may increase the downlink traffic volume of TCP services. As a result, the downlink PDCP-layer and RLC-layer traffic volumes of the cell increase, the downlink transmission duration may increase, the downlink PDSCH RB usage increases, and the average downlink user-perceived rate, average downlink cell throughput, and average MCS index for downlink scheduling may fluctuate.
- After uplink frequency offset range estimation is enabled, the channel estimation mode is different when the **NRDUCellPusch.UlAdditionalDmrsPos** parameter is set to **POS1**, compared with when it is set **POS2**. When channel conditions are favorable, the uplink MCS index may increase and the uplink IBLER may slightly decrease. When channel conditions are poor, due to the accuracy of frequency offset range estimation, the uplink MCS index may slightly decrease, and indicators such as the IBLER, uplink RBLER, and loss rate of uplink packets received by the PDCP layer from the RLC layer may slightly increase.

 NOTE

$$\text{Average uplink MCS index} = \sum(n \times N.\text{ChMeas.PUSCH.MCS}.n) / \sum(N.\text{ChMeas.PUSCH.MCS}.n)$$

8.1.3 Requirements

8.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink frequency offset range estimation requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic functions of uplink scheduling	NRDUCellPusch.UlAdditionalDmrsPos	<i>Uplink Scheduling</i>	Once uplink frequency offset range estimation is enabled, the NRDUCellPusch.UlAdditionalDmrsPos parameter must be set to POS1 .

8.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink channel estimation optimization	UL_CHANNEL_ESTIMATE_OPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	Uplink frequency offset range estimation is mutually exclusive with uplink channel estimation optimization.
Low-frequency TDD	Uplink additional DMRS position adaptation	UL_ADDITIONAL_DMRS_ADAPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	Uplink frequency offset range estimation is mutually exclusive with uplink additional DMRS position adaptation.

8.1.3.2.3 Function Impacts

None

8.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

8.1.3.4 Cells

When uplink frequency offset range estimation is enabled, the **NRDUCellTrp.TxRxMode** parameter cannot be set to **32T32R** or **64T64R**.

8.1.3.5 Others

None

8.1.4 Operation and Maintenance

8.1.4.1 Data Configuration

8.1.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	In high-speed cells, it is recommended that the UL_FREQ_OFS_INTVL_ESTIMATE_SW option be selected to improve the uplink frequency offset estimation accuracy and reduce the DMRS resource consumption.
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	In high-speed cells, it is recommended that the UL_FREQ_OFS_INTVL_EST_OPT_SW option be selected when uplink frequency offset range estimation is enabled so that the frequency offsets of UEs in the cell decrease.

8.1.4.1.2 Using MML Commands

Before using MML commands, refer to **8.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink frequency offset range estimation
MOD NRDUCELHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_FREQ_OFS_INTVL_ESTIMATE_SW-1;
//Enabling "uplink frequency offset range estimation optimization for high-speed cells"
MOD NRDUCELHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_FREQ_OFS_INTVL_EST_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink frequency offset range estimation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_FREQ_OFS_INTVL_ESTIMATE_SW-0;
//Disabling "uplink frequency offset range estimation optimization for high-speed cells"
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_FREQ_OFS_INTVL_EST_OPT_SW-0;
```

8.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.1.4.2 Activation Verification

Compare the average uplink throughput of high-speed UEs before and after the uplink frequency offset range estimation function is enabled. If the throughput increases after this function is enabled, this function has taken effect.

8.1.4.3 Network Monitoring

The indicators listed in [8.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.2 Rank Restriction

8.2.1 Principles

In high speed mobility scenarios, the rank reporting of UEs may be inaccurate, affecting the average downlink throughput of high-speed UEs. The ranks allowed to be reported can be specified using the **NRDUCellPdsch.HighSpeedCellRiRestr** parameter.

It is recommended that this parameter be set to **3** for sections where the train speed exceeds 120 km/h, thereby limiting effective ranks to rank 1 and rank 2 to improve performance.

8.2.2 Network Analysis

8.2.2.1 Benefits

After this function is enabled, the average downlink throughput of high-speed UEs increases.

 NOTE

Average downlink throughput of high-speed UEs = $(N.ThpVol.DL.Hs - N.ThpVol.DL.LastSlot.Hs) / N.ThpTime.DL.RmvLastSlot.Hs$

8.2.2.2 Impacts

The impacts between this function and other functions are described in [8.2.3.2.3 Function Impacts](#).

This function has no impact.

8.2.3 Requirements

8.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeed Flag	<i>High Speed Mobility</i>	Rank restriction requires that the NRDUCell.HighSpeed Flag parameter be set to HIGH_SPEED .

8.2.3.2.2 Mutually Exclusive Functions

None

8.2.3.2.3 Function Impacts

None

8.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

8.2.3.4 Others

None

8.2.4 Operation and Maintenance

8.2.4.1 Data Configuration

8.2.4.1.1 Data Preparation

Table 8-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Cell RI Restriction	NRDUCellPdsch.HighSpeedCellRiRestr	It is recommended that this parameter be set to 3 when rank restriction is enabled in high-speed cells. If a UE does not support rank restriction of high-speed cells, this function does not take effect for the UE.

8.2.4.1.2 Using MML Commands

Before using MML commands, refer to **8.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling rank restriction
MOD NRDUCCELLPDSCH: NrDuCellId=0, HighSpeedCellRiRestr=3;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling rank restriction
MOD NRDUCCELLPDSCH: NrDuCellId=0, HighSpeedCellRiRestr=0;
```

8.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.2.4.2 Activation Verification

Run the **LST NRDUCELLPDSCH** command and check the value of **High Speed Cell RI Restriction** in the command output. If the value is not **0**, the rank restriction function has been enabled.

8.2.4.3 Network Monitoring

The indicators listed in [8.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.3 Multi-TRS Measurement (Trial)

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

8.3.1 Principles

This function allows a UE capable of multi-TRS measurement to measure multiple sets of TRSs at a time, as shown in [Figure 8-1](#) and [Figure 8-2](#). Multi-TRS measurement takes effect only on UEs in high-speed cells in Hyper Cell or Cell Combination networking.

Figure 8-1 Multi-TRS measurement (hyper cells)

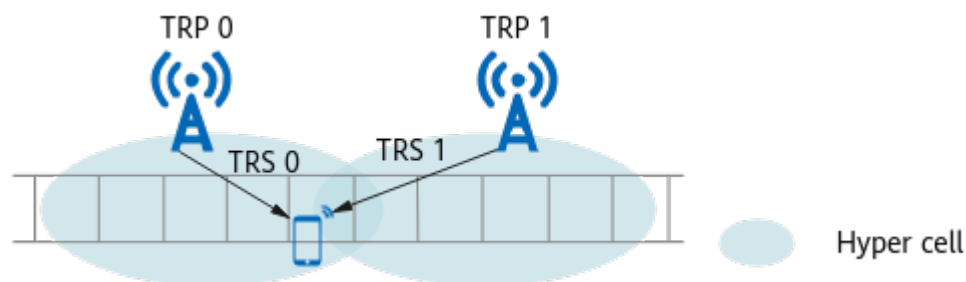
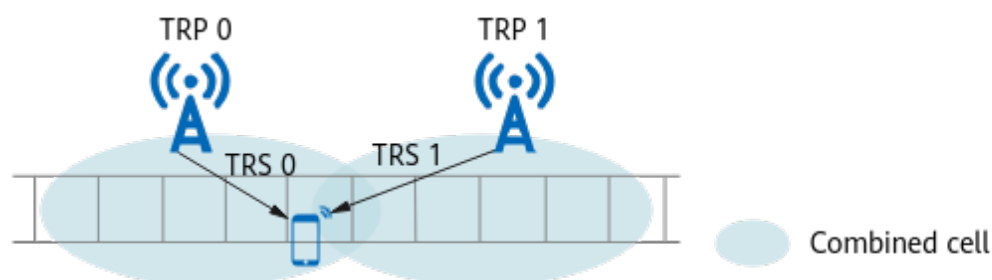


Figure 8-2 Multi-TRS measurement (combined cells)



This function is controlled by both the **NRDUCellAlgoSwitch.MultiTrsMeasSwitch** parameter and the **MULTI_TRS_MULTI_SLOT_MEAS_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. After this function is enabled, the gNodeB configures multiple sets of TRS resources for UEs in the cell. A UE that is capable of measuring multiple sets of TRSs can simultaneously measure multiple sets of TRSs in the serving cell.

8.3.2 Network Analysis

8.3.2.1 Benefits

Enabling this function is recommended when there are UEs capable of multi-TRS measurement. After this function is enabled, the risk of performance deterioration caused by frequency offset adjustment during handovers of high-speed UEs is reduced, and the downlink traffic volume (**Downlink Traffic Volume (DU)**) and average number of UEs (**Average User Number (CU)**) increase.

8.3.2.2 Impacts

The impacts between this function and other functions are described in [8.3.3.2.3 Function Impacts](#).

This function has no impact.

8.3.3 Requirements

8.3.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Multi-TRS measurement requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy	None	Multi-TRS measurement requires that the NRDUCellCsirs.TrsAllocationStrategy parameter be set to MULTI_TRS .
Low-frequency TDD	Hyper Cell Combination	Cell Combination: NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE Hyper Cell: NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell Combination</i>	Multi-TRS measurement requires that the NRDUCell.NrDuCellNetworkingMode parameter be set to HYPER_CELL or HYPER_CELL_COMBINE_MODE .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Data transmission mode	NRDUCellMultiTrp.<i>DataTransMode</i> set to BASIC_MODE or DPS_MODE	None	The MULTI_TRS_MULTI_SLOT_MEAS_SW option of the NRDUCellHighSpeedMob.<i>HighSpeedAlgoOptSw</i> parameter can be selected only when the NRDUCellMultiTrp.<i>DataTransMode</i> parameter is set to BASIC_MODE or DPS_MODE .
Low-frequency TDD	Multi-TRS measurement	NRDUCellAlgoSwitch.<i>MultiTrsMeasSwitch</i> set to ON	<i>High Speed Mobility</i>	The MULTI_TRS_MULTI_SLOT_MEAS_SW option of the NRDUCellHighSpeedMob.<i>HighSpeedAlgoOptSw</i> parameter can be selected only when the NRDUCellAlgoSwitch.<i>MultiTrsMeasSwitch</i> parameter is set to ON .

8.3.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS rate matching optimization	TRS_RATEMATCH_OPT_SW option of the NRDUCellPdsch.<i>RateMatchSwitch</i> parameter	<i>Downlink Scheduling</i>	The TRS_RATEMATCH_OPT_SW option of the NRDUCellPdsch.<i>RateMatchSwitch</i> parameter must be deselected when the MULTI_TRS_MULTI_SLOT_MEAS_SW option of the NRDUCellHighSpeedMob.<i>HighSpeedAlgoOptSw</i> parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS time-frequency alignment	TRS_ALIGNMENT_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	The TRS_ALIGNMENT_SW option of the NRDUCellCsirs.CsiSwitch parameter must be deselected when the MULTI_TRS_MULTI_SLOT_MEAS_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter is selected.
Low-frequency TDD	Periodic TRS transmission	NRDUCellCsirs.TrsPeriod	<i>Channel Management</i>	The MULTI_TRS_MULTI_SLOT_MEAS_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter cannot be selected when the NRDUCellCsirs.TrsPeriod parameter is set to MS10 .

8.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	If intra-base-station DL CoMP (controlled by the INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter) and multi-TRS measurement are enabled in the PCell of a UE and the UE supports multi-TRS measurement, single DCI does not take effect on the UE.

8.3.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

8.3.3.4 Others

UEs must support multi-TRS measurement. Check the **maxNumberActiveTCI-PerBWP** field of the corresponding band in the UE capability information. If the value of this field is greater than or equal to n2, the UE is capable of measuring multiple sets of TRSs. For details, see [8.3.4.2 Activation Verification](#).

8.3.4 Operation and Maintenance

8.3.4.1 Data Configuration

8.3.4.1.1 Data Preparation

[Table 8-3](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Multi-TRS Measurement Switch	NRDUCellAlgoSwitch.MultiTrsMeasSwitch	Set this parameter to ON to enable the multi-TRS measurement function in high-speed cells.
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	Select the MULTI_TRS_MULTI_SLOT_MEAS_SW option to enable the multi-TRS measurement function in high-speed cells.

8.3.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling multi-TRS measurement  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MultiTrsMeasSwitch=ON;  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,  
HighSpeedAlgoOptSw=MULTI_TRS_MULTI_SLOT_MEAS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling multi-TRS measurement  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,  
HighSpeedAlgoOptSw=MULTI_TRS_MULTI_SLOT_MEAS_SW-0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MultiTrsMeasSwitch=OFF;
```

8.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.3.4.2 Activation Verification

1. Run the **LST NRDUCELLALGOSWITCH** command and check the value of **Multi-TRS Measurement Switch** in the command output. If the value is **On**, the multi-TRS measurement function has been enabled.
2. Check whether the value of the **maxNumberActiveTCI-PerBWP** field of the corresponding band in the UE capability information is greater than or equal to n2. If this is true, the UE is capable of measuring multiple sets of TRSs.
3. For a UE capable of multi-TRS measurement, if the **tci-PresentInDCI** field of *PDCCH-config* in the RRC reconfiguration message is **enabled** when the UE accesses the network, multi-TRS measurement has taken effect.

8.3.4.3 Network Monitoring

The indicators listed in [8.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.4 Uplink SINR Adjustment

8.4.1 Principles

In high-speed line-of-sight propagation (LOS) scenarios, signals can be transmitted in a straight line between a base station and a UE without obstacles. In high-speed non-line-of-sight (NLOS) scenarios, such as urban or mountainous areas, channel conditions change rapidly. As a result, the MCS index selected for the uplink data channel based on the SINR may be overly high, increasing the BLER. To address this issue, uplink SINR adjustment is introduced.

This function adjusts the SINR used for MCS index selection based on the channel characteristics to make the MCS index more suitable for the actual channel quality, thereby reducing the BLER. In this way, the average uplink throughput of high-speed UEs can be increased in scenarios where channel conditions change greatly with time. This function is controlled by the **UL_SINR_ADJUST_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

8.4.2 Network Analysis

8.4.2.1 Benefits

After this function is enabled, the average uplink throughput of high-speed UEs increases when channel conditions change greatly with time.

NOTE

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

8.4.2.2 Impacts

The impacts between this function and other functions are described in [8.4.3.2.3 Function Impacts](#).

This function has no impact.

8.4.3 Requirements

8.4.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeed Flag	<i>High Speed Mobility</i>	Uplink SINR adjustment requires the NRDUCell.HighSpeed Flag parameter to be set to HIGH_SPEED .

8.4.3.2.2 Mutually Exclusive Functions

None

8.4.3.2.3 Function Impacts

None

8.4.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

8.4.3.4 Others

None

8.4.4 Operation and Maintenance

8.4.4.1 Data Configuration

8.4.4.1.1 Data Preparation

[Table 8-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	Enable the uplink SINR adjustment function in high-speed cells when the channel condition changes greatly with time. Select the UL_SINR_ADJUST_SW option to enable uplink SINR adjustment.

8.4.4.1.2 Using MML Commands

Before using MML commands, refer to [8.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink SINR adjustment  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=UL_SINR_ADJUST_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink SINR adjustment  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=UL_SINR_ADJUST_SW-0;
```

8.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.4.2 Activation Verification

Run the **LST NRDUCELLHIGHSPEEDMOB** command and check the value of the high speed optimized switch in the command output. If the value of **Uplink SINR Adjustment Sw** is **On** in the command output, uplink SINR adjustment has been enabled.

8.4.4.3 Network Monitoring

The indicators listed in [8.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.5 Uplink MCS Index Selection Optimization based on Narrowband SRS

8.5.1 Principles

Due to the SRS frequency hopping mechanism, only the RB-level SRS SINR information of the current subband is the latest. The timeliness of other subbands' RB-level SRS SINR information is poor. If the RB-level SINRs with poor timeliness are used for average subband value calculation, the calculated SRS SINR for scheduling is inaccurate. As a result, the selected uplink MCS index is inaccurate, decreasing the average uplink UE throughput in high-speed cells.

In high-speed scenarios where the SRS period is prolonged and the frequency hopping multiple is large, poor timeliness of RB-level SRS SINRs will become more significant. Therefore, uplink MCS index selection optimization based on narrowband SRS is introduced. After this function is enabled, filtered SRS SINRs of the frequency hopping subbands are used in MCS index selection for high-speed UEs involved in SRS frequency hopping. This reduces the loss caused by SRS frequency hopping.

This function is controlled by the **UL_SRS_FREQ_HOPPING_MCS_OPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

8.5.2 Network Analysis

8.5.2.1 Benefits

In high-speed cells, this function improves the average uplink throughput of high-speed UEs involved in SRS frequency hopping.

 NOTE

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

8.5.2.2 Impacts

The impacts between this function and other functions are described in [8.5.3.2.3 Function Impacts](#).

The average uplink MCS index may slightly decrease after this function is enabled.

 NOTE

Average uplink MCS index = $\sum(n \times N.ChMeas.PUSCH.MCS.n) / \sum(N.ChMeas.PUSCH.MCS.n)$

8.5.3 Requirements

8.5.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink MCS index selection optimization based on narrowband SRS requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

8.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink dual filtering	UL_SRS_DMRS_DUAL_FILTER_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	Uplink MCS index selection optimization based on narrowband SRS and uplink dual filtering are mutually exclusive.

8.5.3.2.3 Function Impacts

None

8.5.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

8.5.3.4 Others

None

8.5.4 Operation and Maintenance

8.5.4.1 Data Configuration

8.5.4.1.1 Data Preparation

Table 8-5 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	Select the UL_SRS_FREQ_HOPPING_MCS_OPT_SW option to enable uplink MCS index selection optimization based on narrowband SRS in high-speed cells.

8.5.4.1.2 Using MML Commands

Before using MML commands, refer to **8.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink MCS index selection optimization based on narrowband SRS
MOD NRDUCELHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_SRS_FREQ_HOPPING_MCS_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink MCS index selection optimization based on narrowband SRS
MOD NRDUCELHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_SRS_FREQ_HOPPING_MCS_OPT_SW-0;
```

8.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.5.4.2 Activation Verification

Compare the user-perceived rates of high-speed UEs before and after the "uplink MCS index selection optimization based on narrowband SRS" function is enabled. If their user-perceived rate increases after this function is enabled, it indicates that this function has taken effect.

8.5.4.3 Network Monitoring

The indicators listed in [8.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.6 Uplink Frequency Offset Filtering Adaptation

8.6.1 Principles

In high-speed railway scenarios, UEs move fast, channel conditions change rapidly, and the impact of Doppler shift is great. To better meet the performance requirements in high speed mobility scenarios, the uplink frequency offset filtering adaptation function is introduced.

This function is controlled by the **UL_FREQ_OFS_FILTER_ADAPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. When this option is selected, the base station adjusts the uplink frequency offset filtering coefficient based on the frequency offset fluctuation, improving the accuracy of uplink frequency offset measurement results.

8.6.2 Network Analysis

8.6.2.1 Benefits

This function increases the average uplink UE throughput in high-speed cells.

NOTE

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

8.6.2.2 Impacts

The impacts between this function and other functions are described in [8.6.3.2.3 Function Impacts](#).

This function has no impact.

8.6.3 Requirements

8.6.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink frequency offset filtering adaptation requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

8.6.3.2.2 Mutually Exclusive Functions

None

8.6.3.2.3 Function Impacts

None

8.6.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

8.6.3.4 Others

None

8.6.4 Operation and Maintenance

8.6.4.1 Data Configuration

8.6.4.1.1 Data Preparation

Table 8-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	Select the UL_FREQ_OFS_FILTER_ADAPT_SW option to enable uplink frequency offset filtering adaptation in high-speed cells.

8.6.4.1.2 Using MML Commands

Before using MML commands, refer to **8.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink frequency offset filtering adaptation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_FREQ_OFS_FILTER_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink frequency offset filtering adaptation
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=UL_FREQ_OFS_FILTER_ADAPT_SW-0;
```

8.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.6.4.2 Activation Verification

Compare the average uplink throughputs of high-speed UEs before and after the uplink frequency offset filtering adaptation function is enabled. If the throughput increases after this function is enabled, it indicates that this function has taken effect.

8.6.4.3 Network Monitoring

The indicators listed in [8.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.7 Uplink Channel Densified Equalization for High-Speed UEs

8.7.1 Principles

In high-speed scenarios, time-dependent channel variations are significant, which may cause the equalization performance to deteriorate. To address this issue, uplink channel densified equalization for high-speed UEs is introduced. With this function enabled, the number of symbols used for uplink equalization calculation in the time domain is increased to effectively resist time-dependent channel variations and improve uplink performance.

This function is controlled by the **HS_UL_LOW_COST_DENSE_EQUAL_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. The **NRDUCellHighSpeedMob.UIRank1DenseEqualMcsOffset** and **NRDUCellHighSpeedMob.UIRank2DenseEqualMcsOffset** parameters are used to further control the proportion of high-speed UEs for which uplink channel densified equalization takes effect.

8.7.2 Network Analysis

8.7.2.1 Benefits

After "uplink channel densified equalization for high-speed UEs" is enabled, the bit error rate may decrease, the uplink MCS index may increase, and higher-order modulation schemes are used, increasing the average uplink throughput of high-speed UEs.

NOTE

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

8.7.2.2 Impacts

The impacts between this function and other functions are described in [8.7.3.2.3 Function Impacts](#).

- After "uplink channel densified equalization for high-speed UEs" is enabled, the proportion of uplink 256QAM may increase, the proportion of uplink

QPSK may decrease, and the proportions of other modulation schemes may fluctuate.

- If the offset to the MCS index threshold for uplink densified equalization to take effect for high-speed UEs with rank 1 or 2 is set to an excessively large value, the proportion of high-speed UEs with uplink channel densified equalization in effect is small, and the average uplink throughput of high-speed UEs does not increase significantly. If the offset is set to an excessively small value, the average uplink throughput of high-speed UEs may decrease.
- The increase in uplink throughput may increase the downlink traffic volume of TCP services. As a result, the downlink PDCP-layer traffic volume of the cell (indicated by the **Downlink Traffic Volume (DU)**) and the downlink RLC-layer traffic volume of the cell increase, the downlink transmission duration may increase, the downlink PDSCH RB usage increases, and the average downlink user-perceived rate, average downlink cell throughput (indicated by the **Cell Downlink Average Throughput (DU)**), and average MCS index for downlink scheduling may fluctuate.

8.7.3 Requirements

8.7.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

8.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeed Flag	<i>High Speed Mobility</i>	Uplink channel densified equalization for high-speed UEs requires the NRDUCell.HighSpeed Flag parameter to be set to HIGH_SPEED .

8.7.3.2.2 Mutually Exclusive Functions

None

8.7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	"Uplink channel densified equalization for high-speed UEs" does not take effect on UEs involved in PUSCH MU-MIMO.
Low-frequency TDD	Uplink Additional DMRS Position	NRDUCellPusch.UlAdditionalDmrsPos	None	"Uplink channel densified equalization for high-speed UEs" can take effect only on UEs in a cell with the NRDUCellPusch.UlAdditionalDmrsPos parameter set to POS1 .
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch . CompSwitch parameter	<i>CoMP</i>	In high-speed scenarios, "uplink channel densified equalization for high-speed UEs" does not take effect for UL CoMP UEs in cooperating TRPs.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch . CompSwitch parameter	<i>CoMP</i>	In high-speed scenarios, "uplink channel densified equalization for high-speed UEs" does not take effect for DL CoMP UEs in cooperating TRPs.

8.7.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

Uplink channel densified equalization for high-speed UEs is supported by only the following types of RF modules: 1T1R, 2T2R, 4T4R, and 8T8R RF modules.

8.7.3.4 Cells

The **NRDUCellTrp.TxRxMode** parameter must be set to **1T1R**, **2T2R**, **4T4R**, or **8T8R** for the cell.

8.7.3.5 Others

None

8.7.4 Operation and Maintenance

8.7.4.1 Data Configuration

8.7.4.1.1 Data Preparation

[Table 8-7](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-7 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	To enable "uplink channel densified equalization for high-speed UEs" in a high-speed cell, select the HS_UL_LOW_COST_DENSE_EQUAL_SW option.
UL Rank1 Dense Equal MCS Offset	NRDUCellHighSpeedMob.UlRank1DenseEqualMcsOffset	Use the default value.
UL Rank2 Dense Equal MCS Offset	NRDUCellHighSpeedMob.UlRank2DenseEqualMcsOffset	Use the default value.

8.7.4.1.2 Using MML Commands

Before using MML commands, refer to [8.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink channel densified equalization for high-speed UEs
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,
HighSpeedAlgoOptSw=HS_UL_LOW_COST_DENSE_EQUAL_SW-1, UlRank1DenseEqualMcsOffset=0,
UlRank2DenseEqualMcsOffset=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink channel densified equalization for high-speed UEs  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0,  
HighSpeedAlgoOptSw=HS_UL_LOW_COST_DENSE_EQUAL_SW-0;
```

8.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.7.4.2 Activation Verification

Observe the average uplink UE throughput before and after this function is enabled. If the average uplink UE throughput changes, this function has taken effect.

NOTE

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

8.7.4.3 Network Monitoring

The indicators listed in [8.7.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

9 Interference Resistance Optimization for High-Speed Cells

9.1 Inter-TRP Interference Coordinated Scheduling for High-Speed Railways

9.1.1 Principles

To reduce handovers between TRPs and increase cell capacity, high-speed cells are usually combined and the Hyper Cell function is enabled. Independent scheduling in each TRP brings spatial multiplexing gains but also brings interference to uplink and downlink traffic channels. As a result, the bit error rate (BER) increases when UEs pass through the TRP overlapping area, affecting the performance of high-speed UEs. Therefore, inter-TRP interference coordinated scheduling is introduced to reduce interference on traffic channels between TRPs.

This function is controlled by **High Speed Intrf Random Schedule Sw^a** and can be enabled only for the Hyper Cell of combined cells. When **High Speed Intrf Random Schedule Sw^a** is set to **ON**,

- If **Uplink Interference Random Schedule Switch^b** is turned on, uplink inter-TRP interference coordinated scheduling takes effect.
- If **Downlink Interference Random Switch^c** is turned on, downlink inter-TRP interference coordinated scheduling takes effect.

It is recommended that uplink and downlink inter-TRP interference coordinated scheduling both be enabled.

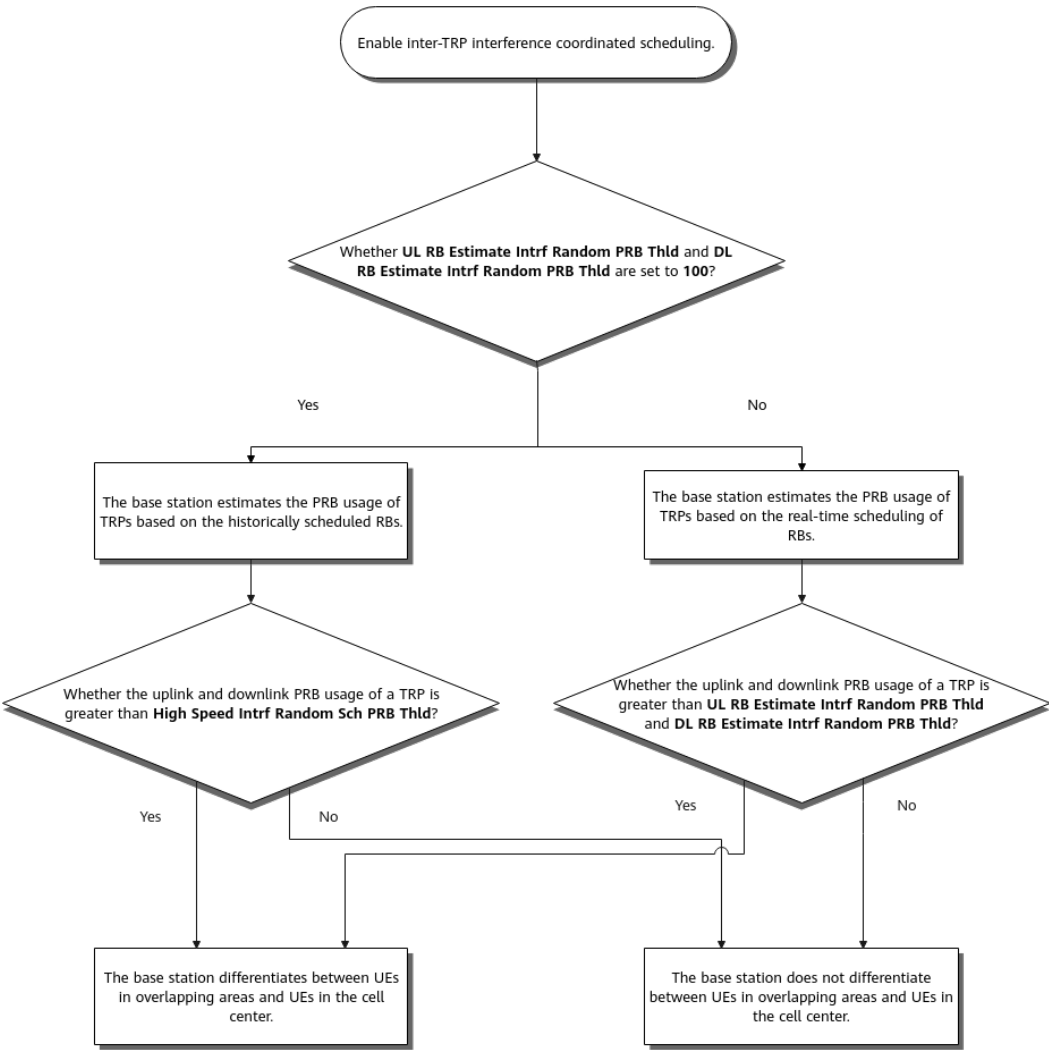
After this function is enabled, the base station staggers resource scheduling based on the TRP ID mod 2 result in Hyper Cell networking. Specifically, the base station allocates RB resources of the low frequency band to TRPs whose TRP ID mod 2 result is 0 and allocates RB resources of the high frequency band to TRPs whose TRP ID mod 2 result is 1. The principles for uplink and downlink inter-TRP interference coordinated scheduling are the same.

- When the threshold expressed as the PRB usage for interference-randomization-based scheduling based on uplink or downlink RB estimation is

set to **100**, the base station estimates the PRB usage of TRPs based on the historically scheduled RBs.

- If the uplink and downlink PRB usages of a TRP are less than or equal to **High Speed Intrf Random Sch PRB Thld^f**, the base station does not differentiate between UEs in overlapping areas and UEs in the cell center. Instead, the base station allocates uplink and downlink scheduling resources to the TRP in a staggered manner based on the TRP ID mod 2 result.
- If the uplink and downlink PRB usages of a TRP are greater than **High Speed Intrf Random Sch PRB Thld^f**, the base station differentiates between UEs in overlapping areas and UEs in the cell center. Specifically, UEs in the overlapping area are scheduled in a staggered manner based on the TRP ID mod 2 result, while those in the center area are not scheduled in a staggered manner.
- When the threshold expressed as the PRB usage for interference-randomization-based scheduling based on uplink or downlink RB estimation is not set to **100**, the base station estimates the PRB usage of TRPs based on the real-time scheduling of RBs.
 - If the uplink and downlink PRB usages of a TRP are less than or equal to **UL RB Estimate Intrf Random PRB Thld^d** and **DL RB Estimate Intrf Random PRB Thld^e**, respectively, the base station does not differentiate between UEs in overlapping areas and UEs in the cell center. Instead, the base station allocates uplink and downlink scheduling resources to the TRP in a staggered manner based on the TRP ID mod 2 result.
 - If the uplink and downlink PRB usages of a TRP are greater than **UL RB Estimate Intrf Random PRB Thld^d** and **DL RB Estimate Intrf Random PRB Thld^e**, respectively, the base station differentiates between UEs in overlapping areas and UEs in the cell center. Specifically, UEs in the overlapping area are scheduled in a staggered manner based on the TRP ID mod 2 result, while those in the center area are not scheduled in a staggered manner.

Figure 9-1 Inter-TRP interference coordinated scheduling for high-speed railways



UEs in the overlapping area are identified based on the absolute SRS RSRP difference between the serving and neighboring TRPs. If this difference, as measured by a UE, is less than or equal to the value of **NRCellComp.OverlappingRsrpOffset**, the UE is identified as being in the overlapping area; otherwise, the UE is identified as being in the center area.

Table 9-1 Parameters related to inter-TRP interference coordinated scheduling for high-speed railways

No.	Parameter Name	Parameter Switch
a	High Speed Intra Random Schedule Sw	NRDUCellAlgoSwitch.HighSpeedIntraRandomSchSw
b	Uplink Interference Random Schedule Switch	NRDUCellAlgoSwitch.UlIntraRandomSchSwitch

No.	Parameter Name	Parameter Switch
c	Downlink Interference Random Switch	DL_INTRF_RANDOM_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter
d	UL RB Estimate Intra Random PRB Thld	NRDUCellHighSpeed-Mob.UlRbEstIntraRandomPrbThld
e	DL RB Estimate Intra Random PRB Thld	NRDUCellHighSpeed-Mob.DlRbEstIntraRandomPrbThld
f	High Speed Intra Random Sch PRB Thld	NRDUCellHighSpeed-Mob.HSIntraRandomSchPrbThld

9.1.2 Network Analysis

9.1.2.1 Benefits

This function is suitable for scenarios where the PRB usage is greater than or equal to 60%, and can decrease inter-TRP interference for high-speed cells and increase the uplink and downlink user-perceived rates of high-speed UEs. If **NRDUCellHighSpeedMob.UlRbEstIntraRandomPrbThld** and **NRDUCellHighSpeedMob.DlRbEstIntraRandomPrbThld** are not set to 100, the user-perceived rates are expected to further increase.

9.1.2.2 Impacts

The impacts between this function and other functions are described in [9.1.3.2.3 Function Impacts](#).

If the inter-TRP interference within a cell is small, enabling this function will decrease user-perceived rates due to increased fragmentation of RB resources. The interference severity can be observed based on the average interference and noise received on each uplink PRB (**N.UL.NI.Avg**) when a train passes by.

9.1.3 Requirements

9.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

9.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	The NRDUCell.HighSpeedFlag parameter must be set to HIGH_SPEED to enable inter-TRP interference coordinated scheduling in high-speed railway scenarios.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell</i>	The NRDUCell.NrDuCellNetworkingMode parameter must be set to HYPER_CELL to enable inter-TRP interference coordinated scheduling in high-speed railway scenarios.
Low-frequency TDD	Interference Randomization-based Scheduling	NRDUCellAlgoSwitch.UlIntrfRandomSwitch NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	<i>Uplink Scheduling</i> <i>Downlink Scheduling</i>	Inter-TRP interference coordinated scheduling for high-speed railway scenarios can be enabled only if the NRDUCellAlgoSwitch.UlIntrfRandomSwitch parameter is set to ON or the DL_INTRF_RANDOM_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter is selected.

9.1.3.2.2 Mutually Exclusive Functions

None

9.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink DCI extension-based scheduling in high-speed cells	DCI_EXTEND_UL_CO NGEST_SW option of the NRDUCellPusch.UIPuschAlgoSwitch parameter	<i>High Speed Mobility</i>	When the uplink DCI extension-based scheduling takes effect in high-speed cells, the estimated number of RBs in some slots may be inaccurate because the scheduling time is earlier than expected. This affects the proportion of UEs in the overlapping area of the center area for which staggered scheduling based on mod 2 result takes effect.
Low-frequency TDD	Avoidance of RedCap downlink scheduling	AVOID_REDCAP_DL_SCH_SW option of the NRDUCellRedCap.RedCapSchAlgoSwitch parameter	<i>RedCap</i>	When inter-TRP interference coordinated scheduling for high-speed railways is in effect, avoidance of RedCap downlink scheduling does not take effect.

9.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

9.1.3.4 Others

None

9.1.4 Operation and Maintenance

9.1.4.1 Data Configuration

9.1.4.1.1 Data Preparation

[Table 9-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 9-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Intrf Random Schedule Sw	NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw	It is recommended that this parameter be set to ON in Hyper Cell networking to control high-speed-specific interference randomization-based scheduling.
High Speed Intrf Random Sch PRB Thld	NRDUCellHighSpeedMob.HSIntrfRandomSchPrbThld	Use the default value.
UL RB Estimate Intrf Random PRB Thld	NRDUCellHighSpeedMob.UIRbEstIntrfRandomPrbThld	The value 60 is recommended.
DL RB Estimate Intrf Random PRB Thld	NRDUCellHighSpeedMob.DIRbEstIntrfRandomPrbThld	The value 60 is recommended.

9.1.4.1.2 Using MML Commands

Before using MML commands, refer to [9.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling Uplink Interference Random Schedule Switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlintrfRandomSchSwitch=ON;
//Enabling Downlink Interference Random Switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_INTRF_RANDOM_SW-1;
//Enabling inter-TRP interference coordinated scheduling for high-speed railways
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighSpeedIntrfRandomSchSw=ON;
//Configuring UL RB Estimate Intrf Random PRB Thld and DL RB Estimate Intrf Random PRB Thld
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, UIRbEstIntrfRandomPrbThld=60,
DIRbEstIntrfRandomPrbThld=60;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling inter-TRP interference coordinated scheduling for high-speed railways
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighSpeedIntrfRandomSchSw=OFF;
```

9.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.1.4.2 Activation Verification

This function has taken effect if the average uplink and downlink UE throughputs change after this function is enabled.

9.1.4.3 Network Monitoring

The indicators listed in [9.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

9.2 Inter-TRP PDCCH Interference Avoidance for VoNR

9.2.1 Principles

In high-speed scenarios, after cell combination networking is used or the Hyper Cell function is enabled, each TRP performs independent scheduling, yielding spatial multiplexing gains. However, this also introduces PDCCH interference and can degrade the MOS of VoNR UEs. To address this issue, the inter-TRP PDCCH interference avoidance for VoNR function is introduced. After this function is enabled, interference between the PDCCHs of VoNR UEs served by different TRPs is reduced by staggering time-domain and frequency-domain resources, thereby reducing the packet loss rate and ensuring VoNR voice quality.

This function is controlled by the **PDCCH_VONR_INTF_AVOID_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter. It is also recommended that the **TOP_VONR_UE_CCE_ALLOC_OPT_SW** option of the same parameter be selected so that CCE allocation for the top-priority VoNR UEs is not constrained by the CCE allocation proportion, further reducing interference between the PDCCHs of VoNR UEs.

9.2.2 Network Analysis

9.2.2.1 Benefits

In high-speed scenarios, after cell combination network is used or the Hyper Cell function is enabled, inter-TRP PDCCH interference avoidance for VoNR can be enabled to reduce interference between the PDCCHs of VoNR UEs, thereby reducing the VoNR packet loss rate and improving VoNR user experience.

9.2.2.2 Impacts

The impacts between this function and other functions are described in [9.2.3.2.3 Function Impacts](#).

After inter-TRP PDCCH interference avoidance for VoNR is enabled, the control-channel resources available for data services decrease, the CCE allocation failure rate increases, and the throughput of data services may decrease.

9.2.3 Requirements

9.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

9.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.2.3.2.1 Prerequisite Functions

None

9.2.3.2.2 Mutually Exclusive Functions

None

9.2.3.2.3 Function Impacts

None

9.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

9.2.3.4 Others

None

9.2.4 Operation and Maintenance

9.2.4.1 Data Configuration

9.2.4.1.1 Data Preparation

[Table 9-3](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 9-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
PDCCH Algorithm Extension Switch	NRDUCellPdcch.PdcchAlgoExtSwitch	It is recommended that the PDCCH_VONR_INTF_AVOID_SW option be selected to enable inter-TRP PDCCH interference avoidance for VoNR.
PDCCH Algorithm Extension Switch	NRDUCellPdcch.PdcchAlgoExtSwitch	It is recommended that the TOP_VONR_UE_CCE_ALLOC_OPT_SW option be selected.

9.2.4.1.2 Using MML Commands

Before using MML commands, refer to [9.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Selecting the PDCCH_VONR_INTF_AVOID_SW and TOP_VONR_UE_CCE_ALLOC_OPT_SW options  
MOD NRDUCELLPDCCH: NrDuCellId=0,  
PdcchAlgoExtSwitch=PDCCH_VONR_INTF_AVOID_SW-1&TOP_VONR_UE_CCE_ALLOC_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deselecting the PDCCH_VONR_INTF_AVOID_SW and TOP_VONR_UE_CCE_ALLOC_OPT_SW options  
MOD NRDUCELLPDCCH: NrDuCellId=0,  
PdcchAlgoExtSwitch=PDCCH_VONR_INTF_AVOID_SW-0&TOP_VONR_UE_CCE_ALLOC_OPT_SW-0;
```

9.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.2.4.2 Activation Verification

This function has taken effect if the downlink BLER of VoNR UEs decreases after this function is enabled. To observe the BLER, perform the following operations:

1. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
2. In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring > BLER Monitoring** to observe the BLER.

9.2.4.3 Network Monitoring

The indicators listed in [9.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

10 High-Speed Cell Scheduling Performance Optimization

10.1 Downlink Robust Weights in High-Speed Cells

10.1.1 Principles

UEs move fast in high-speed railway scenarios, where SRS-based weights in downlink are prone to be aged. As such, PMI-based weights are used by default in high-speed railway scenarios. To avoid reconfigurations of high-speed UEs during inter-TRP handovers, PMI-based weights are configured with PCI scrambling codes and are transmitted through full-width beams. This mechanism results in low beamforming gains and is prone to causing substantial inter-TRP interference.

To further improve downlink performance, the "downlink robust weights in high-speed cells" function is introduced. After this function is enabled, the high-speed cell adopts the beam tracking algorithm and enables downlink robust weights. This function enhances the energy in the main path direction of a beam to improve the matching with TRP-level channels and guarantee the beamforming gains of massive MIMO.

The "downlink robust weights in high-speed cells" function is controlled by the **DL_HS_ROBUST_WEIGHT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter. Further, this function uses the **NRDUCellHighSpeedMob.RobustWeightKFactorThld** and **NRDUCellHighSpeedMob.RobustWeightFilterCoeff** parameters to control how easy robust weights take effect and how fast robust weights are updated for high-speed UEs, respectively.

This function requires that the **NRDUCellPdsch.RobustWtPhaseCalcMethod** parameter be set to **SRS_H_BASED_CALCULATION**.

This function can be enabled only for massive MIMO cells in high-speed railway scenarios. This function takes effect only when the SRS PreSINR of a UE is greater than or equal to the **NRDUCellPdschPrecoding.RobustWtSrsPresinrThld** parameter value.

10.1.2 Network Analysis

10.1.2.1 Benefits

After this function is enabled for massive MIMO cells in high-speed railway scenarios, the bit error rate may decrease, the downlink MCS index may increase, and higher-order modulation schemes are used, increasing the downlink user-perceived rates of high-speed UEs. With an appropriate setting of the **NRDUCellHighSpeedMob.RobustWeightKFactorThld** parameter, this function can provide higher gains if the channel environment is closer to a line-of-sight (LOS) path.

10.1.2.2 Impacts

The impacts between this function and other functions are described in [10.1.3.2.3 Function Impacts](#).

Assume that this function is enabled. If the **NRDUCellHighSpeedMob.RobustWeightKFactorThld** parameter is set to an excessively large or small value, there is a higher probability that the channel type determined by the base station does not match the actual channel type of UEs. As a result, it is more likely that the downlink user-perceived rates of high-speed UEs decrease. In addition, the downlink performance improves, which may result in more uplink TCP response packets. As such, the uplink throughput may decrease.

10.1.3 Requirements

10.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

10.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	The "downlink robust weights in high-speed cells" function requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .
Low-frequency TDD	SRS period and CDM adaptation	SRS_PERIOD_AND_CDM_ADAPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	In high-speed cells, the "downlink robust weights in high-speed cells" function can be enabled only after "SRS period and CDM adaptation" is enabled.
Low-frequency TDD	High Speed Cell RI Restriction	NRDUCellPdsch.HighSpeedCellRiRestr set to 1 or 3	None	The "downlink robust weights in high-speed cells" function can be enabled only when the RI restriction for high-speed cells is limited to rank 1 or rank 2.

10.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink robust weight	DL_ROBUST_WEIGHT_SW option of the NRDUCellPdschPre-code.PrecodingAlgoSwitch parameter	None	The option controlling "downlink robust weights in high-speed cells" and the DL_ROBUST_WEIGHT_SW option of the NRDUCellPdschPre-code.PrecodingAlgoSwitch parameter cannot be both selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference-based robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter	None	Downlink robust weights in high-speed cells and interference-based robust weights cannot be both enabled.
Low-frequency TDD	User CSI-RS Beam Type	NRDUCellCsirs.UserCsirsBeamType set to TYPE8 , TYPE9 , or TYPE10 NRDUCellComp.UserCsirsBeamType set to TYPE8 , TYPE9 , or TYPE10	None	When the "downlink robust weights in high-speed cells" function is enabled, the UE-specific CSI-RS beam type cannot be set to beam type 8, 9, or 10.

10.1.3.2.3 Function Impacts

None

10.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

Only 32T32R and 64T64R RF modules support downlink robust weights in high-speed cells.

10.1.3.4 Cells

The cell must be a high-speed cell with massive MIMO enabled (that is, the **NRDUCellTrp.TxRxMode** parameter must be set to **32T32R** or **64T64R**).

10.1.3.5 Others

None

10.1.4 Operation and Maintenance

10.1.4.1 Data Configuration

10.1.4.1.1 Data Preparation

[Table 10-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 10-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
DL High-Speed Robust Weight Sw	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	It is recommended that the DL_HS_ROBUST_WEIGHT_SW option be selected for massive MIMO cells in high-speed railway scenarios.
Robust Weight K Factor Thld	NRDUCellHighSpeedMob.RobustWeightKFactorThld	It is recommended that this parameter be set to 0 when the "downlink robust weights in high-speed cells" function is enabled.
Robust Weight Filter Coefficient	NRDUCellHighSpeedMob.RobustWeightFilterCoeff	Use the default value.
Robust Weight Phase Calculation Method	NRDUCellPdsch.RobustWtPhaseCalcMethod	To enable the "downlink robust weights in high-speed cells" function, set this parameter to SRS_H_BASED_CALCULATION , meaning that phase calculation is based on the channel matrix H indicated in SRS measurement results.
Robust Weight SRS PreSINR Thld	NRDUCellPdschPrecode.RobustWtSrsPresinrThld	Use the recommended value.

10.1.4.1.2 Using MML Commands

Before using MML commands, refer to [10.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for downlink robust weights in high-speed cells
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=DL_HS_ROBUST_WEIGHT_SW-1;
//Setting the calculation method for the phases corresponding to robust weights
MOD NRDUCELLPDSCH: NrDuCellId=0, RobustWtPhaseCalcMethod=SRS_H_BASED_CALCULATION;
//Configuring the threshold related to robust weights
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, RobustWeightKFactorThld=0, RobustWeightFilterCoeff=10;
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, RobustWtSrsPresinrThld=-8;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for downlink robust weights in high-speed cells  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=DL_HS_ROBUST_WEIGHT_SW-0;
```

10.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.1.4.2 Activation Verification

Observe the average downlink UE throughput before and after this function is enabled. If the average downlink UE throughput changes, this function has taken effect.

10.1.4.3 Network Monitoring

The indicators listed in [10.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

10.2 Downlink Priority Enhancement for Voice Service UEs in a High-Speed Cell

10.2.1 Principles

In high-speed railway scenarios, a large number of UEs are handed over to and from cells at a time, triggering massive signaling configurations. In the traditional scheduling mechanism, the priority of signaling is higher than that of voice packets. As a result, voice packets may not be scheduled in a timely manner. To reduce the scheduling delay of VoNR services, downlink priority enhancement for voice service UEs in a high-speed cell is introduced. After this function is enabled, the downlink scheduling priority of VoNR UEs is increased to prevent voice packets from failing to be scheduled due to signaling impact.

This function is controlled by the **VONR_DL_PRI_SCH_ENH_SW** option of the **NRDUCellDISch.VoiceDISchSwitch** parameter. Before enabling this function, "downlink scheduling with the same priority as signaling scheduling" needs to be enabled (by the selecting the **VONR_DL_PRI_SCH_WITH_SIG_SW** option of the **NRDUCellDISch.VoiceDISchSwitch** parameter). For details about "downlink scheduling with the same priority as signaling scheduling", see "Scheduling Guarantee" in *VoNR*.

10.2.2 Network Analysis

10.2.2.1 Benefits

Downlink priority enhancement for voice service UEs in a high-speed cell is recommended when the maximum number of UEs in a high-speed cell is greater than 100. After this function is enabled, the downlink packet loss rate of voice services and the downlink BLER of voice services decrease in the cell.

Downlink voice packet loss rate = $(\text{N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI} / \text{N.RLC.DL.TrfSDU.RxPackets.DuCell5QI}) \times 100\%$

10.2.2.2 Impacts

The impacts between this function and other functions are described in [10.2.3.2.3 Function Impacts](#).

After this function is enabled, the priority of voice services is higher when there are a large number of VoNR UEs in a cell. As a result, signaling scheduling is not performed in a timely manner, and the service drop rate may increase.

10.2.3 Requirements

10.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

10.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.2.3.2.1 Prerequisite Functions

None

10.2.3.2.2 Mutually Exclusive Functions

None

10.2.3.2.3 Function Impacts

None

10.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

10.2.3.4 Others

None

10.2.4 Operation and Maintenance

10.2.4.1 Data Configuration

10.2.4.1.1 Data Preparation

Table 10-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 10-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Voice DL Schedule Switch	NRDUCellIDISch.VoiceDISchSwitch	It is recommended that the VONR_DL_PRI_SCH_ENH_SW option be selected if the maximum number of UEs in a high speed cell is greater than 100. This option takes effect only when the VONR_DL_PRI_SCH_WITH_SIG_SW option of the NRDUCellIDISch.VoiceDISchSwitch is selected at the same time.

10.2.4.1.2 Using MML Commands

Before using MML commands, refer to **10.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling "downlink priority enhancement for voice service UEs in a high-speed cell"  
MOD NRDUCELLDLSCH: NrDuCellId=0,  
VoiceDISchSwitch=VONR_DL_PRI_SCH_ENH_SW-1&VONR_DL_PRI_SCH_WITH_SIG_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling "downlink priority enhancement for voice service UEs in a high-speed cell"  
MOD NRDUCELLDLSCH: NrDuCellId=0,  
VoiceDISchSwitch=VONR_DL_PRI_SCH_ENH_SW-0&VONR_DL_PRI_SCH_WITH_SIG_SW-0;
```

10.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.2.4.2 Activation Verification

If the maximum number of UEs in a high-speed cell is greater than 100 and this function is enabled, this function has taken effect when the downlink voice packet loss rate changes.

Downlink voice packet loss rate = $(\text{N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI} / \text{N.RLC.DL.TrfSDU.RxPackets.DuCell5QI}) \times 100\%$

10.2.4.3 Network Monitoring

The indicators listed in [10.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

11 Differentiated Scheduling Policy Management for High- and Low-Speed UEs

11.1 Differentiated Scheduling

11.1.1 Principles

Differentiated scheduling increases the scheduling priority of high-speed UEs in high-speed cells to ensure that these UEs are preferentially scheduled, thereby improving their user-perceived rate. This function is controlled by the **HIGH_SPEED_DIFFERENTIAL_SCH_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter.

11.1.2 Network Analysis

11.1.2.1 Benefits

Differentiated scheduling increases the average uplink and downlink throughputs of high-speed UEs in high-speed cells.

11.1.2.2 Impacts

The impacts between this function and other functions are described in [11.1.3.2.3 Function Impacts](#).

After this function is enabled, the uplink and downlink throughputs of high-speed UEs in high-speed cells increase, while the average uplink and downlink throughputs of low-speed UEs in these cells decrease. Therefore, average uplink and downlink throughputs in high-speed cells may fluctuate. If four or fewer UEs are scheduled, differentiated scheduling may not deliver obvious gains because even allocation of uplink RBs takes effect by default.

11.1.3 Requirements

11.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

11.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Differentiated scheduling requires the NRDUCell.HighSpeedFlag parameter to be set to HIGH_SPEED .

11.1.3.2.2 Mutually Exclusive Functions

None

11.1.3.2.3 Function Impacts

None

11.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

11.1.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

11.1.4 Operation and Maintenance

11.1.4.1 Data Configuration

11.1.4.1.1 Data Preparation

Table 11-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the HIGH_SPEED_DIFFERENTIAL_SCH_SW option to enable differentiated scheduling.

11.1.4.1.2 Using MML Commands

Before using MML commands, refer to **11.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the differentiated scheduling function  
MOD NRCELLALGOSWITCH: NrcellId=0, HighSpeedDedPolicySw=HIGH_SPEED_DIFFERENTIAL_SCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the differentiated scheduling function  
MOD NRCELLALGOSWITCH: NrcellId=0, HighSpeedDedPolicySw=HIGH_SPEED_DIFFERENTIAL_SCH_SW-0;
```

11.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.1.4.2 Activation Verification

After this function is enabled, observe the average uplink and downlink throughputs of high-speed UEs in high-speed cells. If the values increase, this function has taken effect.

NOTE

Average downlink throughput of high-speed UEs = $(N.ThpVol.DL.Hs - N.ThpVol.DL.LastSlot.Hs) / N.ThpTime.DL.RmvLastSlot.Hs$

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

11.1.4.3 Network Monitoring

The indicators listed in [11.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

11.2 Differentiated Configuration Between High- and Low-speed UEs

11.2.1 Principles

Differentiated configuration between high- and low-speed UEs provides flexible and differentiated configuration policies for high- and low-speed UEs in high-speed cells. This function includes the following subfunctions:

- [Differentiated Configuration of Handover Parameters](#)
- [Differentiated DRX](#)
- [Differentiated EPS fallback](#)
- [Differentiated UE inactivity timer](#)

Differentiated Configuration of Handover Parameters

- Intra- and inter-frequency handover parameter configuration
Intra- and inter-frequency handover parameters dedicated to low-speed UEs are added to the **NRCeIIIntraFHoMeaGrp** and **NRCeIIInterFHoMeaGrp** MOs. The related parameter group is specified using the **NRCeIIQciBearer.LSueIntraFreqHoMeasGrpId** or **NRCeIIQciBearer.LSueInterFreqHoMeasGrpId** parameter.
After a UE in a high-speed cell is identified as a low-speed UE, the processing mechanism is as follows:
 - If the **NRCeIIQciBearer.LSueIntraFreqHoMeasGrpId** or **NRCeIIQciBearer.LSueInterFreqHoMeasGrpId** parameter of the UE's highest-priority bearer is not set to 255, measurement configurations are delivered to the UE based on the corresponding intra- or inter-frequency handover parameters. Specifically, the parameter group corresponding to both **NRCeIIIntraFHoMeaGrp.IntraFreqHoMeasGroupId** and **NRCeIIQciBearer.LSueIntraFreqHoMeasGrpId**, or both **NRCeIIInterFHoMeaGrp.InterFreqHoMeasGroupId** and **NRCeIIQciBearer.LSueInterFreqHoMeasGrpId**, is delivered.

- If the **NRCellQciBearer.LSUEIntraFreqHoMeasGrpId** or **NRCellQciBearer.LSUEInterFreqHoMeasGrpId** parameter of the UE's highest-priority bearer is set to **255**, intra- or inter-frequency handover parameters are not reconfigured for the UE. Specifically, the intra- or inter-frequency handover parameters corresponding to **NRCellQciBearer.IntraFreqHoMeasGroupId** or **NRCellQciBearer.InterFreqHoMeasGroupId** of the UE's highest-priority bearer take effect.

For details about parameter configuration of the **NRCellIntraFHoMeaGrp** or **NRCellInterFHoMeaGrp** MO, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

Differentiated DRX

If the DRX function is enabled in a high-speed cell (by selecting the **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) and the low-speed UE DRX switch is turned on (by selecting the **LOW_SPEED_UE_DRX_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter), DRX takes effect for identified low-speed UEs in the high-speed cell.

DRX takes effect only when the length of the long DRX cycle (specified by **gNBDrxParamGroup.LongCycle**) is less than or equal to 1/6 of the value of the uplink time alignment timer (specified by **NRDUCellULTaConfig.ULTimeAlignmentTimer**). The default value of the uplink time alignment timer for high-speed cells is 1280 ms. Therefore, the long DRX cycle must be set to a value less than or equal to 160 ms for a high-speed cell.

Differentiated EPS fallback

If differentiated EPS fallback for low-speed UEs is enabled (by selecting the **LOW_SPEED_DIFF_EPS_FALLBACK_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter) in a high-speed cell, an identified low-speed UE will preferentially select a frequency whose **NRCellEutranNFreq.HighSpeedFlag** is set to **LOW_SPEED** to perform EPS fallback.

Differentiated UE inactivity timer

If the **NRDUCellQciBearer.HsCellLowSpeedUeInactTimer** parameter is set to **65535** for all bearers of a high-speed cell, the differentiated UE inactivity timer does not take effect. Otherwise, the following mechanism is used:

- When a UE is not identified as a low-speed UE, the inactivity timer specified by the **NRDUCellQciBearer.UeInactivityTimer** parameter takes effect for the UE.
- When a UE is identified as a low-speed UE:
 - The UE inactivity timer of a bearer whose **NRDUCellQciBearer.HsCellLowSpeedUeInactTimer** parameter is set to a value other than **65535** is used as the value of this parameter.
 - The UE inactivity timer of a bearer whose **NRDUCellQciBearer.HsCellLowSpeedUeInactTimer** parameter is set to

65535 is used as the value of the **NRDUCellQciBearer.UeInactivityTimer** parameter.

If a UE has multiple bearers with different UE inactivity timers, the inactivity timer with the longest length takes effect.

11.2.2 Network Analysis

11.2.2.1 Benefits

Differentiated configuration of handover parameters can improve the handover reliability of low-speed UEs in high-speed cells and reduce ping-pong handovers through parameter optimization. Differentiated DRX can improve the power saving effect of low-speed UEs in high-speed cells.

11.2.2.2 Impacts

The impacts between this function and other functions are described in [11.2.3.2.3 Function Impacts](#).

After differentiated handover parameter configuration or differentiated DRX is enabled, the number of RRC connection reconfigurations in high-speed cells increases. This may slightly decrease the average downlink MCS index and average uplink and downlink cell throughputs, and increase the service drop rate.

11.2.3 Requirements

11.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

11.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Differentiated configuration between high- and low-speed UEs requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .
Low-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter	<i>DRX</i>	The differentiated DRX function in differentiated configuration between high- and low-speed UEs depends on the basic DRX switch.

11.2.3.2.2 Mutually Exclusive Functions

None

11.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter	<i>DRX</i>	After differentiated DRX is enabled, DRX takes effect only on low-speed UEs in high-speed cells.

11.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

11.2.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

11.2.4 Operation and Maintenance

11.2.4.1 Data Configuration

11.2.4.1.1 Data Preparation

11.2.4.1.1 Data Preparation describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Low-Speed UE Intra-Freq HO Measure Group ID	NRCellQciBearer.L SUEIntraFreqHoMeasGrpId	Set this parameter according to the network plan.
Low-Speed UE Inter-Freq HO Measure Group ID	NRCellQciBearer.L SUEInterFreqHoMeasGrpId	Set this parameter according to the network plan.
High Speed Dedicated Policy Switch	NRCellAlgoSwitch .HighSpeedDedPolicySw	Deselect the LOW_SPEED_UE_DRX_SW option.
High Speed Dedicated Policy Switch	NRCellAlgoSwitch .HighSpeedDedPolicySw	Deselect the LOW_SPEED_DIFF_EPS_FALLBACK_SW option.
High Speed Cell Low Speed UE Inactivity Timer	NRDUCellQciBearer.HsCellLowSpeedUeInactTimer	The value 10 is recommended.

11.2.4.1.2 Using MML Commands

Before using MML commands, refer to **11.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Differentiated configuration of handover parameters
 //(Optional) Adding an intra-frequency handover measurement parameter group ID, which is optional for configuring differentiated intra-frequency handover parameters
 ADD NRCELLINTRAFHOMEAGRP: NrCellId=0, IntraFreqHoMeasGrpId=1;
 //(Optional) Adding an inter-frequency handover measurement parameter group ID, which is optional for configuring differentiated inter-frequency handover parameters
 ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGrpId=1;
 //(Optional) Changing the low-speed UE intra-frequency handover measurement group ID and low-speed UE inter-frequency handover measurement group ID (performed when configuring differentiated intra- or inter-frequency handover parameters)
 MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, LSUEIntraFreqHoMeasGrpId=1, LSUEInterFreqHoMeasGrpId=1;
- Differentiated DRX
 //Modifying the long DRX cycle for all DRX parameter groups
 MOD GNBDRXPARAMGROUP: DrxParamGrpId=1, LongCycle=MS160;
 //Turning on the low-speed UE DRX switch
 MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=LOW_SPEED_UE_DRX_SW-1;
- Differentiated EPS fallback
 //Turning on the low-speed differentiated EPS fallback switch
 MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=LOW_SPEED_DIFF_EPS_FALLBACK_SW-1;
- Differentiated UE inactivity timer
 //Configuring the low-speed UE inactivity timer in the high-speed cell
 MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, HsCellLowSpeedUeInactTimer=10;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Differentiated configuration of handover parameters
 //Setting the low-speed UE intra-frequency handover measurement parameter group ID to 255, which indicates an invalid parameter group
 MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, LSUEIntraFreqHoMeasGrpId=255;
 //Setting the low-speed UE inter-frequency handover measurement parameter group ID to 255, which indicates an invalid parameter group
 MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, LSUEInterFreqHoMeasGrpId=255;
- Differentiated DRX
 //Turning off the low-speed UE DRX switch
 MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=LOW_SPEED_UE_DRX_SW-0;
- Differentiated EPS fallback
 //Turning off the low-speed differentiated EPS fallback switch
 MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=LOW_SPEED_DIFF_EPS_FALLBACK_SW-0;
- Differentiated UE inactivity timer
 //Setting the low-speed UE inactivity timer for the high-speed cell to 65535, which indicates an invalid value
 MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, HsCellLowSpeedUeInactTimer=65535;

11.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.2.4.2 Activation Verification

- Differentiated configuration of handover parameters
After this function is enabled, observe the RRC signaling of low-speed UEs to check whether the corresponding handover measurement parameters are delivered based on the configured values in the RRC reconfiguration message.
- Differentiated DRX
After this function is enabled, observe the value of the **N.Cdrx.Sleep.Dur.Total** counter to check whether this function has taken effect. If the value of this counter is not 0, this function has taken effect.
- Differentiated EPS fallback
After this function is enabled, observe the E-UTRAN frequency measurement information delivered in the RRC reconfiguration message during EPS fallback of low-speed UEs. If low-speed E-UTRAN frequencies are preferentially delivered (in the front of the *measObjectToAddModList* IE), this function has taken effect.
- Differentiated UE inactivity timer
After this function is enabled, check whether the duration from the time when a low-speed UE does not perform data transmission to the time when the network initiates a release is consistent with the configuration.

11.2.4.3 Network Monitoring

The indicators listed in [11.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

11.3 Differentiated A2 Threshold Configuration for High-Speed UEs

11.3.1 Principles

Event A2 for inter-frequency handovers triggers gap-assisted measurements, which cause throughput decrease. To address this issue, in scenarios where high-speed cells and adjacent public-network cells use different frequency bands, differentiated A2 thresholds can be configured for high-speed UEs that are newly handed over to high-speed cells. In this way, high-speed UEs that bypass railway station platforms are not likely to start gap-assisted measurements, improving user experience.

This function is controlled by the **NRCellInterFHoMeaGrp.HighSpeedUeA2ThldOfs** parameter. If this parameter is set to a non-zero value, the function takes effect and the gNodeB performs the following operations on the event A1 and A2 thresholds.

- A2 threshold for high-speed UEs handed over to a high-speed cell:
 - Superposed with the value of **NRCellInterFHoMeaGrp.HighSpeedUeA2ThldOfs**.

- The sum of the A2 threshold and the value of **NRCellInterFHoMeaGrp.HighSpeedUeA2ThldOfs** is greater than or equal to -115.

The A2 threshold is the RSRP threshold for triggering event A2 related to coverage-based inter-frequency handover (**NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld**) or the RSRP threshold for triggering event A2 related to event A3-based inter-frequency handover (**NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld**).

- A1 threshold for high-speed UEs handed over to a high-speed cell:
Superposed with the value of **NRCellInterFHoMeaGrp.HighSpeedUeA2ThldOfs**.

The A1 threshold is the RSRP threshold for triggering event A1 related to coverage-based inter-frequency handover (**NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld**) or the RSRP threshold for triggering event A1 related to event A3-based inter-frequency handover (**NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld**).

The mechanisms of reporting A1 and A2 measurement reports based on the superposed A1 and A2 thresholds are as follows:

- If the measured RSRP value of a high-speed UE in the serving cell is always less than the accumulated event A2 threshold within the configured time-to-trigger of events A1/A2 related to inter-frequency measurements (**NRCellInterFHoMeaGrp.InterFreqA1A2TimeToTrig**), the UE sends an event A2 measurement report.
- If the measured RSRP value of a high-speed UE in the serving cell is always greater than the accumulated event A1 threshold within the configured time-to-trigger of events A1/A2 related to inter-frequency measurements (**NRCellInterFHoMeaGrp.InterFreqA1A2TimeToTrig**), the UE sends an event A1 measurement report.

The RSRP thresholds for events A1 and A2 related to coverage-based inter-frequency handovers can be configured separately for each frequency. The function is implemented by setting the **NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs** parameter to different values for events A1, A2, and A5-1.

- RSRP threshold for event A1 related to coverage-based inter-frequency handover = **NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld** + **NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs**.
- RSRP threshold for event A2 related to coverage-based inter-frequency handover = **NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld** + **NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs**.

For details about the measurement gap configuration, RSRP threshold for event A2 related to coverage-based inter-frequency handover, RSRP threshold for event A2 related to event A3-based inter-frequency handover, RSRP threshold for event A1 related to coverage-based inter-frequency handover, RSRP threshold for event A1 related to event A3-based inter-frequency handover, the time-to-trigger of events A1/A2 related to inter-frequency measurements, and frequency offset for event A1/A2/A5-1 related to inter-frequency handover, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

11.3.2 Network Analysis

11.3.2.1 Benefits

In the scenario where high-speed cells and neighboring public-network cells use different frequency bands, the A2 threshold configured in some high-speed cells may be very high to ensure timely handovers for non-high-speed UEs. For example, the A2 threshold of high-speed cells may be greater than -80 dBm in railway platform scenarios. Differentiated A2 threshold configuration for high-speed UEs can be enabled in the preceding scenarios.

This function prevents high-speed UEs from easily starting gap-assisted measurements and thereby increases the average uplink and downlink throughputs of high-speed UEs that are handed over to high-speed cells.

11.3.2.2 Impacts

The impacts between this function and other functions are described in [11.3.3.2.3 Function Impacts](#).

This function has no impact.

11.3.3 Requirements

11.3.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

11.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.Hi ghSpeedFlag	<i>High Speed Mobility</i>	Differentiated A2 threshold configuration for high-speed UEs requires that the NRDUCell.HighSpeed Flag parameter be set to HIGH_SPEED .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cov Inter-freq Handover A2 RSRP Threshold	NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrpThld	<i>SA Mobility Management in Connected Mode or NSA Mobility Management</i>	The sum of the values of NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrpThld and NRCeIIInterFHoMeaGrp.HighSpeedUeA2ThldOfs must be greater than or equal to -115.
Low-frequency TDD	A3 Inter-freq Handover A2 RSRP Threshold	NRCeIIInterFHoMeaGrp.A3InterFreqHoA2RsrpThld	<i>SA Mobility Management in Connected Mode or NSA Mobility Management</i>	The sum of the values of NRCeIIInterFHoMeaGrp.A3InterFreqHoA2RsrpThld and NRCeIIInterFHoMeaGrp.HighSpeedUeA2ThldOfs must be greater than or equal to -115.

11.3.3.2.2 Mutually Exclusive Functions

None

11.3.3.2.3 Function Impacts

None

11.3.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

11.3.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

11.3.4 Operation and Maintenance

11.3.4.1 Data Configuration

11.3.4.1.1 Data Preparation

Table 11-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High-Speed UE A2 Thld Offset	NRCeIIInterFHoMeaGrp.HighSpeedUeA2ThldOfs	To enable differentiated A2 threshold configuration for high-speed UEs, set this parameter to a value other than 0.

11.3.4.1.2 Using MML Commands

Before using MML commands, refer to [11.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the A2 threshold offset for high-speed UEs  
MOD NRCELLINTERFHOMeAGRP: NrcellId=0, InterFreqHoMeasGroupId=1, HighSpeedUeA2ThldOfs=-20;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling differentiated A2 threshold configuration for high-speed UEs  
MOD NRCELLINTERFHOMeAGRP: NrcellId=0, InterFreqHoMeasGroupId=1, HighSpeedUeA2ThldOfs=0;
```

11.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.3.4.2 Activation Verification

Check the RRC_RECFCG signaling message in the Uu interface tracing result.

If the RSRP value after function activation, which is indicated by **a2-Threshold > rsrp** in the RRC_RECFCG message, is equal to the sum of the RSRP value before function activation and the value of **NRCeIIInterFHoMeaGrp.HighSpeedUeA2ThldOfs**, this function has taken effect.

11.3.4.3 Network Monitoring

The indicators listed in [11.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

11.4 Low-Speed UE Performance Optimization

11.4.1 Principles

Low-speed UE performance optimization optimizes the performance of low-speed UEs in high-speed cells, thereby improving the uplink and downlink throughputs of such UEs.

This function is controlled by the **LOW_SPEED_PERFORMANCE_OPT_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter.

11.4.2 Network Analysis

11.4.2.1 Benefits

Low-speed UE performance optimization increases the average uplink and downlink throughputs of low-speed UEs in high-speed cells.

11.4.2.2 Impacts

The impacts between this function and other functions are described in [11.4.3.2.3 Function Impacts](#).

This function has no impact.

11.4.3 Requirements

11.4.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

11.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Performance optimization for low-speed cells requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

11.4.3.2.2 Mutually Exclusive Functions

None

11.4.3.2.3 Function Impacts

None

11.4.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

11.4.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

11.4.4 Operation and Maintenance

11.4.4.1 Data Configuration

11.4.4.1.1 Data Preparation

[Table 11-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCeAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the LOW_SPEED_PERFORMANCE_OPT_SW option to enable low-speed UE performance optimization. It is recommended that this function be enabled in non-suburban areas to improve the performance of low-speed UEs camping on high-speed cells.

11.4.4.1.2 Using MML Commands

Before using MML commands, refer to [11.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the low-speed UE performance optimization function  
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=LOW_SPEED_PERFORMANCE_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the low-speed UE performance optimization function  
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=LOW_SPEED_PERFORMANCE_OPT_SW-0;
```

11.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.4.2 Activation Verification

None

11.4.4.3 Network Monitoring

The indicators listed in [11.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

11.5 In-stop-train UE identification and experience guarantee

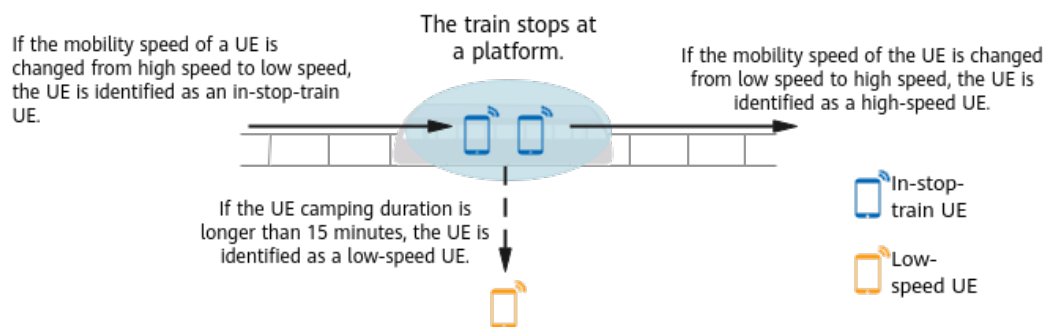
11.5.1 Principles

To ensure user experience in a train that stops at a platform, the function of "in-stop-train UE identification and experience guarantee" is introduced. In-stop-train UE identification is controlled by the **UE_IN_STOP_TRAIN_IDENT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

As illustrated in [Figure 11-1](#), the mechanism of in-stop-train UE identification and experience guarantee is as follows:

1. Identifying UEs in the train that stops at a platform
If this function is enabled, high-speed UEs on a high-speed train are identified as in-stop-train UEs after the train stops at the platform.
2. Starting scheduling guarantee
The scheduling priority of in-stop-train UEs at the platform is the same as that of high-speed UEs.
3. Exiting in-stop-train UE experience guarantee
If either of the following conditions is met: (1) an in-stop-train UE camps on a TRP for more than 15 minutes; (2) the UE speed attribute is changed from low speed to high speed after the train leaves the platform, it indicates that the UE is no longer an in-stop-train UE.

Figure 11-1 In-stop-train UE identification



11.5.2 Network Analysis

11.5.2.1 Benefits

After this function is enabled, the user-perceived rate of UEs on a high-speed train that temporarily stops in a high-speed cell increases.

11.5.2.2 Impacts

The impacts between this function and other functions are described in [11.5.3.2.3 Function Impacts](#).

After this function is enabled, scheduling of in-stop-train UEs at the platform is guaranteed in the same way as high-speed UEs. Therefore, when a train stops temporarily, the experience of low-speed UEs in the high-speed cell deteriorates.

After this function is enabled, in-stop-train UEs at the platform are counted as high-speed UEs. Therefore, the performance counters are affected as follows:

- The average number of high-speed UEs in the RRC_CONNECTED state (indicated by the **N.User.RRConn.Hs.Avg** counter) increases.
- The maximum number of high-speed UEs in the RRC_CONNECTED state (indicated by the **N.User.RRConn.Hs.Max** counter) may increase.
- The uplink and downlink user-perceived rates of high-speed UEs in a cell may fluctuate.
- The uplink and downlink user-perceived rates in a cell may fluctuate.

11.5.3 Requirements

11.5.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

11.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.Hi ghSpeedFlag	<i>High Speed Mobility</i>	In-stop-train UE identification and experience guarantee requires that the NRDUCell.HighSpeed Flag parameter be set to HIGH_SPEED .

11.5.3.2.2 Mutually Exclusive Functions

None

11.5.3.2.3 Function Impacts

None

11.5.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

11.5.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

11.5.4 Operation and Maintenance

11.5.4.1 Data Configuration

11.5.4.1.1 Data Preparation

[Table 11-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Algorithm Optimization Switch	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	To enable in-stop-train UE identification and experience guarantee, select the UE_IN_STOP_TRAIN_IDENT_SW option.

11.5.4.1.2 Using MML Commands

Before using MML commands, refer to [11.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling in-stop-train UE identification and experience guarantee  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=UE_IN_STOP_TRAIN_IDENT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling in-stop-train UE identification and experience guarantee  
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=UE_IN_STOP_TRAIN_IDENT_SW-0;
```

11.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.5.4.2 Activation Verification

After this function is enabled, in-stop-train UEs at the platform are counted as high-speed UEs. Within 15 minutes after the train arrives at the platform, you can check whether this function has taken effect by observing the value of the **N.User.RRConn.Hs.Avg** counter. If the counter value increases, this function has taken effect.

11.5.4.3 Network Monitoring

The indicators listed in [11.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

12 Interference Policy Management for Public and Dedicated Networks

12.1 Interference Coordination Between Public and Dedicated Networks

12.1.1 Principles

When a high-speed cell identifies that a train has arrived at a dedicated-network cell or a TRP of the dedicated-network cell, it instructs adjacent public-network cells to restrict scheduling resources or perform beam coordination. Specifically, the measures include RB restriction, beam avoidance, beam-level power reduction, and cell-level power reduction. This reduces interference from adjacent public-network cells to the dedicated network and minimizes the impact of "coordination between public and dedicated networks" on public networks. For details about the definitions of public and dedicated networks, see [3 Overview](#).

It is recommended that "interference coordination between public and dedicated networks" be enabled in intra-band networking of public and dedicated networks. Public and dedicated networks can use the same SSB frequency or different SSB frequencies.

Interference coordination between public and dedicated networks includes the following subfunctions:

- [RB restriction and power restriction for interference coordination between public and dedicated networks](#)
- [Automatic Identification of Interfering Neighboring Cells for Interference Coordination Between Public and Dedicated Networks](#)
- [Beam avoidance for interference coordination between public and dedicated networks \(low-frequency TDD\)](#)
- [TRP-based Precise Interference Coordination for Interference Coordination Between Public and Dedicated Networks](#)

- **Beam-level Power Reduction for Interference Coordination Between Public and Dedicated Networks (Low-frequency TDD)**

RB restriction and power restriction for interference coordination between public and dedicated networks

This function is controlled by the high-speed cell interference coordination switch (**HIGH_SPEED_INTRF_AVOID_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter) or the interference coordination always-on switch (**INTRF_COORD_ALWAYS_WORKING_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter). The high-speed cell interference coordination switch and the interference coordination always-on switch are mutually exclusive. The interference coordination always-on switch can be turned on for a maximum of three high-speed cells under the same base station.

Interference coordination between public and dedicated networks proceeds as follows:

1. Determining whether a train has arrived at the dedicated-network cell
After the high-speed cell interference coordination switch (the **HIGH_SPEED_INTRF_AVOID_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter) is turned on, a high-speed cell determines whether a train has arrived based on the number of high-speed UEs in the cell. If this number is greater than or equal to the value of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter, it considers the train as having arrived.
If the interference coordination always-on switch is turned on, interference coordination between public and dedicated networks always takes effect. In this case, the dedicated-network cell does not check the train arrival as described in step 1 and the procedure starts from step 2.
2. Determining whether to start interference avoidance
When a train has arrived, the high-speed cell instructs public-network cells (that is, low-speed cells) with **NRCCellRelation.HighSpeedIntrfAvoidFlag** set to **INTRF_AVOID** to start interference avoidance.
3. Taking interference avoidance measures
Low-speed cells with **NRCCellRelation.HighSpeedIntrfAvoidFlag** set to **INTRF_AVOID** start interference avoidance based on the train arrival indication of a high-speed cell and parameter settings of the low-speed cells. Interference avoidance is enabled when the following parameters are set to non-zero values:
 - Downlink power reduction in low-speed cells: The **NRDUCellHighSpeedMob.DlIntrfBasedPwrOffset** parameter specifies the power offset for downlink power decrease in low-speed cells.
 - Uplink power reduction in low-speed cells: The **NRDUCellHighSpeedMob.UlIntrfBasedPwrOffset** parameter specifies the power offset for uplink power decrease in low-speed cells.
 - Downlink RB restriction for low-speed cells: The **NRDUCellHighSpeedMob.DlIntrfLimitRbSchRatio** parameter specifies the proportion of downlink RBs that cannot be scheduled in low-speed cells.

- Uplink RB restriction for low-speed cells: The **NRDUCellHighSpeedMob.UlIntrfLimitRbSchRatio** parameter specifies the proportion of uplink RBs that cannot be scheduled in low-speed cells.

The **NRDUCellHighSpeedMob.DlIntrfLimitRbSchRatio** and **NRDUCellHighSpeedMob.UlIntrfLimitRbSchRatio** parameters specify the proportion of RBs that cannot be scheduled only for the overlapping area of the public and dedicated networks. The proportion of RBs to be scheduled is not limited in non-overlapping areas.

To reduce the impact on neighboring public-network cells after this feature is enabled, set the parameters related to RB restriction and power reduction based on the inter-site distance (ISD) or signal strength for neighboring public-network cells along the high-speed railway. Recommended parameter configurations are listed in [Table 12-1](#).

Table 12-1 Configuration suggestions for RB limitation and power reduction

Scenario	DL Intrf Based Limited RB Sch Ratio	UL Intrf Based Limited RB Sch Ratio	DL Intrf Based Power Offset	UL Intrf Based Power Offset
The ISD is less than or equal to 500 m, or the signal strength difference between neighboring public-network cells and the dedicated-network cell is less than or equal to 15 dB.	75	75	6	6
The ISD is greater than 500 m but less than or equal to 1000 m, or the signal strength difference between neighboring public-network cells and the dedicated-network cell is greater than 15 dB but less than or equal to 20 dB.	50	50	3	3

4. Stopping interference avoidance measures

If the high-speed cell interference coordination switch is turned on, the high-speed cell instructs adjacent public-network cells to restore the normal levels of scheduled RBs and power when the train leaves the high-speed cell.

If the interference coordination always-on switch is turned on, the high-speed cell instructs adjacent public-network cells to restore the normal levels of scheduled RBs and power when this switch is turned off.

Automatic Identification of Interfering Neighboring Cells for Interference Coordination Between Public and Dedicated Networks

This function is controlled by the interfering neighboring cell identification switch (**INTERFERING_NCELL_IDENT_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter). The interfering neighboring cell identification switch depends on the high-speed cell interference coordination switch (**HIGH_SPEED_INTRF_AVOID_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter).

The mechanism of automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks is as follows:

1. Determining whether a train has arrived at the dedicated-network cell
After the interfering neighboring cell identification switch is turned on, a high-speed cell evaluates whether a train has arrived based on the number of high-speed UEs in the cell. If the number of high-speed UEs is greater than or equal to the value of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter, it considers that the train has arrived.
2. Recording low-speed neighboring cells with strong interference to the dedicated network
When a train has arrived at the high-speed cell, the dedicated-network cell randomly selects high-speed UEs to start SSB measurement on intra-frequency or inter-frequency neighboring public-network cells. If the SSB RSRP of a public-network neighboring cell measured by a high-speed UE is greater than or equal to the value of the **NRCCellMeasConfig.IntrfCellIdentSsbRsrpThld** parameter, the high-speed UE reports the measurement results to the dedicated-network cell. In this way, the dedicated-network cell obtains the SSB beam signal strength of the neighboring public-network cells, and then selects and records neighboring cells with strong interference.
3. Determining whether to start interference avoidance
When a train has arrived, the high-speed cell instructs low-speed neighboring cells for which **NRCCellRelation.HighSpeedIntrfAvoidFlag** is set to **INTRF_AVOID** and that are also recorded in the list of neighboring cells with strong interference to dedicated-network cells, to start interference avoidance. In cases where interference coordination between public and dedicated networks has been enabled for multiple cells served by the same base station, if the **INTRF_NCELL_IDENT_ENH_SW** option of the **gNodeBParam.HighSpeedDedPolicySw** parameter has been selected for the dedicated-network base station, the base station will be able to more accurately identify neighboring public-network cells that cause interference, and trigger interference avoidance in these cells.

Beam avoidance for interference coordination between public and dedicated networks (low-frequency TDD)

This function is controlled by the high-speed beam avoidance switch (**HIGH_SPEED_BEAM_AVOID_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter). This function takes effect only when this switch is turned on for both high-speed cells and low-speed

massive MIMO cells. In high-speed cells, the high-speed beam avoidance switch depends on the high-speed cell interference coordination switch (**HIGH_SPEED_INTRF_AVOID_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter).

The processing mechanism of beam avoidance for interference coordination between public and dedicated networks is as follows:

1. Determining whether a train has arrived at the dedicated-network cell
The high-speed cell determines whether a train arrives at the dedicated-network cell based on the number of high-speed UEs in the cell. If the number of high-speed UEs is greater than or equal to the value of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter, it considers that the train has arrived.
2. Obtaining information about interfering cells and SSB beams of the interfering cells
When a train has arrived at the high-speed cell, the dedicated-network cell randomly selects high-speed UEs to start SSB measurement on intra-frequency or inter-frequency neighboring public-network cells. If the SSB RSRP of a public-network neighboring cell measured by a high-speed UE is greater than or equal to the value of the **NRCellMeasConfig.IntrfCellIdentSsbRsrpThld** parameter, the high-speed UE reports the measurement results to the dedicated-network cell. In this way, the dedicated-network cell obtains information about interfering cells and SSB beams of the interfering cells.
3. Determining whether to start interference avoidance
When a train has arrived, the high-speed cell instructs low-speed neighboring cells with **NRCellRelation.HighSpeedIntrfAvoidFlag** set to **INTRF_AVOID** to start interference avoidance.
4. Taking beam avoidance measures
The public-network massive MIMO cells with **NRCellRelation.HighSpeedIntrfAvoidFlag** set to **INTRF_AVOID** and the high-speed beam avoidance switch turned on set the channel estimation for the interfering SSB beams to zero based on the train arrival indication and interfering SSB beam information signaled by the high-speed cell. In cases where interference coordination between public and dedicated networks has been enabled for multiple cells served by the same base station, if the **INTRF_NCELL_IDENT_ENH_SW** option of the **gNodeBParam.HighSpeedDedPolicySw** parameter has been selected for the dedicated-network base station, the base station will be able to more accurately identify neighboring public-network cells that cause interference, and trigger interference avoidance in these cells.
5. Stopping beam avoidance
When the train leaves the high-speed cell, the high-speed cell instructs public-network cells to stop beam avoidance.

TRP-based Precise Interference Coordination for Interference Coordination Between Public and Dedicated Networks

This function is controlled by the **TRP_PRECISE_INTRF_COORD_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter. This function takes effect

only in high-speed cells and requires the **HIGH_SPEED_BEAM_AVOID_SW** or **INTERFERING_NCELL_IDENT_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter to be selected.

The processing mechanism of TRP-based precise interference coordination for interference coordination between public and dedicated networks is as follows:

1. Determining the TRP where the train is located

After this function is enabled, if a high-speed cell determines that a train has arrived, the base station determines whether the train has arrived at a TRP based on the number of high-speed UEs in the TRP. When the number of high-speed UEs in the TRP is greater than or equal to half of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter value, the base station determines that the train has arrived at the TRP.

2. Determining whether to start interference avoidance

When the train has arrived at the TRP, the high-speed cell instructs the interfering low-speed cells around the TRP to start interference avoidance.

Interference avoidance is started in cells for which the **NRCCellRelation.HighSpeedIntrfAvoidFlag** parameter is set to **INTRF_AVOID** on the dedicated network.

3. Starting interference avoidance

Low-speed massive MIMO cells with the **NRCCellRelation.HighSpeedIntrfAvoidFlag** parameter set to **INTRF_AVOID** and the **HIGH_SPEED_BEAM_AVOID_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter selected or low-speed cells with power reduction and RB restriction enabled starts interference avoidance based on the train arrival indication of the high-speed cell.

4. Stopping interference avoidance

When the train leaves the TRP, the high-speed cell instructs the interfering low-speed neighboring cells around the TRP to stop avoidance. When the train leaves the high-speed cell, the high-speed cell instructs all public-network cells to stop avoidance.

Beam-level Power Reduction for Interference Coordination Between Public and Dedicated Networks (Low-frequency TDD)

This function is controlled by the **INTRF_COORD_BEAM_PWR_REDUCE_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter. This function takes effect only when all of the following conditions are met:

- The **INTERFERING_NCELL_IDENT_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter is selected on the dedicated network.
- The **INTRF_COORD_BEAM_PWR_REDUCE_SW** option of the **NRCCellAlgoSwitch.HighSpeedDedPolicySw** parameter is selected on both the dedicated network and public network.
- Low-speed cells on the public network are massive MIMO cells.

Beam-level power reduction for interference coordination between public and dedicated networks can be enabled for a maximum of three high-speed cells served by the same base station.

The processing mechanism of beam-level power reduction for interference coordination between public and dedicated networks is as follows:

1. Determining whether a train has arrived at the dedicated-network cell and determining the TRP where the train locates

After beam-level power reduction is enabled, the high-speed cell determines whether a train has arrived at the dedicated-network cell based on the number of high-speed UEs in the cell. If this number is greater than or equal to the value of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter, it considers the train as having arrived.

If the high-speed cell determines that a train has arrived, the base station further determines whether the train has arrived at a TRP based on the number of high-speed UEs in the TRP. When the number of high-speed UEs in the TRP is greater than or equal to half of the **NRDUCellHighSpeedMob.HighSpeedUserNumThld** parameter value, the base station determines that the train has arrived at the TRP.

2. Determining whether to start interference avoidance

When a train has arrived, the high-speed cell instructs low-speed neighboring cells with **NRCellRelation.HighSpeedIntrfAvoidFlag** set to **INTRF_AVOID** to start interference avoidance.

3. Starting interference avoidance

Low-speed cells with the **NRCellRelation.HighSpeedIntrfAvoidFlag** parameter set to **INTRF_AVOID** and the **INTRF_COORD_BEAM_PWR_REDUCE_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter selected starts interference avoidance based on the train arrival indication of the high-speed cell.

The interference avoidance measure is to reduce the UE-level PDSCH power in each interference beam. The power decrease is equal to the difference between the SSB RSRP of the public network identified by the dedicated network and the absolute value of the **NRCellMeasConfig.IntrfCellIdentSsbRsrpThld** parameter value configured in a low-speed cell. To avoid the impact of excessive power decrease on base stations of the public network, the maximum power decrease allowed by the public network cannot exceed 10 dB. In cases where interference coordination between public and dedicated networks has been enabled for multiple cells served by the same base station, if the **INTRF_NCELL_IDENT_ENH_SW** option of the **gNodeBParam.HighSpeedDedPolicySw** parameter has been selected for the dedicated-network base station, the base station will be able to more accurately identify neighboring public-network cells that cause interference, and trigger interference avoidance in these cells.

4. Stopping interference avoidance

When the train leaves the TRP, the high-speed cell instructs the interfering low-speed neighboring cells around the TRP to stop avoidance. When the train leaves the high-speed cell, the high-speed cell instructs all public-network cells to stop avoidance.

12.1.2 Network Analysis

12.1.2.1 Benefits

- If the dedicated-network cell experiences interference from the public network and the interference avoidance measure is uplink and downlink power reduction or uplink and downlink RB limitation in the public-network cell, this function increases the average uplink and downlink throughputs of high-speed UEs.
If the dedicated-network cell experiences interference from the public network and the interference avoidance measure is high-speed beam avoidance in public-network cells, this function increases the average downlink throughputs of high-speed UEs.
- If the high-speed cell interference coordination switch or the high-speed beam avoidance switch is turned on and automatic identification of interfering neighboring cells takes effect, the dedicated-network cell automatically identifies public-network cells with strong interference and sends messages related to coordination between public and dedicated networks to them. In this case, the number of public-network cells whose performance deteriorates due to coordination between public and dedicated networks greatly decreases. The exact benefit of interference coordination between public and dedicated networks is related to the load of the surrounding public network. If the load of the surrounding public network is low, the benefit may not be significant.
- Compared to when only cell-level interference coordination for interference coordination between public and dedicated networks is enabled, when TRP-based precise interference coordination is enabled, the number of public-network cells whose performance deteriorates due to interference coordination for interference coordination between public and dedicated networks decreases, and the user-perceived rate of public-network UEs increases. If the interference from the adjacent public-network cells to the dedicated network is small, or the power reduction and RB restriction thresholds configured on the public network side are low, enabling TRP-based precise interference coordination may not significantly improve user experience on the public network.
- Compared to when only beam avoidance or only power restriction is enabled, when beam-level power reduction is enabled, the impact of interference coordination between public and dedicated networks on public-network UEs decreases, and the user-perceived rate of public-network UEs increases. If the interference from the adjacent public-network cells to the dedicated network is small, or the power reduction threshold configured on the public network side is low, enabling beam-level power reduction may not significantly improve user experience on the public network.

12.1.2.2 Impacts

The impacts between this function and other functions are described in [12.1.3.2.3 Function Impacts](#).

Interference coordination between public and dedicated networks affects the performance of public networks. Therefore, it is not recommended that this function be enabled in railway stations or scenarios where ToB services are deployed along railways, such as power plants and coal mines. [Table 12-2](#) lists the network impacts of interference coordination between public and dedicated networks.

Table 12-2 Network impacts of interference coordination between public and dedicated networks.

Subfunction	Impact Description
RB restriction and power restriction for interference coordination between public and dedicated networks	After this function is enabled and uplink and downlink power reduction or uplink and downlink RB limitation takes effect for public-network cells, the impacts are as follows: The average uplink throughput (Cell Uplink Average Throughput (DU)) and average downlink throughput (Cell Downlink Average Throughput (DU)) decrease, the RRC connection setup success rate (RRC Setup Success Rate (CU)) may decrease, and the service drop rate may increase in the adjacent public-network cells.
Automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks	After this function is enabled, the average uplink and downlink throughputs of dedicated-network cells (Cell Uplink Average Throughput (DU) and Cell Downlink Average Throughput (DU)) may slightly decrease because the number of public-network cells performing interference coordination significantly decreases.
Beam avoidance for interference coordination between public and dedicated networks	After this function is enabled, the average downlink throughput (Cell Downlink Average Throughput (DU)) decreases, the RRC connection setup success rate (RRC Setup Success Rate (CU)) may decrease, and the service drop rate may increase in the adjacent public-network cells.
TRP-based precise interference coordination for interference coordination between public and dedicated networks	Compared to cell-level interference coordination between public and dedicated networks, this function decreases the number of public-network cells that take avoidance measures for the dedicated network and the avoidance duration. As a result, the uplink and downlink average cell throughputs (indicated by Cell Uplink Average Throughput (DU) and Cell Downlink Average Throughput (DU)) of the dedicated-network cell may decrease slightly, and the user-perceived rate of dedicated-network UEs decreases.
Beam-level power reduction for interference coordination between public and dedicated networks	Compared to beam avoidance or power reduction for interference coordination between public and dedicated networks, this function may increase the interference of the public network to the dedicated network. As a result, the uplink and downlink average cell throughputs (indicated by Cell Uplink Average Throughput (DU) and Cell Downlink Average Throughput (DU)) of the dedicated-network cell may decrease slightly, and the user-perceived rate of dedicated-network UEs decreases.

If public-network neighboring cells cause strong interference to the high-speed railway dedicated network, it is recommended that network optimization be performed first to reduce the interference. For example, you can adjust the azimuth, downtilt, transmit power, and site location of public-network cells to improve user experience on public and dedicated networks. As a supplement to RF optimization, interference coordination between public and dedicated networks can be enabled for interfering neighboring cells at sites where network optimization cannot be performed in the short term, thereby improving high-speed railway user experience. An interfering neighboring cell refers to a public-network neighboring cell whose signal strength difference to the dedicated network is less than 20 dB. Neighboring cells with strong interference can be automatically identified by enabling automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks, using a frequency scanner, or subscribing to external CHRs. After external CHR subscription, UEs will report RSRP measurement results. After interfering neighboring cells are identified, canceling the CHR subscription is recommended to reduce the impact on the network.

- If a cell of the high-speed railway dedicated network and its neighboring cells on the public network have the same SSB frequency, the following fields can be used to identify interfering neighboring cells:

The **SSBResults.RsrpRslt** field of the external CHR event
PERIOD_INTRA_FREQ_MEASUREMENT

The **NeighCellRst.SSBResults.RsrpRslt** field of the external CHR event
PERIOD_INTRA_FREQ_MEASUREMENT

- If a cell of the high-speed railway dedicated network and its neighboring cells on the public network have different SSB frequencies but their frequency bands overlap, the following fields can be used to identify interfering neighboring cells:

The **SSBResults.RsrpRslt** field of the external CHR event
PERIOD_INTER_FREQ_MEASUREMENT

The **NeighCellRst.SSBResults.RsrpRslt** field of the external CHR event
PERIOD_INTER_FREQ_MEASUREMENT

NOTE

The following describes the meanings of the fields in the external CHR event:

- **PhysCellId**: indicates the physical cell identifier of the serving cell or a neighboring cell on the public network.
- **SSBResults.RsrpRslt**: indicates the RSRP of the serving cell, which is an SSB beam-level measurement result.
- **NeighCellRst.SSBResults.RsrpRslt**: indicates the RSRP of a public-network neighboring cell, which is an SSB beam-level measurement result.

12.1.3 Requirements

12.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

12.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Interference coordination between public and dedicated networks requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED . The interference coordination always-on switch (controlled by the INTRF_COORD_ALWAYS_WORKING_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter) can be turned on for a maximum of three high-speed cells under the same base station.
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	Automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks requires that the HIGH_SPEED_INTRF_AVOID_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	For high-speed cells, beam avoidance for interference coordination between public and dedicated networks requires that the HIGH_SPEED_INTRF_AVOID_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter be selected.
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	TRP-based precise interference coordination for interference coordination between public and dedicated networks can be enabled only in high speed cells.
Low-frequency TDD	Hyper Cell Cell Combination	Cell Combination: NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE Hyper Cell: NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell Cell Combination</i>	TRP-based precise interference coordination for interference coordination between public and dedicated networks can be enabled only in Hyper Cell or Cell Combination networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Beam avoidance or automatic identification of interfering neighboring cells	INTERFERING_NCELL_IDENTITY_SW or HIGH_SPEED_BEAM_AVOID_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	TRP-based precise interference coordination for interference coordination between public and dedicated networks can be enabled only when the INTERFERING_NCELL_IDENTITY_SW or HIGH_SPEED_BEAM_AVOID_SW option is selected.
Low-frequency TDD	Automatic identification of interfering neighboring cells	INTERFERING_NCELL_IDENTITY_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	Beam-level power reduction for interference coordination between public and dedicated networks can be enabled only when the INTERFERING_NCELL_IDENTITY_SW option is selected.

12.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference Coord Always Working	INTRF_COORD_ALWAYS_WORKING_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	The HIGH_SPEED_INTRF_AVOID_SW and INTRF_COORD_ALWAYS_WORKING_SW options of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter cannot be selected at the same time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Configuration-based Scheduling for SSB and RMSI Interference Avoidance	CFG_SSB_RMSI_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.ComMChnIntrfAvoidSwitch parameter	<i>High Speed Mobility</i>	Configuration-based scheduling for SSB and RMSI interference avoidance is mutually exclusive with the interference coordination between public and dedicated networks function.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	When NR DU cell resource management between network slices is enabled, the following are true: - If the NRDUCellRes.DlMinRbStartPos parameter is set to 0, the NRDUCellHighSpeedMob.DlIntrfLimitRbSchRatio parameter must be set to 0. In this case, downlink RB restriction for the low-speed cell cannot start. - If the NRDUCellRes.UlMinRbStartPos parameter is set to 0, the NRDUCellHighSpeedMob.UlIntrfLimitRbSchRatio parameter must be set to 0. In this case, uplink RB restriction for the low-speed cell cannot start.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Interference coordination between public and dedicated networks is not compatible with Fusion Cell.
Low-frequency TDD	Beam avoidance	HIGH_SPEED_BEAM_AVOID_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	Beam-level power reduction for interference coordination between public and dedicated networks is mutually exclusive with beam avoidance for interference coordination between public and dedicated networks.
Low-frequency TDD	Power reduction for interfering beams	INTERFERING_BEAM_POWER_REDUCE option of the NRDUCellTmeIntrf.CrossLinkIntrfAvoidMethod parameter	None	Beam-level power reduction for interference coordination between public and dedicated networks is mutually exclusive with power reduction for interfering beams.
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	Downlink RB restriction for the low-speed cell cannot work with user PRB control for FWA.
Low-frequency TDD	TRP-based Precise Interference Coordination for Interference Coordination Between Public and Dedicated Networks	TRP_PRECISE_INTRF_COORD_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter.	<i>High Speed Mobility</i>	The TRP_PRECISE_INTRF_COORD_SW option of NRCCellAlgoSwitch.HighSpeedDedPolicySw and the INTRF_NCELL_IDENT_ENH_SW option of GNodebParam.HighSpeedDedPolicySw cannot be selected at the same time.

12.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power aggregation in unoccupied bandwidths	NRDUCellChnPwr.MaxPdschConvPwrOffset	<i>Power Control</i>	In the interference coordination between public and dedicated networks function, downlink power reduction in low-speed cells is used to reduce interference to high-speed cells. Therefore, when downlink power reduction in low-speed cells takes effect, power aggregation in unoccupied bandwidths does not take effect.
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	- LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter - NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	If both power reduction for low-speed cells in interference coordination between public and dedicated networks and LTE TDD and NR Flash Dynamic Power Sharing are enabled, the power sharing gains on the NR side may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When interference coordination between public and dedicated networks is enabled, the gains provided by NR inter-carrier dynamic power sharing may decrease.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	When interference coordination between public and dedicated networks is enabled and the NRDUCellHighSpeedMob.DlIntrfLimitRbSchRatio parameter is set to a non-zero value, PDSCH MU-MIMO does not take effect.
Low-frequency TDD	Inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When both configuration-based scheduling for SSB and RMSI interference avoidance and inter-frequency SSB interference avoidance are enabled, configuration-based scheduling for SSB and RMSI interference avoidance takes effect while inter-frequency SSB interference avoidance does not.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When both configuration-based scheduling for SSB and RMSI interference avoidance and enhanced inter-frequency SSB interference avoidance are enabled, configuration-based scheduling for SSB and RMSI interference avoidance takes effect while enhanced inter-frequency SSB interference avoidance does not.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When "interference coordination between public and dedicated networks" and NR multi-band power adaptive scheduling are both enabled, the gains provided by NR multi-band power adaptive scheduling may decrease.
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	LNR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE and NR multi-band power adaptive scheduling and "interference coordination between public and dedicated networks" are both enabled, there is a possibility that the benefits of power adaptive scheduling decrease on the NR side.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	- On the LTE side: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter - On the NR side: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When L/NR Inter-band Deep Coordination and "interference coordination between public and dedicated networks" are both enabled, the power sharing gains may decrease on the NR side.

12.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

12.1.3.4 Cells

- Beam avoidance for interference coordination between public and dedicated networks:
 - The public-network cells work in 32T32R or 64T64R mode. Specifically, for low-speed cells, beam avoidance for interference coordination between public and dedicated networks requires that the **NRDUCell.NrDuCellNetworkingMode** parameter be set to **NORMAL_CELL** and the **NRDUCellTrp.TxRxMode** parameter be set to **32T32R** or **64T64R**.
 - The frequency band of public-network cells must be N38, N40, N41, N77, N78, or N79. Specifically, for low-speed cells, the **NRDUCell.FrequencyBand** parameter must be set to **N38, N40, N41, N77, N78, or N79**.

- Beam-level power reduction for interference coordination between public and dedicated networks:
 - The public-network cells work in 32T32R, 64T64R, or 128T128R mode. Specifically, for low-speed cells, beam-level power reduction for interference coordination between public and dedicated networks requires that the **NRDUCell.NrDuCellNetworkingMode** parameter be set to **NORMAL_CELL** and the **NRDUCellTrp.TxRxMode** parameter be set to **32T32R**, **64T64R**, or **128T128R**.
- Other functions: no requirements

12.1.3.5 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

In SA networking, the "interference coordination between public and dedicated networks" function requires that the Xn interface state of gNodeBs be normal.

In NSA networking, the interference coordination between public and dedicated networks function requires that the X2 interface state of gNodeBs be normal.

In intra-frequency band networking of high-speed railway dedicated networks and public networks, "interference coordination between public and dedicated networks" requires that high-speed cells be configured with neighboring public-network cells. For details about neighbor relationship configuration, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

12.1.4 Operation and Maintenance

12.1.4.1 Data Configuration

12.1.4.1.1 Data Preparation

[Table 12-3](#) to [Table 12-7](#) describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 12-3 Parameters used for activation (RB restriction and power restriction for interference coordination between public and dedicated networks)

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option.

Parameter Name	Parameter ID	Setting Notes
High Speed Interference Avoidance Flag	NRCellRelation.HighSpeedIntrfAvoidFlag	When interference coordination for public and dedicated networks is enabled, only neighboring cells with this function enabled can participate in interference coordination for public and dedicated networks.
DL Intrf Based Limited RB Sch Ratio	NRDUCellHighSpeedMob.DLIntrfLimitRbSchRatio	When interference coordination between public and dedicated networks is enabled, this parameter specifies the proportion of downlink RBs that cannot be scheduled when a train has arrived in a neighboring public-network cell of the dedicated network. The value 75 is recommended. ^a
UL Intrf Based Limited RB Sch Ratio	NRDUCellHighSpeedMob.UlIntrfLimitRbSchRatio	When interference coordination between public and dedicated networks is enabled, this parameter specifies the proportion of uplink RBs that cannot be scheduled when a train has arrived in a neighboring public-network cell of the dedicated network. The value 75 is recommended. ^a
DL Intrf Based Power Offset	NRDUCellHighSpeedMob.DLIntrfBasedPwrOffset	When interference coordination between public and dedicated networks is enabled, this parameter specifies the amount of downlink power decrease when a train has arrived in a neighboring public-network cell of the dedicated network. The value 3 is recommended. ^a
UL Intrf Based Power Offset	NRDUCellHighSpeedMob.UlIntrfBasedPwrOffset	When interference coordination between public and dedicated networks is enabled, this parameter specifies the amount of uplink power decrease when a train has arrived in a neighboring public network cell of the dedicated network. The value 3 is recommended. ^a
High Speed User Num Thld	NRDUCellHighSpeedMob.HighSpeedUserNumThld	Set this parameter when the HIGH_SPEED_INTRF_AVOID_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter is selected. Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
a: To reduce the impact on neighboring public-network cells after this feature is enabled, it is recommended that parameters related to RB restriction and power reduction be set based on the ISD or signal strength difference for neighboring public-network cells near the high-speed railway. For details, see 12.1.2.2 Impacts .		

Table 12-4 Parameters used for activation (automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks)

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the INTERFERING_NCELL_IDENT_SW option.
High Speed Interference Avoidance Flag	NRCCellRelation.HighSpeedIntrfAvoidFlag	When interference coordination for public and dedicated networks is enabled, only neighboring cells with this function enabled can participate in interference coordination for public and dedicated networks.
High Speed User Num Thld	NRDUCellHighSpeedMob.HighSpeedUserNumThld	Use the recommended value.
High Speed Dedicated Policy Switch	gNodeBParam.HighSpeedDedPolicySw	It is recommended that the INTRF_NCELL_IDENT_ENH_SW option be selected when interference coordination between public and dedicated networks is enabled in multiple dedicated-network cells served by the same dedicated-network base station.

Table 12-5 Parameters used for activation (beam avoidance for interference coordination between public and dedicated networks)

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the HIGH_SPEED_BEAM_AVOID_SW option.

Parameter Name	Parameter ID	Setting Notes
High Speed Interference Avoidance Flag	NRCellRelation.HighSpeedIntrfAvoidFlag	When interference coordination for public and dedicated networks is enabled, only neighboring cells with this function enabled can participate in interference coordination for public and dedicated networks.
High Speed User Num Thld	NRDUCellHighSpeedMob.HighSpeedUserNumThld	Use the recommended value.
High Speed Dedicated Policy Switch	gNodeBParam.HighSpeedDedPolicySw	It is recommended that the INTRF_NCELL_IDENT_ENH_SW option be selected when interference coordination between public and dedicated networks is enabled in multiple dedicated-network cells served by the same dedicated-network base station.

Table 12-6 Parameters used for activation (TRP-based precise interference coordination for interference coordination between public and dedicated networks)

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCellAlgoSwitch.HighSpeedDedPolicySw	Select the TRP_PRECISE_INTRF_COORD_SW option.
Interfering Cell Ident SSB RSRP Threshold	NRCellMeasConfig.IntrfCellIdentSsbRsrpThld	Use the recommended value.

Table 12-7 Parameters used for activation (beam-level power reduction for interference coordination between public and dedicated networks)

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCellAlgoSwitch.HighSpeedDedPolicySw	Select the INTRF_COORD_BEAM_PWR_REDUCE_SW option.

Parameter Name	Parameter ID	Setting Notes
Interfering Cell Ident SSB RSRP Threshold	NRCellMeasConfig.IntrfCellIdentSsbRsrpThld	Use the recommended value. This parameter can be set on both the public and dedicated networks.
High Speed Dedicated Policy Switch	gNodeBParam.HighSpeedDedPolicySw	It is recommended that the INTRF_NCELL_IDENT_ENH_SW option be selected when interference coordination between public and dedicated networks is enabled in multiple dedicated-network cells served by the same dedicated-network base station.

12.1.4.1.2 Using MML Commands

Before using MML commands, refer to [12.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- RB restriction and power restriction for interference coordination between public and dedicated networks

Before enabling interference coordination between public and dedicated networks, ensure that parameters such as those related to frequency relationships and neighbor relationships with low-speed cells are correctly configured on the dedicated network. Otherwise, this function will not take effect.

In high-speed NR TDD scenarios:

```
//In the high-speed cell, configuring the operating frequency of a low-speed NR cell as a neighboring frequency and setting the low-speed flag for this frequency
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN, FrequencyBand=N41, SubcarrierSpacing=30KHZ, ConnFreqPriority=1,HighSpeedFlag=LOW_SPEED;
//If a public-network cell has not been configured as an external cell, configuring this cell as an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=0, CellId=0, PhysicalCellId=0, Tac=1, SsbFreqPos=6400, HighSpeedFlag=LOW_SPEED;
//If a neighbor relationship with a public-network cell has not been configured, configuring the neighbor relationship with this cell and the interference avoidance flag
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=0, HighSpeedIntrfAvoidFlag=INTRF_AVOID;
//Configuring the proportion of RBs that cannot be scheduled when a train has arrived for neighboring public-network cells
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, DIIntrfLimitRbSchRatio=75, ULIIntrfLimitRbSchRatio=75;
//Configuring the power reduction offset when a train has arrived for neighboring public-network cells
```

```
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, DlIntrfBasedPwrOffset=3, UlIntrfBasedPwrOffset=3;
//Configuring the threshold for the number of high-speed UEs only when the interference
coordination between public and dedicated networks function is enabled
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedUserNumThld=10;
```

- Method 1: Turning on the high-speed cell interference coordination switch for the high-speed cell

```
//The high-speed cell interference coordination switch and the interference coordination always-on
switch are mutually exclusive. They correspond to different function processing mechanisms.
MOD NRCELLALGOSWITCH: NrCellId=0,
HighSpeedDedPolicySw=HIGH_SPEED_INTRF_AVOID_SW-1;
```

- Method 2: Turning on the interference coordination always-on switch for the high-speed cell

```
//The high-speed cell interference coordination switch and the interference coordination always-on
switch are mutually exclusive. They correspond to different function processing mechanisms.
MOD NRCELLALGOSWITCH: NrCellId=0,
HighSpeedDedPolicySw=INTRF_COORD_ALWAYS_WORKING_SW-1;
```

- Automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks

Before enabling this function, check the following items:

(a) On the dedicated network, the operating frequencies of NR public-network cells and the low-speed flag for the frequencies are correctly configured, and the measurement activation flag is set to active.

(b) All public-network cells are configured as external cells, and neighbor relationships with these public-network cells are configured as well.

In high-speed NR TDD scenarios:

```
//If the operating frequency of a public-network cell has not been configured as a neighboring
frequency, configuring this frequency as a neighboring frequency, setting the high speed flag to low
speed, and setting the measurement activation flag to active
ADD NRCELLFREQLATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N41, SubcarrierSpacing=30KHZ,
ConnFreqPriority=1, MeasActvFlag=ACTIVE, HighSpeedFlag=LOW_SPEED;
//If a cell on the preceding public-network low-speed neighboring frequency has not been configured
as an external cell, configuring this cell as an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=0, CellId=0, PhysicalCellId=0, Tac=1,
SsbFreqPos=6400, HighSpeedFlag=LOW_SPEED;
//If a neighbor relationship with a cell on the preceding public-network low-speed neighboring
frequency has not been configured, configuring the neighbor relationship with this cell and the
interference avoidance flag
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
HighSpeedIntrfAvoidFlag=INTRF_AVOID;
//Configuring HighSpeedUserNumThld
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedUserNumThld=10;
//Configuring IntrfCellIdentSsbRsrpThld
MOD NRCELLMEASCONFIG: NrCellId=0, IntrfCellIdentSsbRsrpThld=-110;
//Turning on INTERFERING_NCELL_IDENT_SW for the high-speed cell
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=INTERFERING_NCELL_IDENT_SW-1;
//(Recommended) Turning on INTRF_NCELL_IDENT_ENH_SW when interference coordination
between public and dedicated networks is enabled for multiple dedicated-network cells served by the
same dedicated-network base station
MOD GNODEBPARAM: HighSpeedDedPolicySw=INTRF_NCELL_IDENT_ENH_SW-1;
```

- Beam avoidance for interference coordination between public and dedicated networks

Before enabling this function, check the following items:

(a) On the dedicated network, the operating frequencies of NR public-network cells and the low-speed flag for the frequencies are correctly configured, and the measurement activation flag is set to active.

(b) All public-network cells are configured as external cells, and neighbor relationships with these public-network cells are configured as well.

(c) The Xn interface is set up, and the Xn link is normal. The Xn interface can be manually configured or automatically configured by enabling Xn self-configuration. When enabling Xn self-configuration, make sure that the **NoHoFlag** parameter is set to **PERMIT_HO** for the neighbor relationships. For details about how to configure the Xn interface, see *NG and Xn Self-Management*.

```
//On the public network side: If the operating frequency of a public-network cell has not been
configured as a neighboring frequency, configuring this frequency as a neighboring frequency, setting
the high speed flag to low speed, and setting the measurement activation flag to active
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N41, SubcarrierSpacing=30KHZ,
ConnFreqPriority=1, MeasActvFlag=ACTIVE, HighSpeedFlag=LOW_SPEED;
//On the dedicated network side: If a public-network cell has not been configured as an external cell,
configuring this cell as an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=0, CellId=0, PhysicalCellId=0, Tac=1,
SsbFreqPos=6400, HighSpeedFlag=LOW_SPEED;
//On the dedicated network side: If a neighbor relationship with a public-network cell has not been
configured, configuring the neighbor relationship with this cell and the interference avoidance flag.
When enabling Xn self-configuration, make sure that the NoHoFlag parameter is set to PERMIT_HO.
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
NoHoFlag=PERMIT_HO, HighSpeedIntrfAvoidFlag=INTRF_AVOID;
//Configuring HighSpeedUserNumThld
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedUserNumThld=10;
//Configuring IntrfCellIdSsbRsrpThld
MOD NRCELLMEASCONFIG: NrCellId=0, IntrfCellIdSsbRsrpThld=-110;
//On the dedicated network side: Turning on HIGH_SPEED_BEAM_AVOID_SW
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_BEAM_AVOID_SW-1;
// (Recommended) Turning on INTRF_NCELL_IDENT_ENH_SW when interference coordination
between public and dedicated networks is enabled for multiple dedicated-network cells served by the
same dedicated-network base station
MOD GNODEBPARAM: HighSpeedDedPolicySw=INTRF_NCELL_IDENT_ENH_SW-1;
// (Optional) One day later, performing the following operations to obtain more gains:
//On the dedicated network side: To prevent high-speed UEs from being handed over to the public
network, setting the NoHoFlag parameter to FORBID_HO for the neighbor relationship
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
NoHoFlag=FORBID_HO;
//On the public network side: Turning on the high-speed beam avoidance switch for low-speed cells
that are normal cells and work in 32T32R or 64T64R mode
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_BEAM_AVOID_SW-1;
```

- TRP-based precise interference coordination for interference coordination between public and dedicated networks:

```
//Turning on the TRP_PRECISE_INTRF_COORD_SW on the dedicated network when automatic
identification of interfering neighboring cells or beam avoidance takes effect on the dedicated network
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=TRP_PRECISE_INTRF_COORD_SW-1;
```

- Beam-level power reduction for interference coordination between public and dedicated networks:

Before enabling this function, check the following items:

(a) On the dedicated network, the operating frequencies of NR public-network cells and the low-speed flag for the frequencies are correctly configured, and the measurement activation flag is set to active.

(b) All public-network cells are configured as external cells, and neighbor relationships with these public-network cells are configured as well.

(c) The Xn interface is set up, and the Xn link is normal. The Xn interface can be manually configured or automatically configured by enabling Xn self-configuration. When enabling Xn self-configuration, make sure that the **NoHoFlag** parameter is set to **PERMIT_HO** for the neighbor relationships. For details about how to configure the Xn interface, see *NG and Xn Self-Management*.

```
//On the public network side: If the operating frequency of a public-network cell is not configured as
a neighboring frequency, configuring this frequency as a neighboring frequency, setting the high
```

```
speed flag to low speed, and setting the measurement activation flag to active
ADD NRCELLFREERELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N41, SubcarrierSpacing=30KHZ, ConnFreqPriority=1, MeasActvFlag=ACTIVE,
HighSpeedFlag=LOW_SPEED;
//On the dedicated network side: If a public-network cell has not been configured as an external cell,
configuring this cell as an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=0, CellId=0, PhysicalCellId=0, Tac=1,
SsbFreqPos=6400, HighSpeedFlag=LOW_SPEED;
//On the dedicated network side: If a neighbor relationship with a public-network cell has not been
configured, configuring the neighbor relationship with this cell and the interference avoidance flag.
When enabling Xn self-configuration, make sure that the NoHoFlag parameter is set to PERMIT_HO.
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
NoHoFlag=PERMIT_HO, HighSpeedIntrfAvoidFlag=INTRF_AVOID;
//On the dedicated network side: Configuring HighSpeedUserNumThld
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedUserNumThld=10;
//On the dedicated network side: Configuring IntrfCellIdentSsbRsrpThld
MOD NRCELLMEASCONFIG: NrCellId=0, IntrfCellIdentSsbRsrpThld=-110;
//On the dedicated-network side: Turning on INTRF_COORD_BEAM_PWR_REDUCE_SW for the high-
speed cell
MOD NRCELLALGOSWITCH: NrCellId=0,
HighSpeedDedPolicySw=INTRF_COORD_BEAM_PWR_REDUCE_SW-1;
//(Recommended) Turning on INTRF_NCELL_IDENT_ENH_SW when interference coordination
between public and dedicated networks is enabled for multiple dedicated-network cells served by the
same dedicated-network base station
MOD GNODEBPARAM: HighSpeedDedPolicySw=INTRF_NCELL_IDENT_ENH_SW-1;
//(Optional) One day later, performing the following operations to obtain more gains:
//On the dedicated network side: To prevent high-speed UEs from being handed over to the public
network, setting the NoHoFlag parameter to FORBID_HO for the neighbor relationship
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
NoHoFlag=FORBID_HO;
//On the public network side: Configuring IntrfCellIdentSsbRsrpThld
MOD NRCELLMEASCONFIG: NrCellId=1, IntrfCellIdentSsbRsrpThld=-110;
//On the public network side: Turning on INTRF_COORD_BEAM_PWR_REDUCE_SW
MOD NRCELLALGOSWITCH: NrCellId=1,
HighSpeedDedPolicySw=INTRF_COORD_BEAM_PWR_REDUCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deactivation command examples for interference coordination between public and dedicated networks (RB restriction and power restriction)
//Turning off INTRF_COORD_ALWAYS_WORKING_SW if it has been turned on
MOD NRCELLALGOSWITCH: NrCellId=0,
HighSpeedDedPolicySw=INTRF_COORD_ALWAYS_WORKING_SW-0;
//Turning off HIGH_SPEED_INTRF_AVOID_SW if it has been turned on
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_INTRF_AVOID_SW-0;
- Deactivation command examples for interference coordination between public and dedicated networks (automatic identification of interfering neighboring cells)
//Turning off INTRF_NCELL_IDENT_ENH_SW
MOD GNODEBPARAM: HighSpeedDedPolicySw=INTRF_NCELL_IDENT_ENH_SW-0;
//Turning off INTERFERING_NCELL_IDENT_SW
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=INTERFERING_NCELL_IDENT_SW-0;
- Deactivation command examples for interference coordination between public and dedicated networks (beam avoidance)
//Turning off INTRF_NCELL_IDENT_ENH_SW
MOD GNODEBPARAM: HighSpeedDedPolicySw=INTRF_NCELL_IDENT_ENH_SW-0;
//Turning off HIGH_SPEED_BEAM_AVOID_SW
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_BEAM_AVOID_SW-0;
//Turning off the high-speed beam avoidance switch for low-speed cells that are normal cells and work in 32T32R or 64T64R mode
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_BEAM_AVOID_SW-0;

- Deactivation command examples for interference coordination between public and dedicated networks (TRP-based precise interference coordination)
//Turning off the TRP-based Precise Intrf Coord Sw
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=TRP_PRECISE_INTRF_COORD_SW-0;
- Deactivation command examples for interference coordination between public and dedicated networks (beam-level power reduction)
//Turning off INTRF_NCELL_IDENT_ENH_SW
MOD GNODEBPARAM: HighSpeedDedPolicySw=INTRF_NCELL_IDENT_ENH_SW-0;
//On the dedicated network side: Turning off INTRF_COORD_BEAM_PWR_REDUCE_SW
MOD NRCELLALGOSWITCH: NrCellId=1,
HighSpeedDedPolicySw=INTRF_COORD_BEAM_PWR_REDUCE_SW-0;
//On the public network side: Turning off INTRF_COORD_BEAM_PWR_REDUCE_SW
MOD NRCELLALGOSWITCH: NrCellId=1,
HighSpeedDedPolicySw=INTRF_COORD_BEAM_PWR_REDUCE_SW-0;

12.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.1.4.2 Activation Verification

- RB restriction and power restriction and automatic identification of interfering neighboring cells for interference coordination between public and dedicated networks:
After this function is enabled, check the value of **N.InterfHighSpeed.Trigger.Num**, which indicates the number of times interference coordination is triggered between high-speed railway public and dedicated networks. If the value of this counter is not 0, this function has taken effect.
Effective duration of interference coordination between public and dedicated networks $\approx$ **N.InterfHighSpeed.Trigger.Num** x 2s
- Beam avoidance for interference coordination between public and dedicated networks:
Observe the value of **N.InterfHighSpeed.BeamAvoid.Dur** to check whether the beam avoidance function has taken effect. If the value of this counter is not 0, this function has taken effect.
- TRP-based precise interference coordination for interference coordination between public and dedicated networks:
After this function is enabled, if "RB restriction and power restriction for interference coordination between public and dedicated networks" has taken effect, you can check whether this function has taken effect by observing the value of the **N.InterfHighSpeed.Trigger.Num** counter; if beam avoidance for interference coordination between public and dedicated networks has taken effect, you can check whether this function has taken effect by observing the value of the **N.InterfHighSpeed.BeamAvoid.Dur** counter. If the value of either of the preceding counters decreases, this function has taken effect.
- Beam-level power reduction for interference coordination between public and dedicated networks:
After this function is enabled, check the value of **N.InterfHighSpeed.Trigger.Num**, which indicates the number of times

interference coordination is triggered between high-speed railway public and dedicated networks. If the counter value is not zero, this function has taken effect.

12.1.4.3 Network Monitoring

The indicators listed in [12.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

12.2 Configuration-based Scheduling for SSB and RMSI Interference Avoidance

12.2.1 Principles

When the spectrum resources of a public-network cell and a dedicated-network cell overlap and their SSB frequency-domain positions are different, the public-network SSB and RMSI interfere with the PDSCH of the dedicated-network cell in the overlapping bandwidth.

This function applies to the scenario where the Xn link between the public and dedicated networks is disconnected, the spectrum resources of the public-network cell and dedicated-network cell overlap, and their SSB frequency-domain positions are different. The victim dedicated-network cell cannot obtain the configuration information of the interfering public-network cell through the Xn interface. Therefore, configuration-based scheduling for SSB and RMSI interference avoidance is introduced. The dedicated-network cell provides parameter configuration for the SSB or RMSI information of the interfering public-network cell so that SSB and RMSI interference avoidance can be implemented for the dedicated-network cell.

This function is controlled by the **CFG_SSB_RMSI_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**. *CommChnIntrfAvoidSwitch* parameter. If this option is selected and the **NRDUCellIntrfAvoid**.*IntrfCellIntrfResType* parameter is set to **SSB** or **RMSI** for an interfering cell, configuration-based scheduling for SSB or RMSI interference avoidance is supported in the high-speed cell. This function helps UEs in the high-speed cell avoid certain time-domain and frequency-domain positions for the SSB and RMSI.

When the interference resource type of the interfering cell is SSB or RMSI, the time-frequency domain information of the SSB or RMSI is configured using the following parameters:

- The **NRDUCellIntrfAvoid**.*IntrfCellIntrfSlot* parameter specifies the time-domain position of the SSB or RMSI of the interfering cell.
- The **NRDUCellIntrfAvoid**.*IntrfCellIntrfStartRb* parameter specifies the frequency-domain start RB of the interfering cell.
- The **NRDUCellIntrfAvoid**.*IntrfCellIntrfEndRb* parameter specifies the frequency-domain end RB of the interfering cell.

The time-frequency domain information of the SSB and RMSI can be obtained from the operator or based on the live network configuration and 3GPP specifications.

- For details about the time-frequency domain information of the SSB, see 3GPP TS 38.331 or PBCH channel management in *Channel Management*. The SSB frequency-domain size is 20 RBs.
- For details about the time-frequency domain information of the RMSI, see 3GPP TS 38.213.

12.2.2 Network Analysis

12.2.2.1 Benefits

After configuration-based scheduling for SSB and RMSI interference avoidance is enabled, the increase in the average downlink throughput of high-speed railway dedicated-network cells depends on the number of public-network cells that cause SSB and RMSI interference to dedicated-network cells and the severity of SSB and RMSI interference to dedicated-network cells. The stronger the SSB and RMSI interference and the larger the number of interfering public-network cells, the greater the increase in the average downlink throughput of high-speed railway dedicated-network cells.

Configuration-based scheduling for SSB and RMSI interference avoidance delivers benefits only when the following conditions are met:

- It is recommended that this function be enabled when spectrum resources overlap between public-network cells and dedicated-network cells, their SSB frequency-domain positions are different, and no Xn links are set up between public and dedicated networks (for example, the public network and dedicated network use devices from different vendors).
- The dedicated-network cell is interfered by the SSB of public-network cells. An interfering public-network cell refers to the public-network cell whose signal strength difference between the dedicated-network cell and the public-network cell is less than 5 dB along the high-speed railway. An interfering public-network cell can be identified using a scanner.
- The number of interfering public-network cells must be greater than the number of TRPs in the dedicated-network cell.
- The time-frequency domain positions of SSBs and RMSIs of interfering public-network cells can be accurately configured on the dedicated-network cell.
- The working frequency band, subcarrier spacing, SSB period, and RMSI period of the public-network cells must be consistent with those of the dedicated-network cell. If they are inconsistent, manually configure them to be consistent.

12.2.2.2 Impacts

The impacts between this function and other functions are described in [12.2.3.2.3 Function Impacts](#).

Configuration-based scheduling for SSB and RMSI interference avoidance reduces the time-frequency resources available for downlink scheduling in high-speed dedicated-network cells. Enabling this function decreases the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) on the high-speed railway dedicated network if either of the following conditions is met before this function is enabled: The SSB and RMSI time-frequency domain information of the

interfering cell is incorrectly configured and the dedicated-network cell is not interfered by the SSB and RMSI of the public-network cell; the number of interfering public-network cells is less than the number of TRPs of the dedicated-network cell.

12.2.3 Requirements

12.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

12.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Configuration-based scheduling for SSB and RMSI interference avoidance requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

12.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	Configuration-based scheduling for SSB and RMSI interference avoidance is mutually exclusive with the interference coordination between public and dedicated networks function.

12.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When both configuration-based scheduling for SSB and RMSI interference avoidance and inter-frequency SSB interference avoidance are enabled, configuration-based scheduling for SSB and RMSI interference avoidance takes effect while inter-frequency SSB interference avoidance does not.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.ComMChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When both configuration-based scheduling for SSB and RMSI interference avoidance and enhanced inter-frequency SSB interference avoidance are enabled, configuration-based scheduling for SSB and RMSI interference avoidance takes effect while enhanced inter-frequency SSB interference avoidance does not.

None

12.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

12.2.3.4 Cells

TDD: In high-speed scenarios, the **NRDUCell.SlotAssignment** parameter needs to be set to **8_2_DDDDDDDSUU** when "configuration-based scheduling for SSB and RMSI interference avoidance " is enabled.

12.2.3.5 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

12.2.4 Operation and Maintenance

12.2.4.1 Data Configuration

12.2.4.1.1 Data Preparation

[Table 12-8](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 12-8 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch	Select the CFG_SSB_RMSI_INTRF_AVOID_SW option.
Interfering Cell Intrf Band Index	NRDUCellIntrfAvoidId.IntrfCellIntrfBandIndex	Set this parameter to 0 .
Interfering Cell Intrf Resource Type	NRDUCellIntrfAvoidId.IntrfCellIntrfResType	It is recommended that this parameter be set to SSB or RMSI .
Interfering Cell Intrf Slot	NRDUCellIntrfAvoidId.IntrfCellIntrfSlot	It is recommended that this parameter be set based on the time-domain position of the SSB or RMSI of the interfering cell.
Interfering Cell Intrf Start RB	NRDUCellIntrfAvoidId.IntrfCellIntrfStartRb	It is recommended that this parameter be set based on the frequency-domain position of the SSB or RMSI of the interfering cell.
Interfering Cell Intrf End RB	NRDUCellIntrfAvoidId.IntrfCellIntrfEndRb	It is recommended that this parameter be set based on the frequency-domain position of the SSB or RMSI of the interfering cell.

12.2.4.1.2 Using MML Commands

Before using MML commands, refer to [12.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Adding the SSB time-frequency information of the interfering cell
ADD NRDUCELLINTRFAVOID: NrDuCellId=0, IntrfCellIntrfBandIndex=0, IntrfCellIntrfResType=SSB,
IntrfCellIntrfSlot=15, IntrfCellIntrfStartRb=100, IntrfCellIntrfEndRb=119;
//Adding the RMSI time-frequency information of the interfering cell
ADD NRDUCELLINTRFAVOID: NrDuCellId=0, IntrfCellIntrfBandIndex=0, IntrfCellIntrfResType=RMSI,
IntrfCellIntrfSlot=15, IntrfCellIntrfStartRb=86, IntrfCellIntrfEndRb=133;
//Enabling configuration-based scheduling for SSB and RMSI interference avoidance
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
CommChnIntrfAvoidSwitch=CFG_SSB_RMSI_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling configuration-based scheduling for SSB and RMSI interference avoidance  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
CommChnIntrfAvoidSwitch=CFG_SSB_RMSI_INTRF_AVOID_SW-0;
```

12.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.2.4.2 Activation Verification

After this function is enabled, if the number of downlink RBs scheduled for UEs performing full buffer services decreases, this function has taken effect.

12.2.4.3 Network Monitoring

The indicators listed in [12.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

13 Mobility Enhancement Policy Management

13.1 High-Speed UE Directional Handover

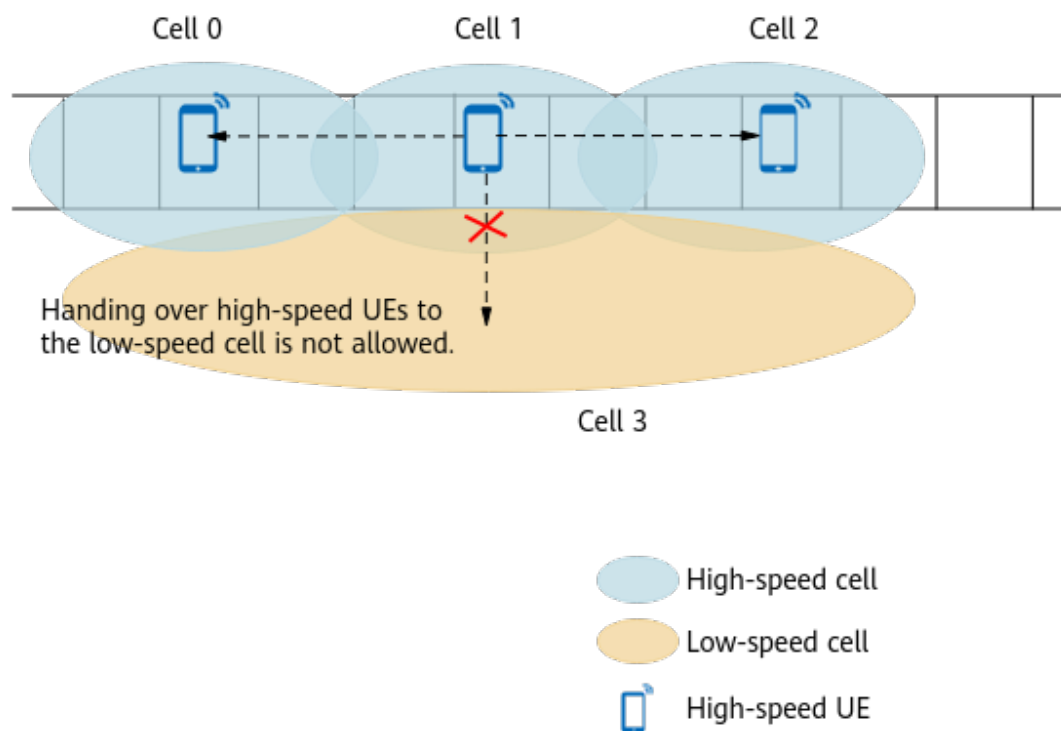
13.1.1 Principles

With the high-speed UE directional handover function, high-speed UEs in a high-speed railway dedicated-network cell with two or more neighboring high-speed cells can only be handed over to high-speed cells. This prevents high-speed UEs from being handed over to low-speed cells, improving their handover success rate. For details about the handover mechanisms in other scenarios, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

As shown in [Figure 13-1](#), when UEs in a high-speed railway dedicated network meet both conditions listed below and the high-speed UE directional handover function is enabled, high-speed UEs in a local cell can be handed over only to high-speed cells in high-speed railway dedicated network (that is, cell 0 and cell 2):

- The local cell serving high-speed UEs is configured with at least two high-speed intra-frequency neighboring cells. An intra-base-station neighboring cell of the local cell is qualified if the **NRDUCell.HighSpeedFlag** parameter is set to **HIGH_SPEED**. An external cell of the local cell is qualified if the corresponding **NRExternalNCell.HighSpeedFlag** parameter is set to **HIGH_SPEED**.
- High-speed UEs meet the conditions for coverage-based intra-frequency and inter-frequency handovers.

Figure 13-1 High-speed UE directional handover



This function is controlled by the **HIGH_SPEED_DIRECTIONAL_HO_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter.

NOTE

For details about neighbor relationship configuration and triggering conditions of coverage-based intra- and inter-frequency handovers, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

13.1.2 Network Analysis

13.1.2.1 Benefits

High-speed UE directional handover increases the handover success rate of high-speed UEs.

NOTE

Handover success rate of high-speed UEs: **Intra-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)**

13.1.2.2 Impacts

The impacts between this function and other functions are described in [13.1.3.2.3 Function Impacts](#).

This function has no impact.

13.1.3 Requirements

13.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

13.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	The high-speed UE directional handover function requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

13.1.3.2.2 Mutually Exclusive Functions

None

13.1.3.2.3 Function Impacts

None

13.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

13.1.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

13.1.4 Operation and Maintenance

13.1.4.1 Data Configuration

13.1.4.1.1 Data Preparation

Table 13-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 13-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the HIGH_SPEED_DIRECTIONAL_HO_SW option to enable high-speed UE directional handover.
High Speed Flag	NRExternalNCell <i>HighSpeedFlag</i>	This parameter must be configured when the high-speed UE directional handover function is enabled and neighbor relationships are configured for the cell. Set this parameter to HIGH_SPEED for high-speed neighboring cells and LOW_SPEED for low-speed neighboring cells.

13.1.4.1.2 Using MML Commands

Before using MML commands, refer to **13.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```

// (Optional) For NR TDD, configuring the high speed flag for neighboring cells
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, PhysicalCellId=0, Tac=0,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6400, NrNetworkingOption=UNLIMITED,
HighSpeedFlag=HIGH_SPEED;
// Turning on the high-speed UE directional handover switch
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_DIRECTIONAL_HO_SW-1;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```

// Turning off the high-speed UE directional handover switch
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_DIRECTIONAL_HO_SW-0;

```

13.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.1.4.2 Activation Verification

After this function is enabled, observe the handover success rates of high-speed UEs (**Intra-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)**) and the RRC connection reestablishment rate to check whether this function has taken effect. If the high-speed UE directional handover function takes effect, the handover success rates increase and the RRC connection reestablishment rate decreases.

13.1.4.3 Network Monitoring

The indicators listed in [13.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

13.2 Inter-Frequency Directional Handover for High-Speed UEs

13.2.1 Principles

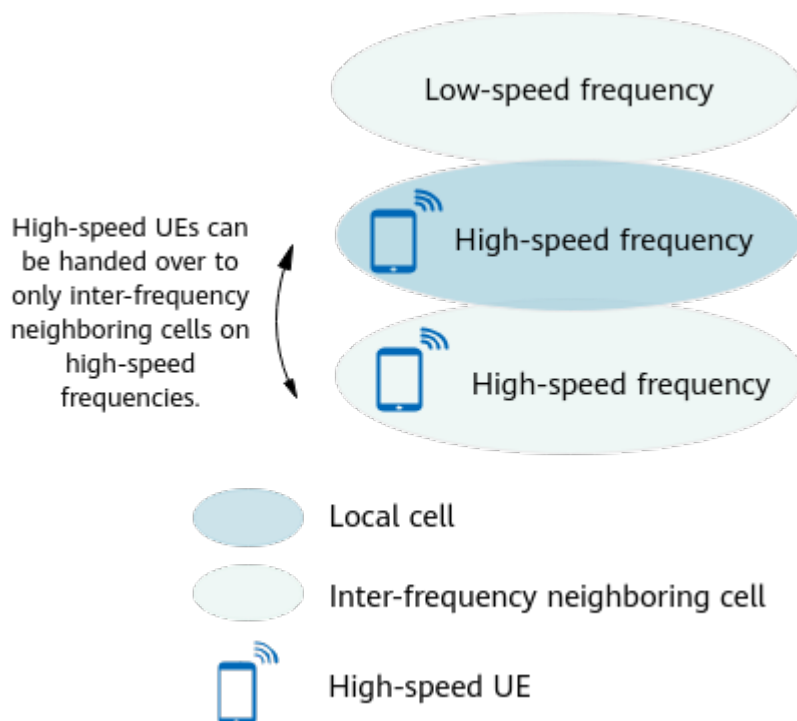
With the "inter-frequency directional handover for high-speed UEs" function, high-speed UEs in a cell on a high-speed-railway dedicated network (that is, high-speed cell) are not allowed to be handed over to inter-frequency neighboring cells on frequencies for low speed mobility. This prevents high-speed UEs from being handed over to low-speed cells. For details about the handover mechanisms in other scenarios, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

Before enabling this function, ensure that the local cell is configured with inter-frequency neighboring cells and at least one of operating frequencies of these neighboring cells is configured as non-low-speed neighboring frequency through the **NRCellFreqRelation.HighSpeedFlag** parameter. If this parameter is set to **HIGH_SPEED** or **NOT_CONFIG**, the frequency is not a low-speed frequency. If this parameter is set to **LOW_SPEED**, the frequency is not a low-speed frequency.

After "inter-frequency directional handover for high-speed UEs" is enabled, as shown in [Figure 13-2](#), the base station delivers measurement configurations only to operating frequencies of neighboring cells for which the **NRCellFreqRelation.HighSpeedFlag** parameter is set to **NOT_CONFIG** or **HIGH_SPEED** not **LOW_SPEED** when high-speed UEs on a high-speed-railway dedicated network meet the triggering conditions of inter-frequency handovers. In this way, handovers to inter-frequency neighboring cells working on only non-low-speed frequencies are performed.

This function is controlled by the **HS_INTER_FREQ_DIR_HO_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter.

Figure 13-2 Inter-frequency directional handover for high-speed UEs



13.2.2 Network Analysis

13.2.2.1 Benefits

After "inter-frequency directional handover for high-speed UEs" is enabled, the following are true:

- Only inter-frequency handovers to frequencies for high speed mobility are allowed, which prevents handovers to frequencies for low speed mobility. This improves the user experience of high-speed UEs and may increase the **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**.
- For high-speed UEs in the downlink CA state, if all non-low-speed neighboring frequencies have been added as SCCs, gap-assisted measurements are not started on added SCCs. This reduces the gap-assisted measurement overhead and improves the user experience of high-speed UEs.

13.2.2.2 Impacts

The impacts between this function and other functions are described in [13.2.3.2.3 Function Impacts](#).

After "inter-frequency directional handover for high-speed UEs" is enabled, the following are true:

- In scenarios where high-speed railways and highways run parallel, if UEs on highway are incorrectly identified as high-speed UEs and high-speed UEs change to low-speed UEs, the UEs may fail to be handed over to low-speed cells because the base station does not deliver neighboring frequencies for which the **NRCellFreqRelation.HighSpeedFlag** parameter set to **LOW_SPEED** for measurement. As a result, the handover failure rate and service drop rate increase. Therefore, you are advised not to enable "inter-frequency directional handover for high-speed UEs" in the preceding scenarios.
- In scenarios where high-speed trains decelerate in urban areas, if the UE attribute changes from high speed to low speed due to train deceleration and weak coverage occurs on the high-speed-railway dedicated network, high-speed UEs may fail to be handed over to low-speed cells because the base station does not deliver neighboring frequencies for which the **NRCellFreqRelation.HighSpeedFlag** parameter is set to **LOW_SPEED** for measurement. As a result, the handover failure rate and service drop rate increase. Therefore, you are advised not to enable "inter-frequency directional handover for high-speed UEs" in the weak coverage scenarios.

13.2.3 Requirements

13.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

13.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	The "inter-frequency directional handover for high-speed UEs" function requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

13.2.3.2.2 Mutually Exclusive Functions

None

13.2.3.2.3 Function Impacts

None

13.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

13.2.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

13.2.4 Operation and Maintenance

13.2.4.1 Data Configuration

13.2.4.1.1 Data Preparation

[Table 13-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 13-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Extend Switch	NRCeIlMobilityCo nfig.InterFreqHoE xtSwitch	To enable "inter-frequency directional handover for high-speed UEs", select the HS_INTER_FREQ_DIR_HO_SW option.

13.2.4.1.2 Using MML Commands

Before using MML commands, refer to [13.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling inter-frequency directional handover for high-speed UEs  
MOD NRCeIlMOBILITYCoNFIG: NrCellId=0, InterFreqHoExtSwitch=HS_INTER_FREQ_DIR_HO_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling inter-frequency directional handover for high-speed UEs
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=HS_INTER_FREQ_DIR_HO_SW-0;
```

13.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.2.4.2 Activation Verification

If low-speed frequencies are configured, check the measurement frequencies delivered in the RRC Reconfiguration message sent by the gNodeB. If the gNodeB delivers only the neighboring frequencies for which the **NRCellFreqRelation.HighSpeedFlag** parameter is set to **NOT_CONFIG** or **HIGH_SPEED** but not **LOW_SPEED**, this function has taken effect.

13.2.4.3 Network Monitoring

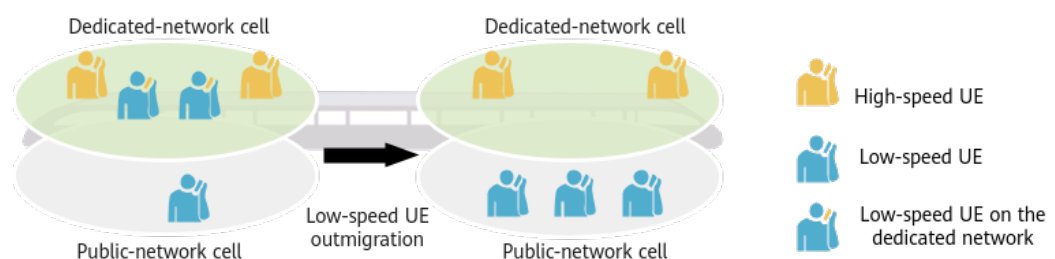
The indicators listed in [13.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

13.3 Low-Speed UE Outmigration

13.3.1 Principles

Low-speed UE outmigration is a process in which low-speed UEs that have accessed the high-speed railway dedicated network are migrated to the public network, as shown in [Figure 13-3](#). In this way, public-network low-speed UEs do not occupy resources on the high-speed railway dedicated network where the train tracks cross urban or suburban areas. This function is controlled by the **NRCellMobilityConfig.LowSpeedUeOutgoingPolicy** parameter.

Figure 13-3 Low-speed UE outmigration



When the low-speed UE outmigration function is enabled, the base station migrates a certain number of low-speed UEs from the high-speed railway dedicated network to the public network periodically before a high-speed train passes by. When a high-speed train passes by the dedicated network, the base station stops the low-speed UE outmigration process. The frequencies of public and dedicated networks are distinguished using the

NRCellFreqRelation.HighSpeedFlag parameter. It is recommended that this parameter be set to **LOW_SPEED** for frequencies of the public network and **HIGH_SPEED** for frequencies of the dedicated network.

Low-speed UE outmigration includes three functions: blind redirection-based outmigration, measurement-based outmigration, and measurement-based outmigration only for SA UEs. [Table 13-3](#) lists the function switches and their differences. To further reduce the probability of incorrect handovers of high-speed UEs to low-speed cells and prevent low-speed UEs from camping on high-speed railway dedicated networks, [Low-Speed-UE-only Outmigration](#) and [Weak-Coverage Low-Speed UE Outmigration](#) are introduced.

- Blind redirection-based outmigration

After this function takes effect, the base station directly migrates identified low-speed UEs to the configured low-speed frequencies, not requiring the UEs to measure the signal quality of these low-speed frequencies. A maximum of 10 UEs can be migrated per second.

Because blind redirection-based outmigration does not require measurement of neighboring cells' signal quality, UEs can be quickly migrated to low frequencies.

- Measurement-based outmigration

After this function takes effect, the base station delivers inter-frequency measurement configurations to the identified low-speed UEs. These measure the signal quality of neighboring cells on the low-speed frequencies. If the signal quality of a neighboring cell on a low-speed frequency measured by a low-speed UE is greater than -104 dBm, the UE is handed over to this public-network cell. If the UE does not support such handovers, it is transferred to the low-speed cell through redirection.

Measurement-based outmigration requires neighboring cell measurement. Therefore, to prevent system overhead increase caused by frequent measurement reporting, a maximum of 15 UEs can be migrated out every 15 seconds.

This function takes effect for both low-speed SA and NSA UEs. If an NSA UE camps on an LTE dedicated-network cell, the NR cell on the public network must be configured as a neighboring cell of the LTE dedicated-network cell to ensure successful handovers of the NSA UE.

- Measurement-based outmigration only for SA UEs

This function is similar to measurement-based outmigration. The difference is that only low-speed SA UEs are migrated to public-network cells. In NSA and SA hybrid networking, it is recommended that this function be enabled when the proportion of low-speed NSA UEs is small, because a small number of low-speed NSA UEs bring only slight impact on the performance and capacity of high-speed cells. This function does not require public-network NR cells to be configured as neighboring cells of dedicated-network LTE cells.

Table 13-3 Function mapping of low-speed UE outmigration

Function Name	Function Switch	Function Differences
Blind redirection-based outmigration	NRCellMobilityConfig.LowSpeedUeOutgoingPolicy set to BLIND_REDIRECTION_BASED	It is recommended that this function be enabled when the frequency bands of public-network and dedicated-network cells do not overlap and the public network provides contiguous coverage along the high-speed railway to ensure that low-speed UEs can access the target frequency.
Measurement-based outmigration	NRCellMobilityConfig.LowSpeedUeOutgoingPolicy set to MEASUREMENT_BASED	This function applies to the scenario where spectrum resources of public-network and dedicated-network cells overlap but their SSB frequency-domain positions are different. It also applies to the scenario where spectrum resources of public-network and dedicated-network cells do not overlap and their SSB frequency-domain positions are different. In scenarios where public network cells along high-speed railways provide discontinuous coverage, low-speed UEs will not be mistakenly migrated to areas without public network coverage since this function required neighboring cell measurement. This function takes effect for both low-speed SA and NSA UEs.
Measurement-based outmigration only for SA UEs	NRCellMobilityConfig.LowSpeedUeOutgoingPolicy set to MEASUREMENT_BASED_SA_UE_ONLY	The application scenario of this function is similar to that of measurement-based outmigration. The difference is that this function migrates only low-speed SA UEs to public-network cells.

Low-Speed-UE-only Outmigration

In high-speed railway scenarios, when the low-speed UE outmigration function is enabled, the probability that high-speed UEs are mistakenly handed over to low-speed cells increases, affecting the user-perceived rate of high-speed UEs. To reduce the probability that high-speed UEs are mistakenly handed over to low-speed cells, when measurement-based outmigration or measurement-based outmigration only for SA UEs is enabled (by setting the **NRCellMobilityConfig.LowSpeedUeOutgoingPolicy** parameter to **MEASUREMENT_BASED** or **MEASUREMENT_BASED_SA_UE_ONLY**) in high-speed cells, it is recommended that the **PERMIT_ONLY_LOW_SPEED_UE_OUT** option of

the **NRCellRelation.HoAlgoSwitch** parameter be selected to enable the function of allowing only low-speed UE outmigration.

Weak-Coverage Low-Speed UE Outmigration

If the **NRCellMobilityConfig.LowSpeedUeOutgoingPolicy** parameter is set to a value other than **OFF** and the **WEAK_COV_LOW_SPEED_UE_OUT_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter is selected, weak-coverage low-speed UE outmigration is enabled.

This function ensures the experience of both high-speed and low-speed UEs in a light-load high-speed cell. After this function is enabled, the base station delivers A2 measurement configurations to low-speed UEs. If the SSB RSRP of the serving cell measured by a low-speed UE is less than the value of

NRCellInterFHoMeaGrp.LowSpeedUeOutA2RsrpThld for a specified period of time, the low-speed UE sends an event A2 measurement report to the base station. Subsequently, the base station migrates such UEs to low-speed cells.

13.3.2 Network Analysis

13.3.2.1 Benefits

It is recommended that this function be enabled when the public and dedicated networks use inter-frequency networking (with different SSB frequency-domain positions). In inter-frequency networking scenarios, it is recommended that unidirectional neighboring cells be configured. That is, public-network cells are configured as neighboring cells of dedicated-network cells while the dedicated-network cells are not configured as neighboring cells of public-network cells. This prevents ping-pong handovers between the public and dedicated networks. You are advised not to enable low-speed UE outmigration when the public and dedicated networks use intra-frequency networking.

Low-speed UE outmigration increases the uplink and downlink throughputs of high-speed UEs in dedicated-network cells.

After the low-speed UE outmigration function is enabled and measurement-based outmigration or measurement-based outmigration only for SA UEs is configured, the following is true: If the **PERMIT_ONLY_LOW_SPEED_UE_OUT** option of the **NRCellRelation.HoAlgoSwitch** parameter is selected, high-speed UEs in high-speed cells will not be handed over to low-speed cells, reducing the service drop rate in low-speed cells caused by frequent handovers of high-speed UEs to low-speed cells.

Assume that low-speed UE outmigration is enabled. Enabling weak-coverage low-speed UE outmigration decreases the number of migrated low-speed UEs and therefore increases the average uplink and downlink throughputs in the high-speed cell.

 NOTE

Handover success rate of high-speed UEs: **Intra-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)**

Average downlink throughput of high-speed UEs = $(N.ThpVol.DL.Hs - N.ThpVol.DL.LastSlot.Hs) / N.ThpTime.DL.RmvLastSlot.Hs$

Average uplink throughput of high-speed UEs = $(N.ThpVol.UL.Hs - N.ThpVol.UE.UL.SmallPkt.Hs) / N.ThpTime.UE.UL.RmvSmallPkt.Hs$

13.3.2.2 Impacts

The impacts between this function and other functions are described in [13.3.3.2.3 Function Impacts](#).

The average uplink throughput (**Cell Uplink Average Throughput (DU)**) and average downlink throughput (**Cell Downlink Average Throughput (DU)**) of low-speed cells decrease in the target cells for low-speed UE outmigration.

After this function takes effect, the proportions of high- and low-speed UEs in high-speed cells change. As a result, the values of indicators related to accessibility and retainability may slightly fluctuate.

In addition, other impacts are as follows:

- If blind redirection-based outmigration is enabled, low-speed UEs may fail to be migrated or even experience service drops if the public network coverage along the high-speed railway is discontinuous.
- If measurement-based outmigration is enabled and there are low-speed NSA UEs in the high-speed cell, low-speed NSA UEs fail to be migrated out if no public-network NR cell is configured as a neighboring cell of the LTE dedicated-network cell.
- If measurement-based outmigration or measurement-based outmigration only for SA UEs is enabled and only low-speed UEs are allowed to be migrated out, the average uplink and downlink throughputs of high-speed UEs in dedicated-network cells may slightly decrease if the capacity of high-speed cells is limited.
- Enabling weak-coverage low-speed UE outmigration decreases the number of migrated low-speed UEs and may cause the average uplink and downlink throughputs of high-speed UEs to fluctuate.

13.3.3 Requirements

13.3.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

13.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Low-speed UE outmigration and "only low-speed UE outmigration" require that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .

13.3.3.2.2 Mutually Exclusive Functions

None

13.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-experience-based inter-frequency handover	UL_EXP_BASED_INTER_FREQ_HO_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	Uplink-experience-based inter-frequency handover is not triggered for low-speed UEs that have been migrated out of the high-speed railway dedicated network.

13.3.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

13.3.3.4 Others

In intra-frequency band networking of high-speed railway dedicated networks and public networks, low-speed UE outmigration requires that high-speed cells be configured with neighboring public-network cells. For details about neighbor relationship configuration, see *SA Mobility Management in Connected Mode* or *NSA Mobility Management*.

13.3.4 Operation and Maintenance

13.3.4.1 Data Configuration

13.3.4.1.1 Data Preparation

Table 13-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 13-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Low Speed UE Outgoing Policy	NRCellMobilityConfig.LowSpeedUEOutgoingPolicy	• If there is public network coverage around, set this parameter to BLIND_REDIRECTION_BASED to implement fast UE outmigration. • It is recommended that this parameter be set to MEASUREMENT_BASED if there are low-speed NSA UEs on the network and it cannot be determined whether there are public network coverage around. • It is recommended that this parameter be set to MEASUREMENT_BASED_SA_UE_ONLY if there are no low-speed NSA UEs on the network or migrating low-speed NSA UEs out of the network is not expected and it cannot be determined whether there are public network coverage around.
High Speed Flag	NRCellFreqRelation.HighSpeedFlag	For low-speed frequencies on the public network, you are advised to set this parameter to LOW_SPEED . For high-speed frequencies, set this parameter to HIGH_SPEED .
High Speed Dedicated Policy Switch	NRCellAlgoSwitch.HighSpeedDedPolicySw	If the load of a high-speed cell is light, it is recommended that the WEAK_COV_LOW_SPEED_UE_OUT_SW option be selected to transfer only low-speed UEs in weak coverage areas, improving the uplink and downlink throughputs of the high-speed cell.

Parameter Name	Parameter ID	Setting Notes
Low-Speed UE Out A2 RSRP Threshold	NRCeInterFHoM eaGrp.LowSpeedU eOutA2RsrpThld	The value -105 is recommended.
Handover Algo Switch	NRCeRelation.H oAlgoSwitch	It is recommended that the PERMIT_ONLY_LOW_SPEED_UE_OUT option be selected to specify whether to enable the function of allowing only low-speed UE outmigration.

13.3.4.1.2 Using MML Commands

Before using MML commands, refer to [13.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Example 1: Blind redirection-based outmigration (The following parameters are configured in high-speed cells.)

For NR TDD:

```
//Configuring the operating frequency of a low-speed NR cell as a neighboring frequency
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N41, SubcarrierSpacing=30KHZ, ConnFreqPriority=1;
//Configuring the low-speed flag for the operating frequency of the corresponding low-speed NR cell
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, HighSpeedFlag=LOW_SPEED;
//Enabling blind redirection-based outmigration
MOD NRCELLMOBILITYCONFIG: NrCellId=0,
LowSpeedUeOutgoingPolicy=BLIND_REDIRECTION_BASED;
```

- Example 2: Measurement-based outmigration (The following parameters are configured in high-speed cells.)

For NR TDD:

```
//Configuring the operating frequency of a low-speed NR cell as a neighboring frequency and setting
the low-speed flag for this frequency
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N41, SubcarrierSpacing=30KHZ, ConnFreqPriority=1,HighSpeedFlag=LOW_SPEED;
//If the adjacent low-speed NR cell and the high-speed cell are not served by the same base station,
configuring the adjacent low-speed NR cell as an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=1, CellName="A", PhysicalCellId=1,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6400, FrequencyBand=N41,
NrNetworkingOption=SA_NSA;
//Configuring the neighbor relationship with the adjacent low-speed NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1,
HoAlgoSwitch=PERMIT_ONLY_LOW_SPEED_UE_OUT-1;
//If there are NSA UEs on the network, in the anchor LTE cell of the high-speed cell, configuring the
operating frequency of a target low-speed NR cell to which UEs are migrated as a neighboring
frequency and the neighbor relationship with this target low-speed NR cell. Otherwise, the PSCell
```

change of NSA UEs on the 5G side will fail. For details, see the configuration on the eNodeB side in *NSA Mobility Management*.

//Enabling measurement-based outmigration

MOD NRCELLMOBILITYCONFIG: NrCellId=0, LowSpeedUeOutgoingPolicy=MEASUREMENT_BASED;
 //(Recommended) Enabling weak-coverage low-speed UE outmigration when the high-speed cell is lightly loaded

MOD NRCELLINTERFHOMEAGRP:

NrCellId=0,InterFreqHoMeasGroupId=1,LowSpeedUeOutA2RsrpThId=-105;

MOD NRCELLALGOSWITCH:

NrCellId=0,HighSpeedDedPolicySw=WEAK_COV_LOW_SPEED_UE_OUT_SW-1;

- Example 3: Measurement-based outmigration for SA UEs only (The following parameters are configured in high-speed cells.)

For NR TDD:

//Configuring the operating frequency of a low-speed NR cell as a neighboring frequency and setting the low-speed flag for this frequency

ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN, FrequencyBand=N41, SubcarrierSpacing=30KHZ, ConnFreqPriority=1, HighSpeedFlag=LOW_SPEED;

//If the adjacent low-speed NR cell and the high-speed cell are not served by the same base station, configuring the adjacent low-speed NR cell as an external cell

ADD NREXTERNALNCELL: Mcc="302",Mnc="220", gNBId=1,CellId=1,CellName="A",PhysicalCellId=1, Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6400, FrequencyBand=N41, NrNetworkingOption=SA_NSA;

//Configuring the neighbor relationship with the adjacent low-speed NR cell

ADD NRCELLRELATION:NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, HoAlgoSwitch=PERMIT_ONLY_LOW_SPEED_UE_OUT-1;

//Enabling measurement-based outmigration only for SA UEs

MOD NRCELLMOBILITYCONFIG:

NrCellId=0,LowSpeedUeOutgoingPolicy=MEASUREMENT_BASED_SA_UE_ONLY;

//(Recommended) Enabling weak-coverage low-speed UE outmigration when the high-speed cell is lightly loaded

MOD NRCELLINTERFHOMEAGRP:

NrCellId=0,InterFreqHoMeasGroupId=1,LowSpeedUeOutA2RsrpThId=-105;

MOD NRCELLALGOSWITCH:

NrCellId=0,HighSpeedDedPolicySw=WEAK_COV_LOW_SPEED_UE_OUT_SW-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

//Disabling weak-coverage low-speed UE outmigration

MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=WEAK_COV_LOW_SPEED_UE_OUT_SW-0;

//Disabling low-speed UE outmigration

MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1,

HoAlgoSwitch=PERMIT_ONLY_LOW_SPEED_UE_OUT-0;

MOD NRCELLMOBILITYCONFIG: NrCellId=0, LowSpeedUeOutgoingPolicy=OFF;

13.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.3.4.2 Activation Verification

External CHR events can be used for observation. This function has taken effect if any of the following conditions is met:

- The CHR event **PRIVATE_INTRA_RAT_HO_OUT.HoCause** contains the cause value **LOW_SPEED_UE_OUT_GOING_BASED**.
- The CHR event **PRIVATE_UE_RELEASE.RedirectionTrigger** contains the cause value **LOW_SPEED_UE_OUT_GOING**.

13.3.4.3 Network Monitoring

The indicators listed in [13.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

13.4 High-Speed UE Return

13.4.1 Principles

The high-speed UE return function is introduced to enable high-speed UEs to camp on high-speed cells for extended periods. After this function is enabled, if the signal quality of a neighboring cell on a high-speed frequency measured by a high-speed UE is greater than -104 dBm, the high-speed UE can be redirected back to a dedicated-network cell.

The high-speed UE return function is controlled by the **HIGH_SPEED_UE_REDIRECT_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter. After this function is enabled in a public-network cell, the base station identifies high-speed UEs in the cell and redirects them to the 4G or 5G dedicated network. This function takes effect only on SA UEs.

4G and 5G high-speed dedicated-network frequencies are configured using the **NRCellEutranNFreq** and **NRCellFreqRelation** MOs, respectively. If the value of **NRCellEutranNFreq.HighSpeedFlag** or **NRCellFreqRelation.HighSpeedFlag** is **HIGH_SPEED** for a frequency, it is a high-speed cell frequency.

For a target frequency with a high-speed flag **HIGH_SPEED**, the following configurations are recommended:

- If the frequency is a 5G dedicated-network frequency and it is available only for this function, select the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter. In this scenario, it is recommended that cells working on this dedicated-network frequency not be configured as neighboring cells on the public network. This prevents low-speed UEs from being handed over from the public network to the dedicated network. Otherwise, the frequency may be used for other functions. In this case, cells working on this dedicated-network frequency must be configured as neighboring cells on the public network. Otherwise, the performance of UEs performing frequency-related services (such as mobility management) is affected.
- If the frequency is a 4G dedicated-network frequency and it is available only for this function, select the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter. In this scenario, it is recommended that cells working on this dedicated-network frequency not be configured as neighboring cells on the public network. This prevents low-speed UEs from being handed over from the public network to the dedicated network. Otherwise, the frequency may be used for other functions. In this

case, cells working on this dedicated-network frequency must be configured as neighboring cells on the public network. Otherwise, the performance of UEs performing frequency-related services (such as mobility management) is affected.

- If the frequency is a 4G or 5G dedicated-network frequency and not dedicated to high-speed UE return, deselect the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter or the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter.

13.4.2 Network Analysis

13.4.2.1 Benefits

High-speed UE return increases the camping duration of high-speed UEs in high-speed cells and decreases the duration when high-speed UEs are not served by the dedicated network. It is recommended that this function be enabled in public-network inter-frequency cells whose coverage overlaps with that of the network dedicated to high-speed railways. It is recommended that coverage-based handover be preferentially enabled when the public and dedicated networks use intra-frequency networking.

13.4.2.2 Impacts

The impacts between this function and other functions are described in [13.4.3.2.3 Function Impacts](#).

After this function is enabled, measurement gaps will be triggered if the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of **NRCellFreqRelation.FrequencyCtrlSwitch** is deselected for a 5G dedicated-network frequency or the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of **NRCellEutranNFreq.FrequencyCtrlSwitch** is deselected for a 4G dedicated-network frequency, and no neighbor relationship is configured between cells on the target high-speed frequency and public-network cells. In this case, there will be throughput performance loss.

After this function takes effect, the proportions of high- and low-speed UEs in high-speed cells change. As a result, the values of indicators related to accessibility and retainability may slightly fluctuate.

13.4.3 Requirements

13.4.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

NOTE

High-speed UE return does not require licenses.

13.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low speed cell	NRDUCell.HighSpeedFlag set to LOW_SPEED	None	High-speed UE return requires that the NRDUCell.HighSpeedFlag parameter be set to LOW_SPEED. The high-speed UE redirection only switch requires that the NRDUCell.HighSpeedFlag parameter be set to LOW_SPEED. High-speed UE redirection only switch: - Configured using the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellFreqRelation.FrequencyCtrlSwitch parameter for a 5G dedicated-network frequency. - Configured using the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellEutranNFreq.FrequencyCtrlSwitch parameter for a 4G dedicated-network frequency.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed cell frequency	NRCellEutranNFreq.HighSpeedFlag or NRCellFreqRelation.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	The high-speed UE redirection only switch requires that the configured frequency is a high-speed cell frequency. High-speed UE redirection only switch: - Configured using the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellFreqRelation.FrequencyCtrlSwitch parameter for a 5G dedicated-network frequency. - Configured using the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellEutranNFreq.FrequencyCtrlSwitch parameter for a 4G dedicated-network frequency. High-speed cell frequency: - Configured using the NRCellFreqRelation.HighSpeedFlag parameter on the 5G public network. - Configured using the NRCellEutranNFreq.HighSpeedFlag parameter on the 4G public network.

13.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-Freq Periodic MR Type	NRCellFreqRelation.InterFreqPrdcMrType	None	The high-speed UE redirection only switch is mutually exclusive with the Inter-Freq Periodic MR Only and Measurement Prohibited settings of the inter-frequency periodic MR type parameter. That is, when the NRCellFreqRelation.InterFreqPrdcMrType parameter is set to PERIODIC_MR_ONLY or FORBID_MEAS , the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellFreqRelation.FrequencyCtrlSwitch parameter must be deselected.
Low-frequency TDD	Measurement Type	NRCellEutranNFreq.MeasureType	None	The high-speed UE redirection only switch is mutually exclusive with the roaming UE measurement setting of the measurement type parameter. That is, when NRCellEutranNFreq.MeasureType is set to ROAMING_MEAS , the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellEutranNFreq.FrequencyCtrlSwitch parameter must be deselected.

13.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-experience-based inter-frequency handover	UL_EXP_BASED_INTER_FREQ_HO_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	Uplink-experience-based inter-frequency handover does not take effect for high-speed UEs.

13.4.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

13.4.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

13.4.4 Operation and Maintenance

13.4.4.1 Data Configuration

13.4.4.1.1 Data Preparation

[Table 13-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 13-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	Select the HIGH_SPEED_UE_REDIRECT_SW option to enable the high-speed UE return function.
High Speed Flag	NRCCellEutranNFreq <i>.HighSpeedFlag</i>	Set this parameter to HIGH_SPEED for a frequency of high-speed 4G cells and to LOW_SPEED for a frequency of low-speed 4G cells.

Parameter Name	Parameter ID	Setting Notes
High Speed Flag	NRCellFreqRelation.HighSpeedFlag	Set this parameter to HIGH_SPEED for a frequency of high-speed 5G cells and to LOW_SPEED for a frequency of low-speed 5G cells.
Frequency Control Switch	NRCellFreqRelation.FrequencyCtrlSwitch	For a 5G dedicated-network frequency whose high-speed flag is HIGH_SPEED, the configuration suggestions are as follows: • If this frequency is dedicated to UEs with high-speed UE return in effect, select the HIGH_SPEED_UE_REDIRECT_ONLY_SW option. In this case, it is recommended that cells of the target high-speed frequency not be configured as neighboring cells of public-network cells to prevent low-speed UEs from being handed over from the public network to dedicated-network cells. • If this frequency is not dedicated to UEs with high-speed UE return in effect, deselect this option. In this case, cells of the target high-speed frequency must be configured as neighboring cells of public-network cells. Otherwise, the performance of UEs performing frequency-related services (such as mobility management) is affected.

Parameter Name	Parameter ID	Setting Notes
Frequency Control Switch	NRCellEutranNFreq.FrequencyCtrlSwitch	For a 4G dedicated-network frequency whose high-speed flag is HIGH_SPEED, the configuration suggestions are as follows: • If this frequency is dedicated to UEs with high-speed UE return in effect, select the HIGH_SPEED_UE_REDIRECT_ONLY_SW option. In this case, it is recommended that cells of the target high-speed frequency not be configured as neighboring cells of public-network cells to prevent low-speed UEs from being handed over from the public network to dedicated-network cells. • If this frequency is not dedicated to UEs with high-speed UE return in effect, deselect this option. In this case, cells of the target high-speed frequency must be configured as neighboring cells on the public network. Otherwise, the performance of UEs performing frequency-related services (such as mobility management) is affected.

13.4.4.1.2 Using MML Commands

Before using MML commands, refer to [13.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

For NR TDD:

```
//On the public network side, enabling the high-speed UE return function for a public-network cell
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_UE_REDIRECT_SW-1;
//On the public network side, configuring the operating frequency of a high-speed NR cell as a neighboring
frequency of the operating frequency of a public-network cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N41, SubcarrierSpacing=30KHZ, ConnFreqPriority=1;
//On the public network side, configuring a 5G dedicated-network frequency as a high-speed cell frequency,
```

and turning on the high-speed UE redirection only switch. In this case, the frequency can be used only by UEs with high-speed UE return in effect.
 MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, HighSpeedFlag=HIGH_SPEED, FrequencyCtrlSwitch=HIGH_SPEED_UE_REDIRECT_ONLY_SW-1;
 //On the public network side, configuring the operating frequency of a high-speed LTE cell as a neighboring frequency of the operating frequency of a public-network cell to allow migration of high-speed UEs back to the 4G dedicated network
 ADD NRCELLEUTRANNFREQ: NrCellId=0, DLearfcn=1000, EutranFreqHighPriReselThld=6, EutranFreqLowPriReselThld=6, EutranFreqReselPriority=2, FreqSpecificOffset=DB1, MeasurementBandwidth=MBW15, MinimumRxLevel=-32, ConnFreqPriority=2, VoltePriority=2;
 //On the public network side, configuring a 4G dedicated-network frequency as a high-speed cell frequency, and turning on the high-speed UE redirection only switch. In this case, the frequency can be used only by UEs with high-speed UE return in effect.
 MOD NRCELLEUTRANNFREQ: NrCellId=0, DLearfcn=1000, HighSpeedFlag=HIGH_SPEED, FrequencyCtrlSwitch=HIGH_SPEED_UE_REDIRECT_ONLY_SW-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling high-speed UE return
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HIGH_SPEED_UE_REDIRECT_SW-0;
```

13.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.4.4.2 Activation Verification

Observe the values of **N.RRCRedirection.HighSpeedUser** and **N.RRCRedirection.N2E.HighSpeedUser** to check whether this function has taken effect. If the value of either counter is not zero, this function has taken effect.

13.4.4.3 Network Monitoring

The indicators listed in [13.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

13.5 Broadcast of R17-compliant HSDN-related IEs

13.5.1 Principles

3GPP TS 38.331 Release 17 now supports broadcast of R17-compliant HSDN-related IEs.

- The base station can broadcast the **hsdn-Cell-r17** IE in SIB1. This IE is used to inform high-speed UEs whether a cell is a high-speed cell.
- The base station can broadcast the **cellEquivalentSize-r17** IE in SIB2. This IE is used to inform high-speed UEs of the number of TRPs in a high-speed cell, and helps these UEs to determine their moving speeds.

- The base station can broadcast the following IEs in SIB3 to SIB5, respectively: *intraFreqNeighHSDN-CellList-r17*, *interFreqNeighHSDN-CellList-r17*, and *eutra-FreqNeighHSDN-CellList-r17*. These IEs are used to inform high-speed UEs whether neighboring cells are high-speed cells, and help these UEs to perform cell reselection.

The "broadcast of R17-compliant HSDN-related IEs" function is introduced based on the preceding protocol definition. After this function is enabled:

- In high-speed cells, the base station broadcasts the *hsdn-Cell-r17* and *cellEquivalentSize-r17* IEs in SIB1 and SIB2, respectively. If an external cell of the local cell is configured as a high-speed cell (by setting the **NRExternalNCell.HighSpeedFlag** or **gNBEutraExternalCell.HighSpeedFlag** parameter to **HIGH_SPEED**) and the **NRCellRelation.NCellReselOffset** parameter is set to a value other than **DB0**, the corresponding neighboring cell will be added to the R17-compliant HSDN neighboring cell list of the local cell, and the base station will broadcast the following IEs in SIB3 to SIB5, respectively: *intraFreqNeighHSDN-CellList-r17*, *interFreqNeighHSDN-CellList-r17*, and *eutra-FreqNeighHSDN-CellList-r17*.
- In low-speed cells, the base station does not broadcast the *hsdn-Cell-r17* and *cellEquivalentSize-r17* IEs in SIB1 and SIB2, respectively. If a neighboring cell of the local cell is configured as a high-speed cell and the neighboring cell reselection offset is not 0, this neighboring cell will be added to the R17-compliant HSDN neighboring cell list of the local cell, and the base station will broadcast the following IEs in SIB3 to SIB5, respectively: *intraFreqNeighHSDN-CellList-r17*, *interFreqNeighHSDN-CellList-r17*, and *eutra-FreqNeighHSDN-CellList-r17*.

This function is controlled by the **HSDN_IE_BROADCAST_SW** option of the **NRCellAlgoSwitch.HighSpeedDedPolicySw** parameter. If broadcast of R17-compliant HSDN-related IEs is disabled, the base station does not broadcast R17-compliant HSDN-related IEs in SIB1 to SIB5.

13.5.2 Network Analysis

13.5.2.1 Benefits

After this function is enabled, the accuracy of reselection to high-speed cells and high-speed neighboring cells for high-speed UEs improves.

This function improves the accuracy of identifying UEs as high-speed UEs in high-speed cells.

13.5.2.2 Impacts

The impacts between this function and other functions are described in [13.5.3.2.3 Function Impacts](#).

For an external NG-RAN cell of the local cell, the **NRExternalNCell.HighSpeedFlag** parameter configured on the local cell and the **NRDUCell.HighSpeedFlag** parameter configured on the external NG-RAN cell must be both set to **HIGH_SPEED**. For an external E-UTRAN cell of the local cell, the **gNBEutraExternalCell.HighSpeedFlag** parameter configured on the local cell and the **Cell.HighSpeedFlag** parameter configured on the external E-UTRAN cell

must be both set to **HIGH_SPEED**. If not, the base station may incorrectly deliver HSDN-related IEs.

After this function is enabled, the local cell broadcasts not only non-HSDN-related IEs but also HSDN-related IEs in SIBs 3 to 5. If the local cell is configured with too many high-speed neighboring cells (for example, more than 16 intra-frequency, inter-frequency, or inter-RAT high-speed neighboring cells), the base station restricts the number of high-speed neighboring cells filled in the HSDN-related IEs in SIBs 3 to 5. In this case, HSDN-capable UEs cannot obtain information about all high-speed neighboring cells. The base station uses this method to limit the bit overhead of system information. If the local cell is configured with a large number of non-high-speed neighboring cells and a large number of high-speed neighboring cells, the number of bits occupied by system information may be too large. In this case, the cell coverage may be affected.

After this function is enabled in a high-speed cell, an R17-compliant UE that receives the *cellEquivalentSize* IE is more likely to be identified as a high-speed UE. The IE used to evaluate the moving speed of a UE in 3GPP TS 38.101 of Release 16 is the same as that of 3GPP Release 17. Therefore, during a version upgrade, you are advised to adjust cell reselection parameters accordingly. Otherwise, UEs moving at a low speed may be identified as high-speed UEs. For details about speed-based cell reselection, see *SA Mobility Management in Idle Mode*.

13.5.3 Requirements

13.5.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

13.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.5.3.2.1 Prerequisite Functions

None

13.5.3.2.2 Mutually Exclusive Functions

None

13.5.3.2.3 Function Impacts

None

13.5.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

13.5.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

Broadcast of R17-compliant HSDN-related IEs takes effect only on R17-compliant HSDN-capable UEs.

13.5.4 Operation and Maintenance

13.5.4.1 Data Configuration

13.5.4.1.1 Data Preparation

[Table 13-6](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 13-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Dedicated Policy Switch	NRCCellAlgoSwitch <i>.HighSpeedDedPolicySw</i>	To enable broadcast of R17-compliant HSDN-related IEs, select the HSDN_IE_BROADCAST_SW option.
Neighboring Cell Reselection Offset	NRCCellRelation.N <i>CellReselOffset</i>	Set this parameter according to the network plan. If broadcast of R17-compliant HSDN-related IEs is required in a high speed cell, set this parameter to a value other than DB0 .

13.5.4.1.2 Using MML Commands

Before using MML commands, refer to [13.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling broadcast of R17-compliant HSDN-related IEs
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HSDN_IE_BROADCAST_SW-1;
//(Optional) If an intra-RAT intra-frequency adjacent cell is a high-speed cell and is not configured as a
neighboring cell, configuring this cell as a neighboring cell and setting the neighboring cell reselection
```

offset to a value other than 0

```
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=0, CellId=0, PhysicalCellId=0, Tac=1,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6400, HighSpeedFlag=HIGH_SPEED;
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=0, NCellReselOffset=DB1;
//(Optional) If an intra-RAT inter-frequency adjacent cell is a high-speed cell and is not configured as a
neighboring cell, configuring this cell as a neighboring cell and setting the neighboring cell reselection
offset to a value other than 0
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=6400, FrequencyBand=N41,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SubcarrierSpacing=30KHZ, ConnFreqPriority=1,
MeasActvFlag=ACTIVE, HighSpeedFlag=HIGH_SPEED;
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=0, CellId=0, PhysicalCellId=0, Tac=1,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6400, HighSpeedFlag=HIGH_SPEED;
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=0, NCellReselOffset=DB1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling broadcast of R17-compliant HSDN-related IEs
MOD NRCELLALGOSWITCH: NrCellId=0, HighSpeedDedPolicySw=HSDN_IE_BROADCAST_SW-0;
```

13.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.5.4.2 Activation Verification

After this function is enabled, observe whether SIB1 to SIB5 carry HSDN-related IEs. If yes, it indicates that this function has taken effect.

13.5.4.3 Network Monitoring

The indicators listed in [13.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

13.6 High-Speed Attribute Indication of Neighboring E-UTRAN Cells

13.6.1 Principles

If a high-speed E-UTRAN cell uses a dedicated frequency, the high speed attribute of this frequency can be configured on the NR side (by setting **NRCellEutranNFreq.HighSpeedFlag** to **HIGH_SPEED**) and the **EUTRA_HIGH_SPEED_IND_SW** option of the **NRCellReselConfig.SibOptionalleInd** parameter can be selected. This configuration means the NR network will broadcast the high speed attribute of this neighboring E-UTRAN frequency in SIB5 to UEs. For details, see section 5.2.2.4 "Actions upon receipt of System Information" in 3GPP TS 38.331.

This helps UEs identify whether a neighboring E-UTRAN frequency is a high-speed cell frequency and optimize the camping policy. The implementation depends on UEs.

13.6.2 Network Analysis

13.6.2.1 Benefits

High-speed attribute indication of neighboring E-UTRAN cells helps UEs identify whether neighboring E-UTRAN frequencies are high-speed cell frequencies and optimize camping policies. The specific gains depend on UE implementation.

13.6.2.2 Impacts

The impacts between this function and other functions are described in [13.6.3.2.3 Function Impacts](#).

This function has no impact.

13.6.3 Requirements

13.6.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

13.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	The "high-speed attribute indication of neighboring E-UTRAN cells" function requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .

13.6.3.2.2 Mutually Exclusive Functions

None

13.6.3.2.3 Function Impacts

None

13.6.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

13.6.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

13.6.4 Operation and Maintenance

13.6.4.1 Data Configuration

13.6.4.1.1 Data Preparation

[Table 13-7](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 13-7 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
High Speed Flag	NRCellEutranNFreq <i>q.HighSpeedFlag</i>	When the function of high-speed attribute indication of neighboring E-UTRAN cells is enabled, configure the high-speed attribute for the frequencies of neighboring high-speed E-UTRAN cells by setting the NRCellEutranNFreq <i>q.HighSpeedFlag</i> parameter of the corresponding frequency to HIGH_SPEED .
SIB Optional IE Indicator	NRCellReselConfig <i>SibOptionalleInd</i>	Select the EUTRA_HIGH_SPEED_IND_SW option to enable high-speed attribute indication of neighboring E-UTRAN cells.

13.6.4.1.2 Using MML Commands

Before using MML commands, refer to [13.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling high-speed attribute indication of neighboring E-UTRAN cells
MOD NRCELLRESELCONFIG: NrCellId=0, SibOptionalleInd=EUTRA_HIGH_SPEED_IND_SW-1;
//Configuring the high-speed attribute for a neighboring E-UTRAN frequency (before the configuration,
ensure that the frequency is dedicated to high-speed E-UTRAN cells)
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000, HighSpeedFlag=HIGH_SPEED;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling high-speed attribute indication of neighboring E-UTRAN cells
MOD NRCELLRESELCONFIG: NrCellId=0, SibOptionalleInd=EUTRA_HIGH_SPEED_IND_SW-0;
```

13.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.6.4.2 Activation Verification

After this function is enabled, start Uu interface tracing for the gNodeB and check whether the *highSpeedEUTRACarrier-r16* IE is delivered in the SIB5 message. If this IE is delivered, this function has taken effect.

13.6.4.3 Network Monitoring

The indicators listed in [13.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

14 Multi-Carrier Coordination Policy Management

14.1 Load Balancing in Fixed Proportions for RRC_IDLE UEs

14.1.1 Principles

Among inter-frequency co-coverage cells in a 5G network, some cells may be heavily loaded while others are lightly loaded, which causes load imbalance. This imbalance will affect UE performance in high-load cells and waste resources in low-load ones.

To resolve this issue, load balancing in fixed proportions for RRC_IDLE UEs is introduced in high-speed railway multi-carrier networking scenarios. This function enables load balancing among multiple carriers and thereby allows the number of UEs served by each carrier to reach the expected proportion.

Load balancing in fixed proportions for RRC_IDLE UEs takes effect only when all of the following conditions are met:

- The **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch.MlbAlgoSwitch** parameter is selected.
- The **NRDUCell.HighSpeedFlag** parameter is set to **HIGH_SPEED**.
- The **NRCellFreqRelation.IdleMlbUeReleaseRatio** parameter is not set to **0** for at least one frequency.
- The **NRCellFreqRelation.CellReselPriority** parameter is not set to **255**.

After this function takes effect, the cell releases UEs to frequencies based on the preset proportions among frequencies, so that the proportions of RRC_IDLE UEs among frequencies approach the expected levels. Expected proportions = 100% – Sum of proportions for all neighboring frequencies

The proportions of UEs to be released for inter-frequency idle mode MLB meet the following conditions:

- The sum of proportions for all neighboring frequencies cannot exceed 90%.
- A maximum of nine neighboring frequencies can be configured.

If a frequency of which the reselection priority parameter **NRCellFreqRelation.CellReselPriority** is set to **255**, the frequency is filtered out by load balancing in fixed proportions for RRC_IDLE UEs in the high-speed cell.

In low-frequency scenarios, if the **CA_UE_FREQ_STEER_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter is selected in the serving cell of a CA-capable UE, the gNodeB determines whether to filter out the neighboring frequencies based on CA capability of the UE, networking continuity, and the frequency of the serving cell on which the UE camps. The frequencies that are not filtered out are delivered to the UE. For details about frequency filtering scenarios and recommended scenarios for CA UE frequency steering, see *SA Mobility Management in Connected Mode*.

14.1.2 Network Analysis

14.1.2.1 Benefits

After load balancing in fixed proportions for RRC_IDLE UEs is enabled, the number of RRC_CONNECTED UEs on each carrier (**N.User.RRCConn.Avg**) is close to the expected proportion.

Load balancing in fixed proportions for RRC_IDLE UEs delivers benefits if the following conditions are met:

- It is recommended that the **NRCellFreqRelation.IdleMlbUeReleaseRatio** parameter be set to the same value for different high-speed frequencies in cells served by one base station along the railway.
- When this function takes effect, it is recommended that the high-speed frequencies with the **NRCellFreqRelation.IdleMlbUeReleaseRatio** parameter specified remain unchanged. Otherwise, load balancing based on the configured proportions may fail in high-speed cells.
- A larger number of times that UEs served by high-speed frequencies are released indicates a longer time for the function to take effect. In this case, the actual load proportion is closer to the expected value.

14.1.2.2 Impacts

None

The impacts between this function and other functions are described in [14.1.3.2.3 Function Impacts](#).

14.1.3 Requirements

14.1.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

14.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

14.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Load balancing in fixed proportions for RRC_IDLE UEs requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .
Low-frequency TDD	Low-frequency UE-number-based idle mode MLB	INTER_FREQ_IDLE_MLB_S option of the NRCCellAlgoSwitch.MlbAlgorithmSwitch parameter	<i>Mobility Load Balancing in Idle Mode</i>	Load balancing in fixed proportions for RRC_IDLE UEs depends on low-frequency UE-number-based idle mode MLB.
Low-frequency TDD	High-speed cell frequency	NRCCellFreqRelation.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Load balancing in fixed proportions for RRC_IDLE UEs requires neighboring frequencies to be high-speed frequencies.

14.1.3.2.2 Mutually Exclusive Functions

None

14.1.3.2.3 Function Impacts

None

14.1.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

14.1.3.4 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

14.1.4 Operation and Maintenance

14.1.4.1 Data Configuration

14.1.4.1.1 Data Preparation

Table 14-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 14-1 Parameters used for activation (load balancing in fixed proportions for RRC_IDLE UEs)

Parameter Name	Parameter ID	Setting Notes
MLB Algorithm Switch	NRCelAlgoSwitch <i>.MlbAlgoSwitch</i>	To enable load balancing in fixed proportions for RRC_IDLE UEs, select the INTER_FREQ_IDLE_MLB_SW option.
Idle MLB UE Release Ratio	NRCelFreqRelatio <i>n.IdleMlbUeRelea</i> <i>seRatio</i>	Set this parameter based on the network plan.

14.1.4.1.2 Using MML Commands

Before using MML commands, refer to [14.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Activation command examples for load balancing in fixed proportions for RRC_IDLE UEs:
//Turning on the inter-frequency MLB switch for idle mode in the high-speed cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
//For NR TDD, configuring the UE release percentages for different neighboring high-speed frequencies
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6400, HighSpeedFlag=HIGH_SPEED, IdleMlbUeReleaseRatio=50;
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6300, HighSpeedFlag=HIGH_SPEED, IdleMlbUeReleaseRatio=20;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the inter-frequency MLB switch for idle mode in the high-speed cell  
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-0;
```

14.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.1.4.2 Activation Verification

After this function is enabled, external CHR events can be used for observation. This function has taken effect if the CHR event

PERIOD_CELL_IDLE_MLB_UE_INFO contains the **HighSpeedIdleMlbUeRelease**, **TargetCarrierFreq**, and **IdleMlbUeReleaseNum** fields and the **TargetCarrierFreq** and **IdleMlbUeReleaseNum** fields have values.

HighSpeedIdleMlbUeRelease is an array. The array size indicates the number of target frequencies to be released. This array contains the target frequency number specified by *TargetCarrierFreq* and the number of UEs released to the frequency specified by *IdleMlbUeReleaseNum*.

14.1.4.3 Network Monitoring

The indicators listed in [14.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

14.2 Uplink-Service-based Carrier Selection in High-Speed Scenarios

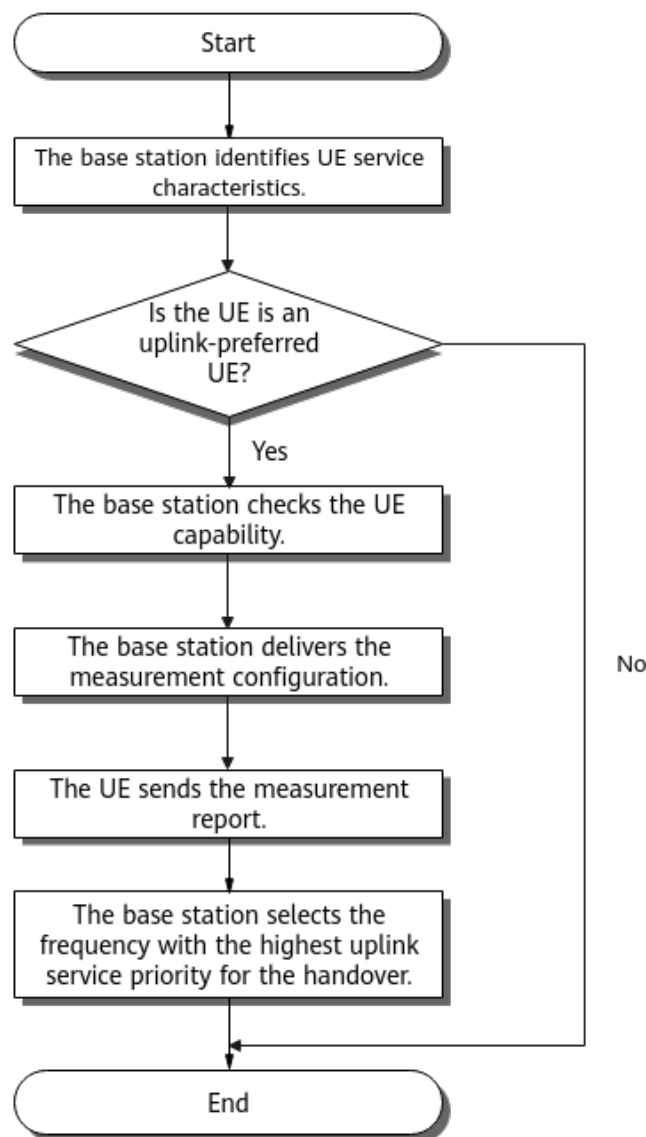
14.2.1 Principles

On a high-speed railway multi-carrier network, a base station triggers handovers or carrier changes for UEs mainly based on cell load and UE-reported RSRP, and selects carriers for UEs to camp on. As carrier difference is not considered, the carriers selected by the base station may not be able to provide the optimal experience for certain UEs running different services. Uplink-service-based carrier selection in high-speed scenarios is introduced to enable differentiated selection of optimal carriers for UEs with different preferred services. This function is controlled by the **HIGH_SPD_SERV_CARR_SEL_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter.

Assume that uplink-service-based carrier selection in high-speed scenarios is enabled. Once identifying uplink-preferred UEs, the base station instructs these

UEs to perform inter-frequency measurements. In this case, the base station performs carrier selection based on the frequency priorities for uplink services. The frequency priority for uplink services is specified by the **NRCellFreqRelation.UlTrafficFreqPriority** parameter. **Figure 14-1** shows the detailed procedure.

Figure 14-1 Uplink-service-based carrier selection in high-speed scenarios



1. Identification of UE service characteristics: After both "uplink-service-based carrier selection in high-speed scenarios" and "service-based carrier selection enhancement in high-speed scenarios" (controlled by the **HIGH_SPD_SERV_CARR_SEL_ENH_SW** option of the **NRCellFeatureSwitch.MultiFreqAlgoSwitch** parameter) are enabled, the base station identifies a UE's service characteristics based on the following conditions, to determine whether it is a uplink-preferred UE:
 - Uplink PRB usage > **NRDUCellServExp.FullBufferUeUlPrbUsageThld**
 - Proportion of slots for uplink scheduling > **NRDUCellServExp.FullBufferUeUlTtiRatioThld**

 NOTE

After service-based carrier selection enhancement in high-speed scenarios is enabled, if a UE is identified as both an uplink-preferred UE and a downlink-preferred UE, downlink-service-based carrier selection in high-speed scenarios preferentially takes effect.

2. UE capability check: The base station obtains the frequency band (specified by the **NRCellFreqRelation.FrequencyBand** parameter) to which the neighboring frequency belongs and the frequency of the local cell and checks the UE capability. The subsequent procedure is triggered if either of the following conditions is met: (1) Uplink CA or SUL is enabled for the base station, but the UE does not support any band combination for uplink CA or SUL; (2) Uplink CA or SUL is disabled for the base station.
3. Measurement configuration delivery: When delivering inter-frequency measurement configurations, the base station selects frequencies with the **NRCellFreqRelation.ULTrafficFreqPriority** parameter set to a value other than 0 for A5-related inter-frequency measurement.

After "uplink-service-based carrier selection in high-speed scenarios" is enabled, the serving cell determines whether to deliver measurement configurations to the UE, based on the configuration of suppression of unnecessary handover for the UE's QCI-specific bearers. This process is referred to as QCI-specific suppression of unnecessary inter-frequency handover. The details are as follows:

- If the UE has a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB does not deliver measurement configurations to the UE.
- If the UE does not have a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB delivers measurement configurations to the UE.

 NOTE

When "uplink-service-based carrier selection in high-speed scenarios" is enabled for the serving cell, you are advised to select the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter for each QCI-specific bearer associated with the target frequency group to prevent ping-pong handovers if service-based inter-frequency handover (specified by the **SERVICE_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter) is enabled for the target cell.

Before the inter-frequency measurement delay timer (defined by **NRCellMobilityConfig.InterFreqMeasDelay**) expires, the base station does not deliver frequency-priority-based inter-frequency measurement configurations and "uplink-service-based carrier selection in high-speed scenarios" does not take effect.

4. Measurement result reporting: Based on the measurement result, the existing inter-frequency handover threshold (specified by the **NRCellInterFHoMeaGrp.MultiFreqSmtSelA5RsrpThld2** parameter) is reused to trigger an inter-frequency handover. If this parameter is set to 255, the RSRP threshold for interference-based inter-frequency handovers (specified by the **NRCellInterFHoMeaGrp.IntrfInterFreqHoRsrpThld** parameter) is used for handover decision.
5. Selection of the frequency with the highest uplink service priority for the handover: If there is a neighboring frequency with the highest uplink service

priority, the base station performs a handover and terminates the inter-frequency measurement. If no neighboring frequency is detected within 3s, the procedure ends.

Assume that frequency-priority-based inter-frequency handover and "uplink-service-based carrier selection in high-speed scenarios" are both enabled. If a UE is running uplink large-packet services, a handover between FDD and TDD cells will be triggered for the UE before the inter-frequency measurement delay timer expires. To avoid ping-pong handovers, the uplink-large-packet-based frequency priority function is introduced. This function is controlled by the **UL_LARGE_PKT_FREQ_PRI_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter. After this function is enabled, it takes effect on the UE only when the UE stops running uplink large-packet services and the inter-frequency measurement delay timer expires. Uplink-large-packet-based frequency priority does not take effect on the UE when the UE is running uplink large-packet services.

14.2.2 Network Analysis

14.2.2.1 Benefits

After uplink-service-based carrier selection in high-speed scenarios is enabled and takes effect in high-speed railway multi-carrier scenarios, the base station identifies UEs with uplink-preferred services and selects frequencies for handovers based on the frequency priorities for uplink services, improving uplink user experience in high-speed railway scenarios. In addition, UE handovers may change the uplink traffic distribution across cells, and the average uplink UE throughput (indicated by the **User Uplink Average Throughput (DU)**) may increase.

After uplink-large-packet-based frequency priority takes effect, the number of handovers decreases.

14.2.2.2 Impacts

The impacts between this function and other functions are described in [14.2.3.2.3 Function Impacts](#).

After uplink-service-based carrier selection in high-speed scenarios takes effect, unnecessary inter-frequency handovers are triggered. As a result, UE distribution among frequency bands changes, air interface signaling overhead increases, and the number of handovers increases. In addition, the values of accessibility and retainability indicators in high-speed cells may slightly fluctuate.

After this function is enabled, the downlink user-perceived data rate in a cell (indicated by the **User Downlink Average Throughput (DU)**) may fluctuate.

To prevent ping-pong handovers and ensure user experience, the following aspects need to be considered: networking of the live network and the coordination between "uplink-service-based carrier selection in high-speed scenarios" and other handover functions (such as frequency-priority-based inter-frequency handover). For details about networking requirements, see [14.2.3.4 Networking](#).

14.2.3 Requirements

14.2.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

14.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

14.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Uplink-service-based carrier selection in high-speed scenarios requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .

14.2.3.2.2 Mutually Exclusive Functions

None

14.2.3.2.3 Function Impacts

None

14.2.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

14.2.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types. Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, ensure that the network plan will not introduce the following situation: After a UE is handed over from cell 1 to cell 2 through "uplink-service-based carrier selection in high-speed scenarios", the UE is

handed over from cell 2 back to cell 1 through "uplink-service-based carrier selection in high-speed scenarios".

- During network planning, ensure that the network plan will not introduce the following situation: After a UE is handed over from cell 1 to cell 2 through "uplink-service-based carrier selection in high-speed scenarios", the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based inter-frequency handover.

NOTE

For details about coverage-based intra-frequency handover, coverage-based inter-frequency handover, and frequency-priority-based inter-frequency handover, see *SA Mobility Management in Connected Mode*.

14.2.3.5 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

14.2.4 Operation and Maintenance

14.2.4.1 Data Configuration

14.2.4.1.1 Data Preparation

[Table 14-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 14-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Multi-Frequency Algorithm Switch	NRCellAlgoSwitch <i>.MultiFreqAlgoSwitch</i>	Uplink-service-based carrier selection in high-speed scenarios requires that the HIGH_SPD_SERV_CARR_SEL_SW option be selected.
Multi-Frequency Algorithm Switch	NRCellFeatureSwitch <i>.MultiFreqAlgoSwitch</i>	To enable "service-based carrier selection enhancement in high-speed scenarios", select the HIGH_SPD_SERV_CARR_SEL_ENH_SW option.
UL Traffic Frequency Priority	NRCellFreqRelation <i>.ULTrafficFreqPriority</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Extend Switch	NRCeIlMobilityCo nfig.InterFreqHoE xtSwitch	It is recommended that the UL_LARGE_PKT_FREQ_PRI_SW option be selected when both frequency-priority-based inter-frequency handover and "uplink-service-based carrier selection in high-speed scenarios" are enabled.

14.2.4.1.2 Using MML Commands

Before using MML commands, refer to [14.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink-service-based carrier selection in high-speed scenarios
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=HIGH_SPD_SERV_CARR_SEL_SW-1;
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=5400, ULTrafficFreqPriority=1;
//Enabling service-based carrier selection enhancement in high-speed scenarios
MOD NRCELLFEATURESWITCH: NrCellId=0, MultiFreqAlgoSwitch=HIGH_SPD_SERV_CARR_SEL_ENH_SW-1;
//Enabling "uplink-large-packet-based frequency priority" when both frequency-priority-based inter-
frequency handover and "uplink-service-based carrier selection in high-speed scenarios" are enabled
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=UL_LARGE_PKT_FREQ_PRI_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink-service-based carrier selection in high-speed scenarios
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=HIGH_SPD_SERV_CARR_SEL_SW-0;
//Disabling service-based carrier selection enhancement in high-speed scenarios
MOD NRCELLFEATURESWITCH: NrCellId=0, MultiFreqAlgoSwitch=HIGH_SPD_SERV_CARR_SEL_ENH_SW-0;
//Disabling uplink-large-packet-based frequency priority
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=UL_LARGE_PKT_FREQ_PRI_SW-0;
```

14.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.2.4.2 Activation Verification

If any of the following counters has a value other than 0, uplink-service-based carrier selection in high-speed scenarios has taken effect:

- **N.HO.InterFreq.IntragNB.ExecSuccOut**
- **N.HO.InterFreq.Xn.IntergNB.ExecSuccOut**

Observe the uplink cell traffic volume (**N.ThpVol.UL.Cell**) before and after this function is enabled. If the value increases, this function has taken effect.

14.2.4.3 Network Monitoring

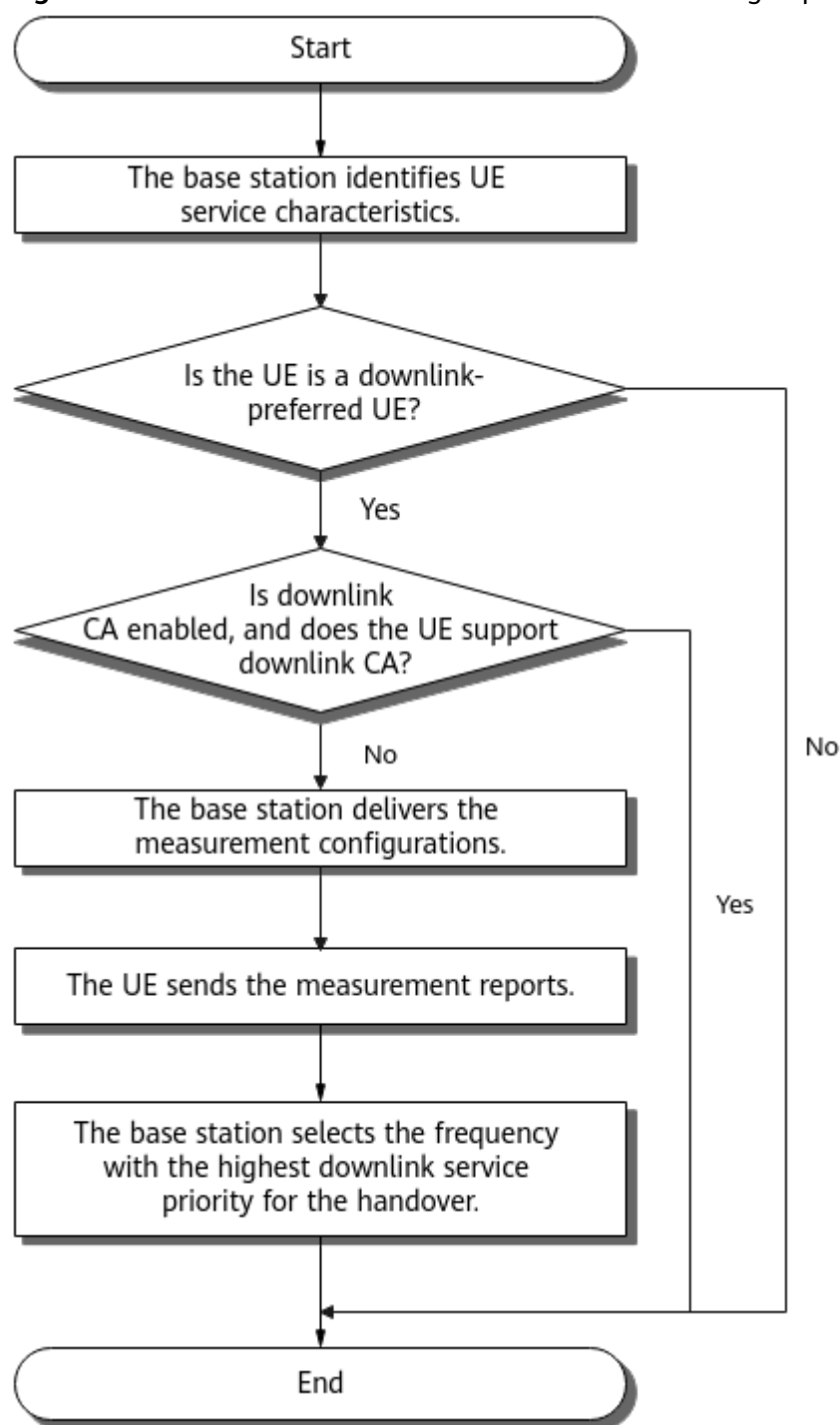
The indicators listed in [14.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

14.3 Downlink-Service-based Carrier Selection in High-Speed Scenarios

14.3.1 Principles

On a high-speed railway multi-carrier network, a base station triggers handovers or carrier changes for UEs mainly based on UE-reported RSRP, and selects carriers for UEs to camp on. As carrier difference is not considered, the carriers selected by the base station may not be able to provide the optimal experience for certain UEs running different services. Downlink-service-based carrier selection in high-speed scenarios is introduced to enable differentiated selection of optimal carriers for UEs with different preferred services. This function is controlled by the **HIGH_SPD_DL_SERV_CARR_SEL_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter.

Assume that downlink-service-based carrier selection in high-speed scenarios is enabled. Once identifying downlink-preferred UEs, the base station instructs these UEs to perform inter-frequency measurements. In this case, the base station performs carrier selection based on the frequency priorities for downlink services. The frequency priority for downlink services is specified by the **NRCellFreqRelation.DlTrafficFreqPriority** parameter. [Figure 14-2](#) shows the detailed procedure.

Figure 14-2 Downlink-service-based carrier selection in high-speed scenarios

1. Identification of UE service characteristics: After this function is enabled, the base station identifies a UE's service characteristics based on the following conditions, to determine whether it is a downlink-preferred UE within a measurement period (specified by the **NRDUCellServExp.FullBufferUeEvalPrd** parameter):
 - Downlink PRB usage > **NRDUCellServExp.FullBufferUeDlPrbUsageThld**
 - Proportion of slots for downlink scheduling > **NRDUCellServExp.FullBufferUeDlTtiRatioThld**

2. UE capability check: The base station obtains the frequency band (specified by the **NRCellFreqRelation.FrequencyBand** parameter) to which the neighboring frequency belongs and the frequency of the local cell and checks the UE capability. The subsequent procedure is triggered if either of the following conditions is met: (1) Downlink CA is enabled for the base station, but the UE does not support any band combination for downlink CA; (2) Downlink CA is disabled for the base station.

NOTE

For details about downlink CA, see *Carrier Aggregation*.

3. Measurement configuration delivery: When delivering inter-frequency measurement configurations, the base station selects frequencies with the **NRCellFreqRelation.DlTrafficFreqPriority** parameter set to a value other than 0 for A5-related inter-frequency measurement.

After "downlink-service-based carrier selection in high-speed scenarios" is enabled, the serving cell determines whether to deliver measurement configurations to the UE, based on the configuration of suppression of unnecessary handover for the UE's QCI-specific bearers. This process is referred to as QCI-specific suppression of unnecessary inter-frequency handover. The details are as follows:

- If the UE has a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB does not deliver measurement configurations to the UE.
- If the UE does not have a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB delivers measurement configurations to the UE.

NOTE

When "downlink-service-based carrier selection in high-speed scenarios" is enabled for the serving cell, you are advised to select the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter for each QCI-specific bearer associated with the target frequency group to prevent ping-pong handover if service-based inter-frequency handover (specified by the **SERVICE_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter) is enabled for the target cell.

4. Measurement result reporting: Based on the measurement result, the existing inter-frequency handover threshold (specified by the **NRCellInterFHoMeaGrp.MultiFreqSmtSelA5RsrpThld2** parameter) is reused to trigger an inter-frequency handover. If this parameter is set to 255, the RSRP threshold for interference-based inter-frequency handovers (specified by the **NRCellInterFHoMeaGrp.IntrfInterFreqHoRsrpThld** parameter) is used for handover decision.
5. Selection of the frequency with the highest downlink service priority for the handover: If there is a neighboring frequency with the highest downlink service priority, the base station performs a handover and terminates the inter-frequency measurement. If no neighboring frequency is detected within 3s, the procedure ends. The preceding procedure is started again after the UE is handed over to the next cell with this function enabled.

14.3.2 Network Analysis

14.3.2.1 Benefits

After downlink-service-based carrier selection in high-speed scenarios is enabled and takes effect in high-speed railway multi-carrier scenarios, the base station identifies UEs with downlink-preferred services and selects frequencies for handovers based on the frequency priorities for downlink services, improving downlink user experience in high-speed railway scenarios. In addition, UE handovers may change the downlink traffic distribution across cells, and the average downlink UE throughput (indicated by the **User Downlink Average Throughput (DU)**) may increase.

14.3.2.2 Impacts

The impacts between this function and other functions are described in [14.3.3.2.3 Function Impacts](#).

After downlink-service-based carrier selection in high-speed scenarios takes effect, unnecessary inter-frequency handovers are triggered. As a result, UE distribution among frequency bands changes, air interface signaling overhead increases, and the number of handovers increases. In addition, the values of accessibility and retainability indicators in high-speed cells may slightly fluctuate.

After this function is enabled, the uplink user-perceived data rate in a cell (indicated by the **User Uplink Average Throughput (DU)**) may fluctuate.

After this function is enabled, if the target neighboring frequency does not meet the coverage requirements and a downlink-preferred UE does not detect the target frequency within 3 seconds, the carrier selection procedure in the local cell is terminated. As a result, the downlink-preferred UE cannot obtain gains in the local cell, and the 3-second inter-frequency measurement is performed in vain. The downlink-preferred UE needs to wait until it is handed over to the next intra-frequency cell before the base station starts the carrier selection procedure again.

To prevent ping-pong handovers and ensure user experience, the following aspects need to be considered: networking of the live network and the coordination between "downlink-service-based carrier selection in high-speed scenarios" and other handover functions (such as frequency-priority-based inter-frequency handover). For details about networking requirements, see [14.3.3.4 Networking](#).

14.3.3 Requirements

14.3.3.1 Licenses

For details about the license requirements, see [4.3.1 Licenses](#).

14.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

14.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Downlink-service-based carrier selection in high-speed scenarios requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .

14.3.3.2.2 Mutually Exclusive Functions

None

14.3.3.2.3 Function Impacts

None

14.3.3.3 Hardware

The hardware requirements are the same as those for high-speed cells. For details, see [4.3.3 Hardware](#).

14.3.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types. Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, ensure that the network plan will not introduce the following situation: After a UE is handed over from cell 1 to cell 2 through "downlink-service-based carrier selection in high-speed scenarios", the UE is handed over from cell 2 back to cell 1 through "downlink-service-based carrier selection in high-speed scenarios".
- During network planning, ensure that the network plan will not introduce the following situation: After a UE is handed over from cell 1 to cell 2 through "downlink-service-based carrier selection in high-speed scenarios", the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based inter-frequency handover.

NOTE

For details about coverage-based intra-frequency handover, coverage-based inter-frequency handover, and frequency-priority-based inter-frequency handover, see *SA Mobility Management in Connected Mode*.

14.3.3.5 Others

The requirements are the same as those for high-speed cells. For details, see [4.3.6 Others](#).

14.3.4 Operation and Maintenance

14.3.4.1 Data Configuration

14.3.4.1.1 Data Preparation

Table 14-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 14-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Extend Switch	NRCellMobilityConfig . <i>InterFreqHoExtSwitch</i>	Downlink-service-based carrier selection in high-speed scenarios requires that the HIGH_SPD_DL_SERV_CARR_SEL_SW option be selected.
DL Traffic Frequency Priority	NRCellFreqRelation . <i>DLTrafficFreqPriority</i>	Set this parameter based on the network plan.

14.3.4.1.2 Using MML Commands

Before using MML commands, refer to [14.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink-service-based carrier selection in high-speed scenarios
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=HIGH_SPD_DL_SERV_CARR_SEL_SW-1;
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=5400, DLTrafficFreqPriority=1;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink-service-based carrier selection in high-speed scenarios  
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=HIGH_SPD_DL_SERV_CARR_SEL_SW-0;
```

14.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.3.4.2 Activation Verification

If any of the following counters has a value other than 0, downlink-service-based carrier selection in high-speed scenarios has taken effect:

- **N.HO.InterFreq.IntragNB.ExecSuccOut**
- **N.HO.InterFreq.Xn.IntergNB.ExecSuccOut**

14.3.4.3 Network Monitoring

The indicators listed in [14.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

15 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

16 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

17 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

18 Reference Documents

1. 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
2. 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
3. 3GPP TS 38.213: "NR;Physical layer procedures for control"
4. 3GPP TS 38.101: "NR;User Equipment (UE) radio transmission and reception"
5. *5G Networking and Signaling*
6. *CoMP*
7. *DRX*
8. *Hyper Cell*
9. *LTE and NR Power Allocation*
10. *MIMO (TDD)*
11. *Massive MIMO AHR (Low-Frequency TDD)*
12. *NR Inter-Carrier Dynamic Power Allocation*
13. *Turbo Uplink (Low-Frequency TDD)*
14. *URLLC*
15. *VoNR*
16. *Extended Cell Range*
17. *Super Uplink*
18. *Positioning*
19. *Uplink Scheduling*
20. *Downlink Scheduling*
21. *Modulation Schemes*
22. *Multi-Frequency Smart Selection*
23. *Multi-Frequency Smart Aggregation*
24. *3D Networking Experience Improvement*
25. *Power Control*
26. *Virtual Grid-based Multi-Frequency Coordination*
27. *Energy Conservation and Emission Reduction*
28. *3D Networking Experience Improvement*

29. *UL and DL Decoupling*
30. *Cell Management*
31. *Channel Management*
32. *Carrier Aggregation*
33. *Remote Interference Management (Low-Frequency TDD)*
34. *Cell Combination*
35. *SA Mobility Management in Connected Mode or NSA Mobility Management*
36. *SA Mobility Management in Idle Mode or NSA Mobility Management*
37. *UE Power Saving*
38. *3900 & 5900 Series Base Station Product Documentation*
39. *Co-MM (Low-Frequency TDD)*
40. *Mobility Load Balancing in Idle Mode*
41. *Fusion Cell (Low-Frequency TDD)*
42. *Beam Management*
43. *Distributed Antenna Solutions (Low-Frequency TDD)*
44. *RedCap*
45. *Coordinated Interference Management (Low-Frequency TDD)*
46. *Massive MIMO iBeam (Low-Frequency TDD)*
47. *ViNR*
48. *Multi-RAT Coordinated Energy Saving*
49. *Random Access Control*
50. *FWA*
51. *MRO*
52. *Network Slicing*
53. *Uplink Flexible Spectrum Scheduling*
54. *License Control Item Lists*
55. *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
56. *NG and Xn Self-Management*